PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

# Challenges in Digital Marketing and Client Communication for Real Estate Properties

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## ABSTRACT

This study developed Iloilo Top Property Homes (ITPH) to address challenges in digital marketing, property management, and client communication. This study focused on issues such as limited online property visibility, outdated property information, manual appointment management, delayed responses to client inquiries, and difficulties in organizing property records. The system was designed to provide a centralized platform for managing property listings, appointments, client inquiries, and agent activities. It also incorporated property search and filtering, map-based property location, and chat functionality to improve communication between clients and real estate personnel. The system was developed using the Agile software development methodology, which involved planning, system design, development, testing, deployment, and user evaluation.

The developed ITPH system provided clients with an accessible platform for browsing available properties, viewing property details, checking property availability, locating properties through a map, and scheduling appointments online. The system also provided administrators with tools for managing property listings, monitoring appointments, handling inquiries, and managing agents. The chat functionality supported faster communication by allowing clients to send inquiries and receive automated responses. These features helped organize property information and client transactions within a centralized, web-based system.

The study concluded that ITPH provided a structured and accessible platform for real estate property management and client communication. By integrating property listings, appointment scheduling, mapping, and chat functionality, the system supported more organized business processes and reduced reliance on manual methods. The system also provided real estate personnel with a more efficient way to manage property information and client interactions, supporting improved service delivery and day-to-day operational efficiency.

# CHAPTER I

# INTRODUCTION

## 1.1 Background of the Study

In today's real estate industry, digital transformation has significantly changed how businesses operate and how clients search for properties. Most modern clients now rely heavily on online platforms such as websites, social media, and property listing applications to find available real estate options. They expect real-time updates, high-quality images, detailed descriptions, and fast communication from agents or property sellers. Because of this shift in consumer behavior, the real estate industry is becoming more dependent on digital tools and online accessibility.

However, despite this growing demand for digital convenience, many real estate businesses still rely on traditional and outdated marketing approaches. These include printed advertisements, manual posting of listings, and physical inquiries handled in person or through basic messaging applications. Such methods are no longer sufficient in a highly competitive, and technology-driven market. The gap between customer expectations and business practices continues to widen, making it difficult for traditional real estate agencies to keep up. As a result, businesses that fail to adapt to digital systems often experience reduced customer engagement and lower conversion rates. This situation highlights the urgent need for modernization in real estate operations through the integration of web-based systems that can streamline communication, improve marketing strategies, and enhance overall service efficiency.

In many real estate companies today, client communication remains largely manual and inefficient. Agents typically handle inquiries through messaging applications, phone calls, or face-to-face interactions. While these methods are functional, they are not scalable, especially when agents are managing multiple clients simultaneously. Each inquiry must be answered individually, which consumes a significant amount of time and effort. This manual approach often leads to delayed responses, missed messages, and inconsistent communication quality. Clients, especially those searching urgently for properties, may become frustrated when responses are not immediate. In a competitive market, even small delays can result in losing potential buyers to other agencies that offer faster and, more responsive systems. Additionally, manual communication does not provide a structured way of tracking conversations, making it difficult for agents to follow up effectively. Without proper organization, important client inquiries may be overlooked or forgotten. This inefficiency reduces productivity and affects the overall performance of real estate businesses. Therefore, there is a strong need for an automated communication system that can manage inquiries more efficiently, provide instant responses, and ensure that no client interaction is missed or delayed.

Another major challenge in the real estate industry is the management of property listings and marketing materials. Many businesses still rely on outdated or inconsistent updates when posting properties online. In some cases, properties that have already been sold or reserved remain visible on listings, which creates confusion among potential buyers. Additionally, property descriptions may lack detail, and images may not accurately represent the actual condition of the property. This reduces the credibility of the business and negatively affects customer trust. Modern clients expect visually appealing and well-organized property presentations that allow them to make informed decisions without physically visiting the location. Without proper digital marketing tools, real estate companies struggle to maintain accurate and engaging listings. This issue also affects branding, as poorly maintained listings give the impression of unprofessional. In contrast, businesses that utilize updated digital platforms can attract more clients and improve engagement. Therefore, there is a need for a centralized system that allows real-time updates of property listings, ensuring accuracy, consistency, and visual appeal across all available properties.

The rise of Overseas Filipino Workers (OFWs) as property investors has also influenced the need for improved real estate systems. OFWs often rely entirely on online platforms when searching for properties because they are physically located outside the country. They cannot easily visit properties or conduct in-person inspections, making digital information their primary basis for decision-making. Because of this, real estate businesses must

ensure that their online platforms are complete, accurate, and responsive. Delays in communication or incomplete property details can lead OFW clients to lose interest and seek alternative providers. This highlights the importance of having a reliable web-based system that supports remote property browsing, appointment, and communication. By providing accessible and updated information, real estate businesses can better serve OFW clients and increase their chances of securing international buyers. The demand from overseas investors further emphasizes the need for automation and digital transformation in the real estate sector.

Agents in real estate businesses often experience high workloads due to the volume of inquiries and follow-ups they must manage daily. Without an automated system, agents must manually track each client interaction, respond to messages individually, and continuously monitor interest levels. This repetitive process consumes valuable time that could otherwise be used for closing deals or improving client relationships. Over time, this workload leads to inefficiency, stress, and reduced productivity among agents. In some cases, important inquiries may be missed due to human error or lack of proper tracking systems. This directly affects sales performance and customer satisfaction. A structured and automated system would allow agents to organize inquiries systematically, prioritize urgent requests, and automate follow-up messages. This would significantly reduce workload while improving overall efficiency in handling client transactions.

The absence of an automated inquiry tracking system is another critical issue in many real estate operations. Without a centralized platform, it becomes difficult to monitor client interactions and determine the status of each inquiry. Agents often rely on memory, manual notes, or scattered messaging platforms to manage their communication history.

This increases the risk of miscommunication and lost opportunities. Additionally, management teams find it difficult to evaluate performance or identify which properties are generating the most interest. A web-based system with inquiry tracking capabilities would solve this issue by storing all client interactions in a structured database. This allows for better monitoring, reporting, and decision-making based on real-time data.

Customer expectations in the real estate industry continue to evolve rapidly due to advancements in technology. Modern clients expect convenience, speed, and transparency in all transactions. They prefer platforms that allow them to browse properties, compare options, and communicate instantly with agents. Businesses that fail to meet these expectations risk losing competitiveness in the market. Traditional methods are no longer sufficient in providing the level of service that modern customers demand. Therefore, real estate companies must adopt digital solutions that enhance user experience and streamline operations.

Digital marketing plays a crucial role in improving the visibility of real estate properties. Through online platforms, businesses can reach a wider audience beyond their physical location. Social media integration, search engine optimization, and online advertisements significantly increase exposure. However, many real estate companies still lack proper digital marketing strategies, limiting their reach and effectiveness. A web-based system with integrated marketing features can help solve this issue by automating property promotion and ensuring consistent online presence.

The integration of automation in real estate systems improves efficiency by reducing manual tasks and human errors. Features such as automated notifications, inquiry responses, and real-time updates allow businesses to operate more smoothly. Automation also ensures that clients receive timely responses and accurate information. This improves customer satisfaction and builds trust in the business. In addition, automation allows agents to focus more on important tasks such as closing deals and building client relationships.

Overall, the real estate industry is undergoing a major transformation driven by digital technology and changing consumer behavior. Businesses that continue to rely on traditional methods face challenges in competitiveness, efficiency, and customer satisfaction. The adoption of a web-based system with digital marketing and automated

communication features is essential in addressing these challenges. Such a system improves communication, enhances property visibility, and streamlines operations for both clients and agents. Ultimately, it supports the goal of modernizing real estate services and improving overall business performance in a digital-first environment.

## 1.2 Statement of the Problem

The real estate business plays a vital role in providing housing opportunities, investment options, and property services for individuals, families, and investors. However, despite its importance in the economy and society, many real estate businesses are currently experiencing significant operational challenges. These challenges are mainly caused by the continued use of outdated marketing practices, manual processing of transactions, and inefficient handling of client inquiries. As the industry continues to evolve in a digital-first environment, businesses that fail to adopt modern technologies face difficulties in remaining competitive and relevant in the market. In today's digital age, clients expect real estate services to be fast, accessible, and technology driven. Most property seekers now rely on online platforms to search for listings, compare options, and communicate with agents. They expect real-time responses, updated property information, and visually appealing presentations of available properties.

However, many real estate businesses are still dependent on traditional methods that are no longer aligned with modern consumer behavior. This gap between customer expectations and business capabilities creates inefficiencies that negatively affect customer satisfaction and business performance. The general problem addressed by this study is: How can a web-based real estate management system with integrated digital marketing and automated communication features be developed and implemented to improve the operational efficiency, client engagement, and overall business performance of real estate businesses in San Miguel, Iloilo City?

This general problem encompasses the fundamental challenge of bridging the gap between traditional real estate operational practices and the digital capabilities required to compete effectively in the modern market. It recognizes that technological solutions must be not merely technically functional but effectively integrated into business operations to produce meaningful improvements in performance outcomes.

This study aimed to address the following problems:

1. The business is not reaching enough new customers, especially its intended target audience.
2. Marketing materials and property listing appear outdated and less appealing to clients.
3. Responses to client inquiries are slow because most inquiries are handled manually.
4. Agents are overload with follow-ups and inquiry management tasks.
5. There is no automated system for tracking and updating client inquiries.
6. There is a need to determine how business can effectively attract more customer through improved marketing strategies.

## 1.3 Objectives of the Study

The rapid advancement of digital technology has fundamentally altered the landscape of business operations across all industries, with real estate experiencing particularly significant impacts due to the Information intensive and relationship-dependent nature of property transactions. Clients now demand services that are faster, more accessible, more transparent, and more responsive than traditional real estate operational models can consistently deliver. The expectations shaped by digital experiences in e-commerce, social media and on demand services have transferred to
real estate, creating both challenges for businesses that fail to adapt and opportunities for those that successfully embrace digital transformation.

In response to these evolving market realities, this study undertakes the development of a comprehensive web based real estate management system specifically designed to address the operational challenges identified in the San Miguel, Iloilo City real estate market. The system's design is informed by thorough analysis of current operational practices, examination of best practices in property technology from both local and international contexts, and direct engagement with stakeholders including real estate business owners, agents, and prospective clients. The objectives are formulated to be specific, measurable, achievable, relevant, and time-bound (SMART), ensuring that the study's outcomes can be clearly evaluated against defined success criteria.

### 1.5.1 General Objective

The general objective of this study is to develop and implement a web-based real estate management system called ILOILO TOP PROPERTY HOMES (ITPH) that enables clients to conveniently browse and reserve real estate properties online while providing automated communication features such as chat functionality. The system also aims to assist administrators in efficiently managing property listings, monitoring appointments and handling client inquiries in a more organized, systematic, and data-driven manner.

Through the strategic integration of digital tools, automated processes, and user-centered design principles, the proposed system seeks to reduce manual workloads, improve response times, enhance the quality and consistency of client service and modernize traditional real estate operations. The system aims to shift real estate businesses from fragmented, manual processes to a centralized, technology-driven platform that supports improved client engagement, operational efficiency, and business competitiveness in the San Miguel, Iloilo City real estate market.

### 1.5.2 Specific Objective

Specifically, this study aimed to:

**Objective 1: Develop a user-friendly interface for property browsing.**
This objective focuses on the design and implementation of an intuitive, responsive, and visually appealing user interface that allows clients to easily browse available real estate properties in San Miguel, Iloilo City.

**Objective 2: Implement an Online Appointment System**
This objective involves the development of a secure, efficient, and reliable appointment system that enables clients to schedule appointments with real estate agents directly through the web-based platform. The system will automatically record appointment details in a centralized database, ensuring data accuracy, minimizing manual encoding errors and providing instant confirmation to both clients and administrators.

**Objective 3: Provide automated communication through chat functionality.**
This objective addresses the critical need for improved client communication by integrating an automated communication feature that enables clients to send inquiries and receive timely, consistent responses Regarding property details, availability, pricing, and appointment status. The chat functionality will accessible directly within the web platform, providing a seamless communication channel that does not require clients to use external messaging applications or email.

**Objective 4: Develop an administrative dashboard for system management.**
This objective focuses on the creation of a comprehensive administrative dashboard that provides Administrators with centralized control over all system functions and real-time visibility into business operations.

**Objective 5: Ensure secure data handling and system protection.**
This objective addresses the critical importance of security in any system handling sensitive personal and business information. The study will implement comprehensive security measures including robust login authentication with password encryption, role-based access control that restricts system functionality based on user roles, input validation to prevent injection attacks and data corruption, and defensive programming techniques to protect against common web application vulnerabilities.

## 1.4 Scope and Limitations

**Scope:**
This project encompasses the complete development of a web-based real estate management system designed specifically for real estate properties located in San Miguel, Iloilo City. The system provides a centralized digital platform accessible through standard web browsers, enabling multiple categories of users to perform their respective functions within a unified, integrated environment.

- **Geographic Scope:** The system focuses exclusively on real estate properties situated within the municipality of San Miguel, Iloilo City. This geographic focus ensures that property data, location information, market context, and system features are specifically relevant and accurate for this defined area. The system will support property listings across all barangays and development areas within San Miguel, encompassing residential, commercial, and mixed-use properties available in the local market.

- **Functional Scope, Client-Facing Features:** The system provides comprehensive property browsing capabilities allowing clients to view available properties, access detailed property information including images, descriptions, specifications, pricing, and availability status, and utilize search and filtering functions to identify properties matching their criteria. Clients can create personal accounts to manage their browsing history, saved properties, and appointment requests. The appointment management feature enables authenticated clients to submit appointment requests for properties of interest. These appointment are automatically recorded in the system database with time stamp and client information, generating notifications to administrators and initiating the appointment management work flow. Clients can track their appointment status through their account dashboard and receive updates as their appointment progresses. The integrated communication feature provides clients with the ability to send inquiries directly through the system interface, initiating structured conversations with assigned agents. All communications are stored within the system, creating comprehensive interaction records accessible to both clients and administrators.

- **Functional Scope, Administrative Features:** The administrative module provides comprehensive property listing management capabilities, enabling authorized administrators and agents to add new properties, upload images, enter detailed specifications, set pricing and availability status, and update or remove listings as needed. The system ensures that all client-facing information reflects current, accurate data through centralized content management. The inquiry and communication management features provide administrators with tools to view, organize, assign, and respond to client inquiries through the integrated chat interface. Comprehensive communication histories enable quality monitoring, agent performance assessment, and service continuity across staff changes. The appointment monitoring and management features enable administrators to view all pending and confirmed appointment, update appointment statuses, coordinate appointment scheduling, and generate reports on appointment activity. The dashboard provides real-time visibility into key business metrics including inquiry volumes, response times, appointment rates, and property interest levels. User Scope: The system serves three primary categories of users. Clients are individuals seeking to browse, inquire about, or reserve real estate properties. Agents are authorized real estate professionals assigned by administrators to manage client inquiries, update property information, and coordinate appointments. Administrators are senior authorized users responsible for overall system management, user administration, property oversight, and business intelligence. Technical Scope: The system is developed using established web technologies including HTML, CSS, and JavaScript for front end user interface development, PHP for server-side processing and business logic implementation, and MySQL for relational database management. This technology stack was selected based on widespread availability, extensive documentation, proven reliability, and compatibility with standard web hosting infrastructure commonly used by small to medium-sized businesses in the Philippines.The system implements responsive web design principles to ensure optimal user experience across desktop computers, tablets, and smart phones. Security measures include password-based authentication, session management, input validation, SQL injection prevention, and role-based access control.

**Delimitation:**
While the ILOILO TOP PROPERTY HOMES (ITPH) system provides comprehensive functionality for its intended scope, certain features and capabilities are explicitly excluded from the current development phase. These delimitation are not permanent restrictions but rather boundaries that define the current capstone project scope, with the understanding that future development iterations may expand system capabilities beyond these boundaries.

- **Financial Transaction Processing:** The system does not include online payment processing, bank integration, e-wallet transactions, or any form of electronic financial transaction handling. All financial dealings, including appointment fees, down payments, and full property payments, are assumed to be handled externally between the client and the real estate office through traditional payment methods. This delimitation reflects the significant technical complexity, regulatory requirements, and security considerations involved in implementing secure online payment systems, which extend beyond the scope and resources of the current capstone project. The system does, however, record appointment intentions and facilitate the scheduling of property viewings and consultations that may lead to offline financial transactions.

- **Advanced Digital Marketing Automation:** While the system improves digital marketing effectiveness through professional property presentation and web presence, it does not include advanced digital marketing automation features such as automated social media posting, search engine optimization tools, paid advertising management, email marketing campaigns, or customer segmentation for targeted marketing. The system provides the foundation digital platform upon which such marketing capabilities could be built in future development phases, but the current implementation focuses on core operational functionality rather than sophisticated marketing automation.

- **Legal Documentation Processing:** The system does not handle legal documentation such as contract generation, deed preparation, ownership transfer processing, title verification, or any other legally binding document management. These processes remain outside the system and are handled manually by authorized legal and real estate professionals. The inclusion of legal document processing would require specialized legal expertise, compliance with Philippine real estate law and regulations, and integration with government registry systems, all of which are beyond the current project scope.

- **Property Valuation and Automated Pricing:** The system does not provide automated property valuation, pricing recommendations, or market analysis tools. Property pricing decisions remain the exclusive responsibility of the real estate business based on their market knowledge, comparative analysis, and internal policies. Automated valuation models require extensive historical transaction data, sophisticated analytical algorithms, and continuous market data feeds that are not available within the scope of this project.

- **External Communication Integration:** The system does not integrate with external real-time communication channels such as direct phone systems, SMS messaging services, email servers, or third-party messaging platforms. All client-agent communication is contained within the system's integrated chat interface to maintain security, ensure comprehensive record-keeping, and provide a consistent user experience. While this design choice ensures data integrity and security, it means that clients must access the web platform to initiate and review communications rather than using their preferred external messaging applications.

- **Mobile Application Development:** The system is implemented as a responsive web application accessible through standard web browsers rather than as a native mobile application for IOS or Android platforms. While the responsive design ensures usability across mobile devices, the system does not leverage native mobile capabilities such as push notifications, offline storage, camera integration, or GPS functionality that would be available in a dedicated mobile application.

- **Offline Functionality:** The system requires active internet connectivity for all functions and does not include offline capabilities or data synchronization features for environments with intermittent or unreliable

internet access. Users must have access to a stable internet connection and a compatible device to utilize the platform fully.

- **Multi-Location Expansion:** While the system architecture could theoretically support expansion to additional locations, the current implementation is configured exclusively for San Miguel, Iloilo City. The system does not include features for managing properties across multiple municipalities, cities, or regions, nor does it support franchise or multi-access operational models.

## 1.5 Definition of Terms

1. **Administrator.** Conceptually, an administrator refers to an individual with the highest level of authority and responsibility within an organizational system. Operationally, in the context of the ILOILO TOP PROPERTY HOMES (ITPH) system, an Administrator is the highest-level authorized user who possesses full control over all system features, functions, and configurations. Administrators are responsible for managing the overall operation of the real estate platform, including creating and managing user accounts for agents and other staff, overseeing property listing management, monitoring all appointment activities, supervising agent performance, configuring system settings, and ensuring data security and integrity.

2. **Agent.** Conceptually, an agent is an individual authorized to act on behalf of another party in business transactions. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, an Agent is an authorized user assigned by the Administrator to perform specific operational responsibilities including handling client inquiries, updating property information, following up on appointments and appointment, and communicating with clients through the integrated chat interface. Agents have access to a defined subset of system features relevant to their designated responsibilities but do not possess full administrative control.

3. **Appointment.** Conceptually, appointment refers to a scheduled meeting or arrangement. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system.

4. **Automated Communication.** Conceptually, automated communication refers to the use of technology to generate and deliver messages without requiring continuous human intervention. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, automated communication encompasses system-generated responses and notifications that assist in handling client inquiries and managing operational work flows.

5. **Client.** Conceptually, a client refers to an individual or organization that uses the professional services of another. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, a Client is an individual or potential property buyer who accesses the system to view available real estate listings, submit inquiries, make appointment and communicate with agents.

6. **Digital Marketing.** Conceptually, digital marketing refers to the use of digital channels, platforms, and technologies to promote products and services to consumers. Operationally, in this study, digital marketing encompasses the strategies and tools used to promote real estate properties through the ILOILO TOP PROPERTY HOMES (ITPH) web platform and associated online channels.

7. **MySQL.** Conceptually, MySQL is an open-source relational database management system. Operationally, MySQL serves as the database management system for the ILOILO TOP PROPERTY HOMES (ITPH) system, providing reliable, scalable, and efficient storage and retrieval of all system data. Overseas Filipino Workers (OFW's). Conceptually, OFW's are Filipino nationals who are employed in foreign countries. Operationally, in this study, OFW's represent a significant target market segment for the ILOILO TOP PROPERTY HOMES (ITPH) system, characterized by their reliance on digital platforms for property research and transactions due to physical absence from the Philippines.**PHP (Hypertext Preprocessor).** Conceptually, PHP is a widely used open-source server-side scripting language designed for web development. Operationally, PHP serves as the

primary programming language for the server-side components of the ILOILO TOP PROPERTY HOMES (ITPH) system.

8. **Property Listing.** Conceptually, a property listing is a published advertisement or record of a real estate property available for sale, lease, or appointment. Operationally, in the ILOILO TOP PROPERTY HOMES (ITPH) system, a Property Listing is a digital record within the database that represents an individual real estate property.

9. **PropTech (Property Technology).** Conceptually, PropTech refers to the application of technology to the real estate industry to improve efficiency, transparency, and user experience. Operationally, PropTech encompasses the digital tools, platforms, and systems used to modernize real estate operations.

10. **Role-Based Access Control (RBAC).** Conceptually, RBAC is a security approach that restricts system access based on the roles of individual users. Operationally, the ILOILO TOP PROPERTY HOMES (ITPH) system implements RBAC by defining three primary user roles, Administrator, Agent, and Client, each with specific permissions and access rights.

11. **User Acceptance Testing (UAT).** Conceptually, UAT is the final phase of software testing in which intended end users test the system. Operationally, UAT for the ILOILO TOP PROPERTY HOMES (ITPH) system involves representative users from each user category evaluating the system's functionality, usability, and overall performance.**Web-Based System.** Conceptually, a web-based system is a software application that operates through a web browser and can be accessed via the internet. Operationally, the web-based system refers to the ILOILO TOP PROPERTY HOMES (ITPH) platform, which serves as the main interface where clients can browse properties, make appointment, and communicate with agents.

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# CHAPTER II

# REVIEW OF RELATED LITERATURE

## 2.1 Foreign Studies

**FOREIGN RELATED STUDIES**

**A. Digital Marketing in Residential Real Estate**

Rabby and Chimhundu [1] conducted a study entitled "Beyond the Binary: Deciphering the Nexus and Nuances of Digital Marketing on Customer Purchase Behaviour in the Residential Real Estate Market of Sydney." The researchers surveyed 399 residential real estate consumers in Australia to examine the relationship between digital marketing and consumer purchase behavior. Their findings indicated that digital marketing positively influences customer attitudes, trust, and satisfaction, which contribute to purchase intention. This study is relevant to the ITPH project because it supports the use of digital platforms to promote real estate properties, present property photographs and descriptions, and provide prospective buyers with accessible property information.

**B. Online Real Estate Platforms**

Ullah et al. [2] conducted a study entitled "Modelling Users' Perception of the Online Real Estate Platforms in a Digitally Disruptive Environment: An Integrated KANO-SISQual Approach." The study examined users' perceptions of online real estate platforms and the factors that influence their experiences. This research is relevant to the ITPH project because the proposed website provides online property listings, photographs, descriptions, and search functions. The study provides a basis for considering platform usability, information quality, and reliability when developing a real estate website.

**C. Customer Service Chatbots**

Hsu and Lin [3] conducted a study entitled "Understanding the User Satisfaction and Loyalty of Customer Service Chatbots." The researchers analyzed survey responses from 219 chatbot users to examine how chatbot service quality affects user satisfaction and loyalty. Their findings indicated that conversational quality and service recovery quality significantly influenced user satisfaction, while core service quality and satisfaction influenced chatbot loyalty. This study is relevant to the ITPH project because the ITPH Assistant chatbot is intended to help customers obtain property information and answers to common inquiries. The findings emphasize the importance of providing clear, useful, and appropriate responses to improve customer experience.

**D. Chatbot Satisfaction and Continued Use**

Ashfaq et al. [4] conducted a study entitled "I, Chatbot: Modeling the Determinants of Users' Satisfaction and Continuance Intention of AI-Powered Service Agents." The researchers examined the factors affecting user satisfaction and continued use of AI-powered chatbots. Based on responses from 370 chatbot users, the study identified information quality and service quality as important factors influencing user satisfaction. Perceived usefulness and ease of use were also associated with users' intention to continue using chatbot services. This study is relevant to the ITPH project because the website includes a chatbot designed to assist customers with property inquiries and related information. It highlights the importance of making chatbot interactions understandable, useful, and reliable.

## E. Chatbots and User Experience

Cheng and Jiang [5] conducted a study entitled "How Do AI-Driven Chatbots Impact User Experience? Examining Gratifications, Perceived Privacy Risk, Satisfaction, Loyalty, and Continued Use." The researchers surveyed 1,064 consumers in the United States to examine the relationship between chatbot use, user

satisfaction, perceived privacy risk, and continued use. The study associated informational benefits with chatbot satisfaction and found that perceived privacy risk negatively affected satisfaction. Satisfaction was also associated with continued use and customer loyalty. This study is relevant to the ITPH project because the website aims to provide helpful automated customer support while maintaining the privacy and security of customer information.

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### 2.1.1 Web-Based Real Estate Management Systems and Property Platforms

The development of web-based real estate platforms has changed how buyers search for properties and how real estate businesses present their listings and communicate with potential customers. Traditional methods, such as printed brochures, physical advertisements, and face-to-face inquiries, may limit the accessibility of property information and make communication more time-consuming. Through web-based platforms, real estate businesses can provide property listings, photographs, descriptions, location information, and inquiry channels in a centralized online environment. These features allow prospective buyers to explore available properties and review relevant information before deciding whether to contact an agent or schedule an appointment. Consequently, the effectiveness of a real estate website depends not only on its ability to display listings but also on the accuracy of its information, ease of navigation, and support for customer inquiries.

Ullah et al. [1] examined users' perceptions of real estate online platforms through an integrated KANO-SISQual approach. Their study collected questionnaire responses from 407 users in Sydney and identified reliability, information accuracy, and platform credibility as particularly important factors influencing users' perceptions. The researchers also identified website design, content structure, and informative visual features as elements that can improve the user experience. These findings demonstrate that real estate platforms should provide reliable property information and present it in an organized and understandable manner. This study is relevant to the development of the Iloilo Top Property Homes (ITPH) website because its property listings include photographs, descriptions, and other property details intended to help customers evaluate available housing options. By organizing these details in a centralized platform, ITPH aims to make property information more accessible to prospective buyers and reduce the need to rely exclusively on printed promotional materials or direct inquiries.

Sing and Zou [2] investigated how online property listings affect search efficiency in the housing market. Using property listing and transaction data from Singapore's private residential property market, the researchers found that online listing portals help buyers search through more properties and identify options that better match their preferences. Their findings also indicated that listing descriptions that contain more information and are easy to read are associated with more efficient property transactions. This research highlights the importance of providing clear descriptions and useful property details instead of presenting listings with limited information. In relation to ITPH, the website allows users to browse property listings and review information about the

available properties before proceeding with an inquiry or appointment request. Search and filtering features can further help users narrow down their choices according to their needs. These functions support the objective of making property discovery more convenient and helping prospective buyers identify properties they wish to explore further.

Lee [3] examined how homebuyers seek information through real estate platforms in Taiwan. The study emphasized the importance of factors such as brand perception, search functionality, and search results in users' experiences with online property platforms. It also discussed the potential of artificial intelligence-based search features to support property information discovery. These findings suggest that a real estate website should make it easy for users to locate relevant listings and understand the information presented. This is applicable to ITPH because the website provides a digital space where customers can explore properties and obtain information without immediately visiting a physical office. Its property browsing and search functions are intended to simplify the process of locating potential housing options. Although ITPH does not necessarily implement every advanced feature discussed in the study, the research provides a basis for prioritizing clear navigation, accessible property details, and effective search functionality.

In addition to property information and search functionality, electronic service quality is an important consideration in the development of real estate websites. Yalçın and Çatlı [4] studied the electronic service quality of digital marketplaces in real estate marketing. Their research examined the relationship between electronic service quality, customer satisfaction, and customer loyalty among users in Ankara, Türkiye. The findings indicated that electronic service quality is significantly associated with customer satisfaction and loyalty. This suggests that a digital platform should not only display information but also provide a satisfactory and convenient experience throughout the user's interaction with the website. For ITPH, this principle is relevant to the process of browsing properties, submitting inquiries, and requesting appointments. Clear instructions, understandable page layouts, and accessible contact channels can help customers complete these activities with less confusion. The study therefore supports the importance of evaluating ITPH from the perspective of usability and customer experience rather than considering website functionality alone.

The reviewed studies collectively demonstrate that effective real estate platforms require accurate property information, organized listings, convenient search functionality, and satisfactory electronic services. These factors provide a foundation for the development of the ITPH website, which is designed to support property discovery and customer communication within its intended service area in Iloilo. Unlike a platform that focuses only on property listing and display, ITPH also provides inquiry and appointment-related functions intended to help customers move from browsing a property to communicating with the real estate team. Its website presents property photographs and descriptions, allows users to explore available listings, and provides channels through which customers can express their interest in a property.

Furthermore, ITPH incorporates customer communication and appointment-related functions to support interactions among customers, agents, and administrators. These functions are relevant to the communication challenges addressed by the present study, particularly the need to manage customer inquiries and appointment requests more systematically. The website is intended to provide a more organized alternative to relying solely on printed promotional materials and informal communication. However, the reviewed studies also indicate that the usefulness of a real estate platform depends on the quality and reliability of the information and services it provides. Therefore, the development and evaluation of ITPH should consider whether users can locate property information, understand listing details, submit inquiries, and request appointments with minimal difficulty.

Overall, these international studies provide a research basis for the development of a web-based real estate platform that emphasizes accessible property information, efficient searching, and improved electronic service quality. The findings support the design priorities of ITPH while also establishing the need to evaluate its usability and effectiveness in addressing the identified digital marketing and client communication challenges. Although the cited studies were conducted in other countries and market settings, their findings offer relevant concepts that can be considered in developing and evaluating a real estate website for the intended users in Iloilo.

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### 2.1.2 Digital Marketing Strategies and Automation in Real Estate

The transformation of real estate marketing from traditional to digital channels has been extensively studied in international contexts, with researchers examining the effectiveness of various digital strategies, the role of social media, search engine optimization, and the integration of comprehensive marketing automation tools. The shift from print media and physical signage to digital platforms represents more than a change in medium; it constitutes a reconceptualization of how properties are discovered, evaluated, and transacted. Digital marketing enables persistent, searchable, and shareable property presentations that extend the temporal and geographic reach of listings far beyond the limitations of traditional methods. Research on digital marketing strategy for property developers in Indonesia [21] examined the case of PT Widyaka Propertindo to analyze how digital marketing strategies could improve property sales and market reach.

The study grounded its analysis in established marketing theories, specifically the 7Ps Marketing Mix Model originally developed by Booms and Bitner [3] and further elaborated by marketing scholars including Kotler and Armstrong [18]. The research also drew from digital marketing frameworks by Chaffey and Chadwick [5], who defined digital marketing as achieving marketing objectives through applying digital technologies and media. The study's findings revealed that property developers implementing comprehensive digital marketing strategies experienced significantly improved market reach and customer engagement compared to those relying solely on traditional methods such as printed brochures, billboards, and personal networking. The integration of social media
marketing, search engine optimization, and online property listings created multiple touchpoints for potential buyers, particularly younger demographics and overseas investors who primarily rely on digital channels for property research and decision-making.

The research emphasized that digital marketing in real estate involves strategic content creation, targeted advertising based on demographic and behavioral data, and data-driven decision making to optimize marketing effectiveness and return on investment. The study concluded that the transition from traditional promotional methods to digital marketing through web-based systems is essential for competitive advantage in modern real estate markets. This finding directly supports the current study's objective of integrating digital marketing features into the real estate management platform, ensuring that properties listed in San Miguel, Iloilo City can reach broader audiences including OFW investors and younger local buyers.

The Indonesian context is particularly analogous to the Philippine context because both markets feature rapidly growing middle classes, significant overseas worker populations investing in domestic real estate, and high mobile device penetration rates that make digital channels the most efficient means of market access. Expanding on digital marketing automation, Birdeye [2] conducted extensive research on real estate marketing automation solutions, analyzing how automated work flows transform property marketing and client communication across diverse market contexts. The study examined various automation tools including CRM integrations, social media automation platforms, online review management systems, and lead nurturing workflows that operate across multiple digital channels simultaneously. The research defined real estate marketing automation as the use of software and technology to streamline and automate various marketing tasks, including lead generation, email campaigns, social media management, online reviews, and client follow-ups.

The study found that automation enables agents to create scalable, repeatable, and measurable marketing strategies that attract high-quality leads, nurture relationships throughout the sales cycle, and ultimately shorten the time from initial inquiry to closed transaction. From drip email campaigns and automated workflows to predictive analytics and AI-driven follow-ups, marketing automation empowers agents to work smarter and grow faster. A particularly compelling case study within this research examined Moss & Company, a property management firm managing over 300 properties.

After implementing comprehensive automation tools including automated review responses, AI-generated social media posts, and omnichannel messaging, the company achieved remarkable results: a 761% increase in map views, 218% rise in search views, and 290% boost in web traffic. These quantitative improvements translated directly into increased lead generation and client engagement, demonstrating the tangible business impact of marketing automation.The study emphasized that automation does not replace personal relationships but enhances them by allowing agents to maintain timely and relevant communication across every stage of the client journey.

This insight is crucial for the current study, as it validates the approach of automating routine marketing and communication tasks while preserving the human elements of real estate transactions that require empathy, negotiation skills, and personal judgment. The Moss & Company case study provides particularly strong evidence because it demonstrates results across multiple metrics simultaneously, suggesting that automation creates synergistic effects where improvements in one channel (map visibility) reinforce gains in others (web traffic and search visibility). Further supporting the role of automation, NetSuite [14] conducted comprehensive research on automating real estate processes, examining how automation technology encompasses property listings across multiple platforms, chatbots for customer inquiries, simplified contract management, and AI-powered lead generation tools.

The study found that by automating repetitive and time-consuming tasks, real estate professionals can redirect their efforts from administrative paperwork to high-value activities such as building client relationships, conducting property showings, and closing transactions. Citing the National Association of Realtors Research Group's 2023 Technology Survey, the study noted that 92% of real estate businesses now use at least one social media network to promote listings, maintain relationships with existing clients, generate leads, and track industry trends. Automated CRM systems collect contact information and preferences from customer interactions across social media posts,property listing websites, and other digital touch points, organizing this data into centralized databases. Agents can then use these unified profiles to more easily track and follow up with potential clients, identifying those most likely to proceed to the next stage of the transaction process.

The research concluded that automation helps businesses reduce operational costs and scale their operations without losing their ability to deliver personalized attention to clients. As technology continues to evolve, automation will likely bring even more benefits, but only for businesses that strategically implement, apply, and continuously update these tools.

This finding underscores the importance of building automation capabilities into the proposed system from the outset, rather than treating them as optional add-ons. For the San Miguel, Iloilo City context, where local real estate agencies may lack dedicated marketing staff, built-in automation features level the competitive playing field by enabling small agencies to maintain consistent digital presence comparable to larger firms with dedicated marketing departments.

### 2.1.3 Artificial Intelligence and Automated Communication in Real Estate

The integration of artificial intelligence and automated communication systems into real estate operations represents one of the most significant technological advances in the industry, with international research demonstrating substantial improvements in efficiency, client satisfaction, and conversion rates. AI in real estate is not limited to futuristic applications but encompasses practical, deployable tools that address specific operational bottlenecks, particularly in client communication and lead management. Research on AI-enhanced

CRM with chatbots for real estate lead optimization [20] investigated the integration of artificial intelligence and conversational agent technology into customer relationship management systems specifically designed for real estate applications. The study focused on three critical automation functions: inquiry handling through natural language processing, lead scoring based on prospect behavior and stated preferences, and automatic tour scheduling integrated with calendar systems.

The chatbot integration process followed a structured workflow: users send queries regarding property details, pricing, or location; the chatbot processes these queries using NLP techniques to understand intent and extract key parameters; the system communicates with the CRM database through APIs to retrieve relevant information; and the chatbot provides instant, contextual responses to the user. This architecture ensures that routine inquiries receive immediate attention regardless of time zones or agent availability, while complex or high-value inquiries can be seamlessly escalated to human agents.

The research employed case studies from five real estate companies to conduct rigorous before-and-after comparisons of system performance. Key performance indicators included containment rate (percentage of queries handled without human intervention), conversion rate (percentage of leads converted to customers), and return on investment (calculatedfrom cost savings and revenue increases). The comparative results demonstrated dramatic improvements across all metrics: response time decreased from high to low (specifically, from hours or days to seconds), conversion rates doubled from 20% to 40%, query handling capacity improved from 40% to 85%, and operational costs associated with initial inquiry management were significantly reduced. The study concluded that AI chatbot integration transforms CRM systems from passive databases into active engagement tools that operate continuously.

This research is particularly relevant to the current study as it provides empirical validation for the proposed chat functionality, establishes performance benchmarks that the current system can aim to achieve, and demonstrates that automated communication does not degrade service quality but rather enhances it by ensuring immediate responsiveness. The doubling of conversion rates is especially significant for the current study because it suggests that the inclusion of automated chat functionality in the proposed system could directly improve business outcomes for the target agency, not merely provide technological novelty.

Complementing this research, Whippy [25] investigated AI automation applications in real estate, focusing on specific use cases including automated follow-ups, appointment scheduling, and predictive lead prioritization. The study found that timely follow-ups build rapport and significantly boost referral rates, with AI assistants automatically sending thank-you messages after property showings, sharing property recaps, gathering client feedback, and delivering personalized new listings based on stated preferences and browsing history.

The research demonstrated that AI-driven scheduling tools can consider multiple variables simultaneously, including travel time between showings, client preferences for specific times or days, and priority levels assigned to different leads. This intelligent scheduling helps agents plan more efficient days with fewer conflicts and less unproductive travel time. Predictive lead scoring analyzed prospect behavior across multiple touch points including email opens, website property views, virtual tour engagement, and responses to SMS or chatbot prompts, creating composite scores that indicate likelihood to convert. Real-time alerts notified agents immediately when leads clicked on listings, viewed 3D tours, liked or commented on social media posts, or submitted website forms. This immediacy enabled agents to follow up during moments of peak interest, dramatically improving conversion rates compared to delayed responses.

The study concluded that AI tools help schedule outreach at ideal post-sale moments, maximizing response rates and ensuring satisfied clients share their experiences online, thereby generating valuable social proof and referral business. This body of research on AI and automated communication provides robust empirical support for the current study's planned chat functionality, automated inquiry responses, and appointment management features. The consistent finding across studies is that automation enhances rather than replaces human agent capabilities, allowing professionals to focus on high-value relationship-building activities while technology handles routine interactions. For the proposed system serving San Miguel, Iloilo City these findings suggest that

even a basic automated inquiry acknowledgment and follow-up system can yield measurable improvements in client satisfaction and conversion efficiency. Nylas [15] examined how integrated communication platforms enhance real estate software experiences, with a compelling case study of LionDesk, which became the #1 Real Estate CRM in 2022.

By integrating email and calendar functionalities through APIs rather than building custom solutions, LionDesk saved 28,242 development hours and $1.8 million while efficiently serving over 165,000 real estate professionals with centralized communication tools. The study found that streamlined agent-client interactions through integrated email and messaging, automated virtual property tour scheduling, and AI-driven personalization significantly improve productivity and client satisfaction. The LionDesk case study is instructive for the current research because it demonstrates the economic and operational value of communication integration. While the proposed system may not initially integrate external email and calendar APIs at the same scale, the principle of centralized communication—where all client interactions are logged, searchable, and actionable within a single interface—directly informs the proposed system's messaging and inquiry management modules. The quantified savings (development hours and costs) provide a business case for prioritizing communication integration over peripheral features.

## 2.2 Local Studies

This section examines research, market analyses, and system development projects conducted within the Philippine context. These studies provide essential localized insights into market conditions, user behaviors, regulatory environments, and technological infrastructures that directly influence the design, functionality, and potential success of the proposed web-based real estate management system for San Miguel, Iloilo City. The Philippine context presents unique characteristics that distinguish it from international markets: high mobile-first internet usage,
significant OFW investment flows, fragmented local property markets, and government initiatives specifically supporting digital transformation for overseas workers.

### 2.2.1 Philippine Real Estate Digital Platforms Market Analysis:

Ken Research [13] conducted a comprehensive market analysis of the Philippines Real Estate Digital Platforms market, providing detailed segmentation and identifying major players in the expanding digital property ecosystem. The report segmented the market by type (residential platforms, commercial platforms, rental platforms, investment platforms, auction platforms, property management platforms, industrial platforms, agricultural land platforms, vacation rental platforms, foreclosed property platforms), end-user (first-time home buyers, real estate agents, property developers, investors, renters, corporates), application (property listing, virtual tours, market analysis tools, customer relationship management, online auctions, digital payment integration), and distribution mode (mobile applications, web platforms, social media channels).

The research identified major players in the Philippine digital real estate market, including dedicated property platforms such as Lamudi Philippines, Property24 Philippines, MyProperty.ph, Dot Property Philippines, and Hoppler; general marketplace platforms including Carousell Philippines; and major property developers with digital presences such as Ayala Land, Megaworld Corporation, DMCI Homes, Century Properties, Vista Land, SM Development Corporation, Rockwell Land, and Filinvest Land. The presence of both dedicated real estate platforms and general marketplace sites indicates a fragmented but rapidly expanding digital ecosystem with varying levels of specialization and service quality.

The study found that the Philippine real estate digital platform market is experiencing rapid growth driven by several factors: increasing internet penetration across the archipelago, widespread smartphone adoption, changing consumer preferences toward online property research, and the growing importance of digital channels for reaching overseas Filipino investors. Web platforms and mobile applications serve as primary distribution channels, with property listing and virtual tours being the most utilized applications by consumers. This market analysis establishes the commercial viability and competitive context for the proposed system. While major platforms dominate national-level property searches, the study reveals opportunities for localized, specialized

platforms serving specific geographic markets such as San Miguel, Iloilo City. The current study addresses this market gap by providing a dedicated platform for a defined local area, potentially offering more relevant content, better community context, and more personalized service than national platforms that treat all locations uniformly. The identification of specific market segments (residential, commercial, rental, investment) also suggests that the proposed system should architecturally support multiple property types and user roles from the outset, even if initial implementation focuses on residential sales.

### 2.2.2 Local Real Estate Platform Development: The Tokkatok Case Study

Codica [6] documented the development of Tokkatok, a real estate marketplace specifically designed to address the unique challenges of the Philippine property market. This case study is particularly valuable because it emerged directly from the founder's personal experience of attempting to find rental properties in the Philippines through Facebook groups, where listings were frequently outdated, unverifiable, poorly structured, and lacking accountability mechanisms. The study identified several structural challenges in the Philippine real estate market that Tokkatok was designed to address: Unstructured market environment: The absence of reliable, dedicated digital platforms specifically built for property rentals and sales in the Philippines forced property seekers to rely on informal channels such as Facebook groups, word-of-mouth referrals, and physical signage. These channels lack verification mechanisms, standardized listing formats, and reliable communication tools. Technical barriers for non-technical founders: Many local real estate entrepreneurs lack technical backgrounds but recognize the need for digital platforms. They require development partners who can guide product decisions, technology choices, and implementation strategies without overwhelming them with technical complexity.

1. **Regional integration challenges:** Connecting Philippine platforms with local services such as GCash for payments and Facebook Business for marketing requires navigating local regulatory constraints, international compliance requirements, and technical compatibility issues.

2. **Accessibility requirements:** Given the diversity of digital literacy levels in the Philippines, platforms must be simple and usable for people unfamiliar with sophisticated digital tools, including older users and those in rural areas with limited technology exposure.

3. **Distributed team collaboration:** Technology development for Philippine markets often involves distributed teams working remotely across different time zones, requiring strong communication protocols and comprehensive documentation.

4. **Tokkatok was designed around three foundational pillars:** security, usability, and performance. The secure and flexible user management system implemented email verification, social login options, multi-factor authentication via email, JWT-based session handling with refresh token rotation, secure time-limited password recovery, step-by-step profile setup, and profile photo uploads stored in AWS S3 with automatic optimization.

The streamlined listing management system featured a multi-step wizard to guide users through property creation, context-aware fields that adjusted based on property category (residential, commercial, rental),batch drag-and-drop image uploads, map-based address selection with validation, draft mode and publishing controls, admin moderation queue with approve/reject logic, and optional paid features to highlight premium listings.

Advanced search and discovery capabilities were powered by Elastic search, providing faceted filters, price range selectors, category-specific filters for precision searching, proximity-based search by location, recommendations for similar listings on property detail pages, and emphasized featured listings in search results. The study placed particular emphasis on mobile responsiveness and performance optimization, recognizing that smartphones serve as the primary internet access device for most Filipinos. The interface prioritized mobile fluidity through responsive layouts for all screen sizes, optimized image loading for faster rendering, stable performance across older devices and slower networks, and intuitive mobile UX designed for non-technical users. This case study directly informs multiple aspects of the current research. The emphasis on mobile-first design validates the proposed system's responsive design approach. The identified market challenges

(fragmented listings, lack of verification, informal communication channels) mirror the problems identified in the current study's problem statement. The technology choices and feature priorities provide practical benchmarks for the proposed system's development. The Tokkatok experience also serves as a cautionary illustration of the consequences ofinadequate digital infrastructure: without structured platforms, markets devolve into inefficient, trust-deficient environments where buyers and sellers waste resources on verification and coordination tasks that automated systems could handle seamlessly.

### 2.2.3 Government Support for Digital Real Estate Platforms Serving OFWs

The Philippine News Agency [16] reported on significant government backing for Bahay Deals, an AI-powered online platform specifically designed to help Overseas Filipino Workers invest in real estate in the Philippines. This government endorsement represents a notable development in national policy, recognizing the critical economic role of OFW remittances and investments in the domestic property market while acknowledging the need for reliable digital tools to facilitate remote investment decisions.

The report detailed the specific challenges that OFWs face when investing in Philippine real estate from abroad: difficulty verifying property legitimacy and developer credibility, lack of transparent and standardized pricing information, limited communication channels with developers and agents across time zones, inability to physically inspect properties before making purchase decisions, and vulnerability to fraudulent schemes targeting overseas investors. Government support for digital platforms addresses these challenges by providing implicit credibility and trust markers for OFW investors. When government agencies endorse or support specific platforms, overseas investors gain confidence that these platforms meet certain standards of transparency, security, and reliability. This endorsement also signals national recognition of the importance of digital transformation in real estate to serve overseas investors effectively.

This study is directly relevant to the current research as it validates the importance of web-based real estate systems in serving OFW clients, a primary target demographic identified in the proposal. The government's active support for such platforms indicates that the current study aligns with national digitalization priorities and economic development strategies. Furthermore, the documented challenges faced by OFWs (verification, transparency, communication, remote inspection) directly inform the feature requirements for the proposed system, suggesting that comprehensive property information, high-quality imagery, automated communication, and transparent availability status are essential rather than optional features. The current study can position itself as contributing to national development goals by providing municipal-level digital infrastructure that complements national initiatives like Bahay Deals.

### 2.2.4 OFW Investment Behavior and Digital Platform Preferences

Unicapital Inc. [22] conducted research on how OFWs build wealth and invest their money, revealing that real estate remains the most favored investment option among overseas workers due to its tangible nature, perceived stability, and potential for generating rental income. The study found that OFWs are increasingly taking proactive approaches to financial growth, moving beyond simply sending remittances to actively building diversified investment portfolios. The research identified traditional investment preferences among OFWs: real estate for tangible stability and rental income potential; small business ventures or franchises for creating sustainable income streams upon return to the Philippines; and savings accounts or mutual funds for lower-risk, steady returns. However, the study also identified an emerging trend toward stock market investment, driven by higher growth potential, better liquidity compared to real estate, and easy diversification across industries and markets.

Crucially, the research found that modern digital platforms have made investing significantly easier for Filipinos working abroad. Platforms providing 24/7 access to investment opportunities using only smartphones or laptops have democratized access to financial markets. OFWs prefer platforms that offer convenience (access anytime, anywhere), low-cost entry (allowing small initial investments), and educational resources (guides, tips, and real-time data to support confident decision-making). The study concluded that the best wealth-building

strategy for OFWs involves diversification, balancing stable traditional assets like real estate with higher-growth, liquid investments such as stocks.

This research supports the current study's focus on creating an accessible, user-friendly web-based platform for property browsing and appointment. The findings emphasize that OFW clients require platforms that operate across time zones, provide comprehensive information to compensate for inability to visit properties physically, and offer immediate, reliable communication channels to address questions and concerns promptly. The identification of convenience, low-cost entry, and educational resources as preferred platform characteristics directly informs the proposed system's design priorities: 24/7 availability, minimal barriers to browsing and inquiry submission, and potentially informative content about the San Miguel market and property acquisition processes.

### 2.2.5 Home Financing and Real Estate Platforms for Non-Traditional Income Earners

Reelist8 [19] explored home financing solutions specifically for OFWs and freelancers in the Philippine property market, highlighting how digital platforms can serve as intermediaries connecting buyers with appropriate financing options. The study found that navigating home financing in the Philippines is particularly complex for those with non-traditional income sources, as standard employment documentation may not be available or may not adequately represent their earning capacity. The research cataloged various financing options available to OFWs and freelancers: home loans from local banks requiring proof of income and employment documentation; government-sponsored housing loans through Pag-IBIG Fund and the Social Security System offering low-interest rates and extended repayment terms; in-house financing from property developers with less stringent documentation requirements and faster approval times; private lending institutions and non-bank financial companies providing alternative options with more flexible requirements;and rent-to-own programs allowing lease-to-purchase arrangements ideal for those without immediate full purchase capacity.

The study emphasized that digital platforms serving OFWs must extend beyond simple property listings to include comprehensive educational resources about financing options, requirements, and application processes. By providing this guidance within the platform, OFWs can explore financing options concurrently with property browsing, making more informed decisions about affordability and purchase feasibility. This research informs the current study by suggesting that the proposed system could enhance its value to OFW clients by including property financing information, connecting buyers with financing resources, or at minimum providing educational content about the various paths to property acquisition. While the current study's scope excludes direct financial transaction processing, the inclusion of financing guidance aligns with the broader goal of serving OFW clients comprehensively. Even basic informational content about Pag-IBIG requirements, typical down payment percentages, or developer financing options could significantly reduce the cognitive load on OFW buyers who otherwise must research financing separately from property searching.

### 2.2.6 Philippine Real Estate Industry Challenges and Digital Imperatives

Multiple local studies collectively establish the challenging context of the Philippine real estate industry and the imperative for digital transformation. Research by the Philippine Institute for Development Studies indicates that housing affordability remains a critical national issue, as property prices continue to rise faster than household incomes, limiting home ownership capacity for many Filipino families, particularly in urban areas where prices are significantly higher than rural regions. Studies on Philippine real estate industry practices identify persistent problems including unethical practices by some developers and agents, unlicensed real estate practitioners operating without proper accreditation, documentation issues in property transactions, and insufficient regulatory enforcement.

These challenges affect buyer trust and decision-making confidence when investing in real estate properties, creating an environment where transparency and accountability mechanisms are desperately needed. Despite these challenges, the real estate sector contributes significantly to national economic development by providing employment across construction, sales, property management, and ancillary services; generating housing

solutions for growing populations; and contributing government revenue through taxes, transfer fees, and registration costs.

This economic importance underscores the national interest in improving industry practices and transaction efficiency. The Philippine Statistics Authority [17] documented the increasing use of internet and mobile devices across the Philippines, fundamentally changing how businesses promote services and how consumers research purchases. This digital shift affects all sectors, including real estate, where traditional methods of property promotion through printed flyers, newspaper advertisements, and physical signage are rapidly losing effectiveness compared to digital channels.

The Department of Trade and Industry [7] formally emphasized that small and medium enterprises, including local real estate agencies, must shift from traditional marketing methods to digital platforms to remain competitive and reach broader customer bases. This government guidance establishes digital transformation not merely as an option but as a necessity for business survival and growth. University of the Philippines studies demonstrate that online property listings and web-based systems measurably improve accessibility for property seekers, enhance customer engagement through interactive features and immediate feedback, and support better-informed decision-making through comprehensive, easily comparable property information.

These findings support the transition from traditional estate promotion to digital marketing strategies using web-based systems in the local real estate industry. The convergence of these local studies creates a compelling narrative of necessity: the Philippine real estate industry faces structural challenges (affordability, unethical practices, documentation issues) that digital systems can partially mitigate through transparency, standardization, and automated record-keeping. Simultaneously, national digital adoption trends and government policy create favorable conditions for system implementation. The proposed system for San Miguel, Iloilo City enters this environment as a timely intervention that addresses documented market failures while aligning with national digitalization trajectories.

### 2.2.7 Digital Investment Education and OFW Empowerment

Bria Homes [4] researched smart investment options for OFWs, emphasizing the critical role of online trading platforms and digital educational tools in facilitating property investment decisions. The study found that numerous online platforms now assist OFW investors in learning fundamentals of real estate investment, property valuation techniques, market analysis methodologies, and legal considerations in property acquisition. The research highlighted that digital platforms serving OFW investors must prioritize three core principles: transparency in pricing and property details, comprehensive property information including virtual tours and neighborhood data, and responsive customer support capable of addressing inquiries across international time zones.

The availability of online resources and virtual property tours has become essential for OFWs who cannot easily visit properties before making purchase commitments. The study concluded that empowering OFWs with digital tools and educational resources enables them to make more confident, informed investment decisions while reducing their vulnerability to misinformation, fraudulent schemes, and high-pressure sales tactics. This research reinforces the current study's emphasis on providing detailed property information, high-quality imagery, location details and automated communication features as essential components for serving OFW clients effectively. The three principles identified—transparency, comprehensiveness, and responsiveness—can serve as direct design requirements for the proposed system's user interface and feature set.

## 2.3 Related Literature

This section presents the theoretical foundations, conceptual frameworks, and technological principles that underpin the proposed web-based real estate management system. The literature is organized into thematic areas covering information systems theory, digital marketing concepts, web development architectures, customer relationship management, appointment system design, security engineering, specific technology platforms, software development methodologies, user experience design, database principles, and geographic information

systems. Each thematic area is developed with sufficient depth to support scholarly analysis while maintaining clear relevance to the practical requirements of real estate system development.

### 2.3.1 Information Systems Theory and Technology Acceptance Frameworks

Information Systems (IS) theory provides the fundamental conceptual lens through which the proposed real estate management system can be understood as more than a technical artifact, but as a socio-technical intervention designed to transform organizational processes and improve stakeholder outcomes. An information system is formally defined as an organized combination of people, hardware, software, communication networks, data resources, and policies that stores, retrieves, transforms,and disseminates information in an organization [12]. In the context of real estate, information systems serve as the digital backbone for managing property portfolios, client relationships, marketing campaigns, and transaction workflows.

The Socio-Technical Systems theory, originally developed by Trist and Bamforth [26] in their seminal study of coal mining operations and subsequently expanded by numerous organizational theorists, posits that effective organizational systems must jointly optimize both social and technical subsystems. The theory argues that optimizing technical components alone while ignoring social factors such as user behaviors, organizational culture, power dynamics, and workflow patterns leads to sub optimal outcomes and system rejection. Conversely, focusing solely on social aspects without adequate technical capability results in inefficient, manual processes that cannot scale. This theory is particularly relevant to real estate management systems because successful implementation requires careful attention to how agents currently handle client relationships, how administrators manage property listings, and how clients prefer to search for and inquire about properties.

The proposed system must not merely digitize existing processes but reimagine them in ways that respect established professional relationships while introducing efficiencies that stakeholders find genuinely valuable. For example, automating inquiry responses must be implemented in ways that do not make clients feel ignored or depersonalized, and administrative dashboards must align with how managers actually oversee operations rather than imposing alien workflows. In the San Miguel context, where real estate agencies may be family-owned businesses with long-established personal networks, the socio-technical balance is especially delicate; the system must enhance rather than disrupt the personal trust relationships that drive local real estate transactions.

The Technology Acceptance Model (TAM), developed by Fred Davis [9] at the MIT Sloan School of Management, provides the most widely cited theoretical framework for understanding how users come to accept and use new information technologies. TAM posits that two primary beliefs determine user acceptance: perceived usefulness and perceived ease of use. Perceived usefulness is defined as the degree to which a person believes that using a particular system would enhance his or her job performance or effectiveness. Perceived ease of use refers to the degree to which a person believes that using a system would be free of physical and mental effort. TAM further proposes that perceived usefulness and perceived ease of use are influenced by external variables such as system characteristics, training, documentation, and user support. These beliefs then shape behavioral intentions to use the system, which in turn determine actual system usage.

Finally, actual usage influences job performance, productivity gains, and user satisfaction, creating feedback loops that reinforce or diminish continued use. In the context of the proposed real estate system, TAM provides specific guidance for design decisions. To maximize perceived usefulness, the system must clearly demonstrate benefits in terms of time savings (faster inquiry handling, automated listing updates), improved client management (organized contact histories, automated follow-ups), and increased sales effectiveness (better lead tracking, faster response times). To maximize perceived ease of use, the interface must be intuitive enough that agents and administrators can perform core tasks with minimal training, navigation must follow familiar patterns, and help resources must be readily accessible. For the San Miguel market, where agents may have varying levels of digital literacy, perceived ease of use is not merely a convenience factor but a potential adoption barrier that must be aggressively addressed through simple interfaces, clear labeling, and minimal required steps for common tasks.

The Unified Theory of Acceptance and Use of Technology (UTAUT), proposed by Venkatesh et al. [23], extends TAM by incorporating additional determinants of technology acceptance and identifying moderating variables that influence the strength of relationships in the model. UTAUT identifies four key constructs that directly influence behavioral intention and use behavior: Performance expectancy refers to the degree to which users believe the system will help them attain gains in job performance.

For real estate agents, this translates to beliefs about whether the system will help them close more deals, manage more clients efficiently, or provide better service quality. Effort expectancy is the degree of ease associated with the use of the system. This encompasses not just interface simplicity but also the learning curve required to become proficient, the availability of training resources, and the clarity of system documentation.Social influence is the extent to which users perceive that important others (managers, colleagues, clients) believe they should use the system. In real estate agencies, if management actively encourages system use and colleagues demonstrate successful outcomes, individual agents are more likely to adopt the platform enthusiastically.

Facilitating conditions refer to the degree to which users believe organizational and technical infrastructure exists to support system use. This includes reliable internet connectivity, adequate hardware, technical support availability, and organizational policies that accommodate the time needed to learn and use the system effectively. UTAUT also identifies four key moderating variables: gender, age, experience, and voluntariness of use. For example, research suggests that performance expectancy may be more influential for men, while effort expectancy may be more important for women; that social influence may be stronger for older users; and that facilitating conditions matter more when system use is mandatory rather than voluntary. For the proposed real estate system, UTAUT suggests that successful adoption depends on a holistic approach addressing all four direct determinants.

The system must demonstrably improve agent and administrator performance, minimize learning effort, garner visible support from agency leadership, and be supported by adequate technical infrastructure and organizational policies. The moderating variables are particularly relevant in the Philippine context, where real estate agencies often employ agents across wide age ranges and varying experience levels, necessitating flexible training approaches and interface designs that accommodate diverse user profiles.

The Information Systems Success Model, originally developed by DeLone and McLean [24] and significantly updated in 2003, provides a comprehensive framework for evaluating IS effectiveness across six interrelated dimensions:

System quality refers to technical performance aspects including response time, system reliability, usability, navigability, and visual appeal. For web-based real estate systems, high system quality means pages load quickly even with multiple high-resolution property images, the system remains available during peak usage periods, and the interface is visually professional and easy to navigate. Information quality encompasses the accuracy, completeness, timeliness, relevance, and consistency of information provided by the system. In real estate applications, information quality requires that property listings accurately reflect current availability and pricing, that property descriptions are comprehensive and honest, that images faithfully represent actual conditions, and that status updates (available, reserved, sold) are applied immediately.

Service quality addresses the quality of support services provided to system users, including technical support responsiveness, training quality, and documentation completeness. Even well-designed systems require ongoing support to address edge cases, user questions, and technical issues. Use encompasses the actual consumption of system features by intended users, including frequency of use, duration of sessions, and variety of features utilized. High use indicates that users find the system valuable enough to integrate into their regular workflows. User satisfaction represents users' overall positive or negative feelings about the system, measured through surveys, interviews, or behavioral indicators such as continued use and recommendation to others. Net benefits capture the ultimate positive impacts of the system on individual productivity, organizational effectiveness, and

broader stakeholder outcomes. For real estate systems, net benefits include increased sales, reduced administrative costs, improved client satisfaction, and enhanced competitive positioning.

These theoretical frameworks collectively establish that the success of the proposed web-based real estate system depends not merely on technical functionality but on how users perceive, adopt, and derive value from the system in their professional contexts. The frameworks also provide validated instruments and constructs that the current study can use to evaluate system success beyond simple technical checklists, enabling a more nuanced and academically rigorous assessment of project outcomes.

### 2.3.2 Digital Marketing Theory and Real Estate Promotion

Digital marketing represents the strategic application of internet-based technologies, platforms, and channels to achieve marketing objectives and create value for both businesses and consumers. Chaffey and Chadwick [5], in their comprehensive text on digital marketing strategy, define digital marketing as "achieving marketing objectives through applying digital technologies and media." This definition emphasizes that digital marketing is not about technology for its own sake but about leveraging technology to accomplish concrete business goals such as lead generation, brand awareness, customer engagement, and transaction facilitation. In real estate contexts, digital marketing encompasses a broad spectrum of activities: property listings on dedicated websites and general marketplace platforms; social media promotion through organic posts and paid advertising; search engine optimization to improve visibility in organic search results; email marketing campaigns targeting specific buyer segments; content marketing through blog posts, market reports, and neighborhood guides; and online advertising through pay-per-click campaigns, display ads, and retargeting strategies. The 7Ps Marketing Mix Model, originally developed by Booms and Bitner [3] as a services marketing extension of McCarthy's traditional 4Ps (Product, Price, Place, Promotion), adds three additional elements critical for service contexts: People, Process, and Physical Evidence. This expanded framework provides comprehensive guidance for real estate digital marketing:

**Product** in real estate digital marketing refers to the properties being marketed, encompassing not just physical structures but also locations, neighborhoods, amenities, and lifestyle associations. Effective digital presentation requires comprehensive property details, accurate specifications, and compelling descriptions that help potential buyers envision living or investing in the property.

**Price** encompasses listing prices, payment terms, financing options, and value propositions. Digital platforms enable dynamic pricing displays, comparison tools, and transparent cost breakdowns that help buyers assess affordability and value. The ability to filter and sort properties by price range is one of the most frequently used features in real estate websites.

**Place** in digital contexts refers to the channels through which properties are marketed and discovered. This includes the real estate agency's own website, national property portals (Lamudi, Property24), social media platforms (Facebook, Instagram), search engines (Google), and email inboxes. Effective digital marketing requires strategic presence across multiple channels where target buyers are likely to search.

**Promotion** includes all digital communications designed to attract attention, generate interest, and drive action. This encompasses social media posts showcasing properties, email newsletters featuring new listings, blog content addressing buyer concerns, paid advertising targeting specific demographics, and search engine optimization to improve organic visibility.

**People** refers to the human element of real estate service delivery, including agents, administrators, and support staff who interact with clients through digital channels. In digital marketing, people are represented through agent profiles, biography pages, video introductions, and responsive communication that projects professionalism, expertise, and trustworthiness.

**Process** involves the workflows and procedures for handling inquiries, managing listings, processing appointment, and completing transactions. Digital marketing must accurately represent these processes, setting

clear expectations about response times, viewing procedures, documentation requirements, and appointment protocols.

**Physical** Evidence in digital real estate marketing translates to the tangible proof of service quality that users can observe through screens. This includes website design quality, property photograph resolution and composition, virtual tour production values, the professionalism of listing descriptions, and the overall user experience of interacting with the digital platform. High-quality physical evidence builds trust and credibility, while poor-quality presentation undermines confidence regardless of actual property quality.

Kotler and Armstrong [18], in their foundational marketing text, emphasize that the marketing mix must be adapted to specific target markets, competitive environments, and service characteristics. For real estate in San Miguel, Iloilo City, this means tailoring digital marketing strategies to local market conditions, the specific demographics of likely buyers (local families, investors, OFWs), and the competitive landscape of existing property marketing channels. Search Engine Optimization (SEO) theory provides systematic frameworks for improving the visibility of web-based real estate platforms in organic search engine results. SEO encompasses technical optimization (site structure, page speed, mobile responsiveness, secure connections), on-page optimization (keyword-rich content, meta tags, header structure, image alt text), and off-page optimization (back link building, social signals, local citations).

For real estate systems, effective SEO ensures that potential buyers searching for location-specific terms such as "house for sale San Miguel Iloilo," "property investment Iloilo City," or "affordable homes San Miguel" can easily discover the platform. Local SEO is particularly important for real estate, as property searches are inherently location-based. Local SEO strategies include claiming andoptimizing Google Business Profile listings, ensuring consistent Name-Address-Phone (NAP) information across directories, generating local reviews, and creating location-specific content that signals geographic relevance to search engines. For the proposed system, local SEO strategies should target not just "Iloilo City" but specific barangays within San Miguel, neighborhood landmarks, and proximity to key infrastructure (schools, markets, churches) that local buyers search for.

Content marketing theory, articulated by Pulizzi [27] and others, emphasizes the creation and distribution of valuable, relevant, and consistent content to attract and retain a clearly defined audience. Unlike traditional advertising that interrupts consumers with promotional messages, content marketing attracts consumers by providing genuinely useful information. In real estate, effective content marketing includes neighborhood guides that describe local amenities, schools, and transportation; market trend reports that analyze pricing patterns and investment potential; home buying guides that explain financing options and legal procedures; and property investment case studies that demonstrate successful outcomes.

The theory posits that providing valuable content builds trust and credibility with potential clients over time, establishing the real estate business as a knowledgeable authority in the local market. This trust translates into preferential consideration when content consumers are ready to buy or sell property, effectively shortening sales cycles and improving conversion rates. For the San Miguel context, content marketing might include guides to barangay characteristics, flood risk assessments, transportation access to Iloilo City proper, and development projections that help buyers understand long-term property value trajectories.

Social media marketing theory leverages platforms such as Facebook, Instagram, TikTok, and LinkedIn to engage with potential clients, showcase properties, build brand awareness, and foster community relationships. The theory emphasizes the visual nature of real estate marketing, as property photographs and videos generate significantly higher engagement rates than text-only posts. Research consistently shows that social media posts with images receive substantially more interactions (likes, comments, shares) than text-only content, and video content achieves even higher engagement levels.

Social media algorithms favor content that generates meaningful interactions, making it essential for real estate platforms to create engaging, shareable content that encourages user participation. This includes asking

questions in posts, creating polls about neighborhood preferences, sharing behind-the-scenes content about property preparation, and encouraging satisfied clients to share their experiences and tag the agency. For Philippine markets, Facebook remains the dominant social platform, making Facebook integration and sharing capabilities particularly important features for the proposed system.

### 2.3.3 Web-Based System Architectures and E-Commerce Principles

Web-based systems, also known as web applications, represent a fundamental category of software applications that run on remote web servers and are accessed by users through web browsers over internet connections. Unlike traditional desktop applications that require installation on local computers, web-based systems offer the critical advantage of platform independence, enabling access from any device with a browser and internet connectivity. This characteristic makes web-based systems particularly suitable for real estate applications where clients may search for properties from home computers, work laptops, or mobile devices; where agents may need to update listings while away from the office; and where administrators may require access during travel or after hours.

The three-tier architecture represents the most widely adopted design pattern for robust web-based systems, separating concerns across three distinct logical and often physical layers: The Presentation Tier, also known as the Client Layer or User Interface Layer, handles all user interaction functions including interface rendering, input capture, and client-side validation. Technologies employed at this tier typically include HTML for document structure, CSS for visual styling and layout, and JavaScript for interactive behavior and dynamic content updates. In modern web development, this tier may also incorporate frontend frameworks such as React, Vue, or Angular, though the proposed system will utilize foundational technologies to maintain simplicity and compatibility.

The Application Tier, variously called the Business Logic Layer, Logic Tier, or Middle Tier, processes business rules, manages workflows, handles data processing, and coordinates communication between the presentation and data tiers. This tier implements the core functionality of the system, enforcing rules such as "only administrators can approve property listings," "clients cannot reserve already-sold properties," and"inquiry responses must be logged for accountability." The application tier is typically implemented using server-side languages such as PHP, Python, Java, or Node.js. The Data Tier, also referred to as the Database Layer or Persistence Layer, stores and retrieves data, manages transactions, ensures data integrity, and handles backup and recovery operations. Relational database management systems such as MySQL, PostgreSQL, or Microsoft SQL Server typically serve this tier, though NoSQL databases such as MongoDB are increasingly used for specific application requirements.

This three-tier architecture provides several significant advantages for web-based real estate systems. Separation of concerns enables independent development, testing, and maintenance of each tier, allowing frontend designers to modify user interfaces without affecting business logic, and database administrators to optimize queries without disrupting application functionality. Scalability is enhanced because each tier can be scaled independently based on demand; for example, additional web servers can be added to the presentation tier during high-traffic periods without modifying the database tier. Maintainability improves because changes to one tier's technology or implementation can often be made with minimal impact on other tiers, provided that inter-tier interfaces remain consistent.

Client-server architecture forms the foundational communication model for web-based systems. In this model, clients (typically web browsers running on user devices) request services and resources from centralized servers. Clients initiate communication by sending HTTP requests containing specific methods (GET for retrieving data, POST for submitting data, PUT for updating resources, DELETE for removing resources) along with relevant parameters, headers, and body content. Servers process these requests, interact with databases or other backend services as needed, and return HTTP responses containing status codes (200 for success, 404 for not found, 500 for server errors), response headers, and response bodies containing requested data or confirmation messages.

In real estate systems, this client-server interaction occurs continuously: when a client searches for properties, the browser sends a GET request with search parameters; the server queries the database for matching properties and returns formatted results. When a client submits an inquiry, the browser sends a POST request with form data; the server validates the data, stores it in the database, triggers any automated responses, and returns a confirmation. When an administrator updates a property status, the browser sends a PUT request; the server updates the database record and may broadcast the change to other users viewing that property. E-commerce principles, while developed primarily for online retail transactions, provide valuable frameworks for designing real estate web-based workflows.

E-commerce research has extensively studied product catalog management, which directly parallels property listing
management in real estate. Principles such as comprehensive product descriptions, high-quality imagery, customer reviews, and related product recommendations all have direct analogues in real estate systems.The e-commerce concept of shopping cart functionality parallels property shortlisting or favorites features in real estate platforms, where clients can save interesting properties for later comparison and review. The checkout process in e-commerce, involving review of selected items, confirmation of details, and final submission, parallels the appointment submission process in real estate, where clients review property details, confirm their contact information, and submit appointment requests.

Trust theory in e-commerce, developed by researchers such as McKnight, Choudhury, and Kacmar [28], identifies factors that build consumer confidence in online platforms. These factors include website quality and professional design, security indicators such as SSL certificates and trust badges, privacy policies explaining how personal data is used, clear return and refund policies, and social proof through customer reviews and testimonials. Applied to real estate systems, these trust factors translate to professional website design, visible security measures protecting client data, clear privacy policies, transparent terms for appointments and testimonials or success stories from satisfied clients.

Responsive web design theory, articulated by Ethan Marcotte [29] in his seminal article, advocates for creating websites that provide optimal viewing and interaction experiences across a wide range of devices, from desktop computer monitors to mobile phone screens. Responsive design employs three core technical components: fluid grids that use relative units (percentages) rather than fixed units (pixels) for layout dimensions; flexible images that scale within their containing elements to prevent overflow on smaller screens; and CSS media queries that apply different styling rules based on device characteristics such as screen width, height, orientation, and resolution.

Given the high prevalence of mobile device usage in the Philippines, where smartphones often serve as the primary or sole internet access device for many users, responsive design is not merely a nice-to-have feature but an essential requirement. OFW clients accessing the platform from abroad may predominantly use mobile devices due to convenience or limited access to desktop computers. Local clients in San Miguel may similarly rely on smartphones for property browsing. A mobile-first design approach, which prioritizes the mobile experience during design and progressively enhances features for larger screens, ensures that the broadest possible audience can effectively use the platform.

### 2.3.4 Customer Relationship Management and Communication Automation

Customer Relationship Management (CRM) theory encompasses the strategic approaches, processes, and technologies that organizations use to manage and analyze customer interactions and data throughout the customer lifecycle. The fundamental goal of CRM is to improve business relationships with customers, assist in customer retention, and drive sales growth. CRM systems consolidate customer information from various communication channels and touch points to provide a comprehensive, unified view of each customer's history, preferences, interactions, and potential value to the organization.

Buttle [30] proposed the CRM value chain model, which identifies five primary stages of CRM implementation that create value for both the organization and its customers: Customer portfolio analysis involves examining

the customer base to identify high-value prospects, segmenting customers by characteristics and behaviors, and understanding the distribution of customer value across the portfolio. In real estate applications, this translates to analyzing which types of properties attract the most inquiries, which client segments (local buyers, OFWs, investors) generate the highest conversion rates, and which geographic areas or price points show the strongest demand.

Customer intimacy refers to building a deep understanding of individual customer needs, preferences, timelines, and constraints. For real estate agents, this means maintaining detailed profiles of client preferences including desired locations, property types, budget ranges, financing needs, and decision timelines. Digital CRM systems enable systematic capture and retrieval of this information, ensuring that agents can personalize interactions even when managing large numbers of prospects. Network development involves creating and maintaining relationships with partners, suppliers, and complementary service providers who can enhance value delivery to customers. In real estate, networks include property developers, financial institutions offering mortgages, home inspectors, contractors, and legal professionals. A comprehensive CRM system may track these relationships and facilitate referrals or coordinated service delivery.

Value proposition development entails creating compelling offers and experiences tailored to specific customer segments. For real estate platforms, value propositions might include exclusive early access to new listings for registered users, personalized property recommendations based on search history, or bundled services such as financing consultation and legal referral.Managing the customer lifecycle addresses the progression of customer relationships from initial awareness through consideration, purchase, ownership, and eventual resale or upgrade. In real estate, this lifecycle is typically longer than in many other industries, with clients potentially returning years later for additional investments or referring family members. CRM systems must support long-term relationship maintenance, including periodic check-ins, market updates, and anniversary acknowledgments.

Automated communication theory examines how technology can streamline, standardize, and enhance communication processes while maintaining sufficient personalization to avoid impersonal or robotic interactions. Automation in client communication involves using software to send triggered messages based on specific client actions, predefined timelines, or system events. The goal is not to eliminate human communication but to ensure that routine communications occur consistently and immediately, freeing human agents to focus on complex, emotionally nuanced, or high-value interactions.

In real estate applications, automated communication workflows might include: immediate acknowledgment messages when inquiries are submitted, confirming receipt and setting expectations for response time; scheduled follow-up messages if no agent response occurs within a specified timeframe (e.g., 24 hours); appointment reminder messages sent 24 hours before scheduled property viewings; status update notifications when reserved properties change status; and post-transaction follow-up sequences requesting feedback, offering assistance, and maintaining relationships for future referrals.

Chatbot technology and conversational agents represent increasingly sophisticated components of automated communication systems. Rule-based chatbots operate using predefined scripts, decision trees, and pattern matching to respond to common inquiries with structured answers. For example, a rule-based real estate chatbot might recognize questions about property availability, pricing, or location and provide standardized responses drawn from the property database. While limited in their ability to handle unanticipated queries, rule-based chatbots are relatively simple to implement, maintain, and control.

AI-powered chatbots leverage natural language processing (NLP), machine learning, and sometimes large language models to understand and respond to more complex, conversational queries. These systems can interpret intent even when queries are phrased in varied ways, maintain context across multi-turn conversations, and generate human-like responses. In real estate applications, AI chatbots can handle nuanced inquiries about neighborhood characteristics, financing options, or comparison between properties. However, AI chatbots require more extensive training data, ongoing refinement, and careful monitoring to ensure accurate, appropriate responses.

The theory of service quality in digital communication, derived from the SERVQUAL model originally developed by Parasuraman, Zeithaml, and Berry [31] and subsequently adapted for online contexts, identifies five key dimensions of service quality that apply to digital real estate communication:

**Responsiveness** refers to the willingness and speed with which service providers respond to customer requests and inquiries. In digital contexts, this translates to response time for chat messages, email replies, and inquiry submissions. Research consistently shows that response time significantly impacts customer satisfaction and conversion rates in real estate, with faster responses correlating strongly with higher engagement and transaction completion.

**Reliability** concerns the ability to perform promised services dependably and accurately. For real estate digital communication, reliability means that automated responses contain accurate information, that promised follow-ups occur as scheduled, and that system-generated messages reflect current property status rather than outdated data.

**Assurance** involves the knowledge, courtesy, and ability of service providers to convey trust and confidence. In automated systems, assurance is communicated through professional language, clear identification of the sender (system vs. human agent), transparent disclosure of automation, and easy escalation paths to human agents when needed.

**Empathy** refers to the caring, individualized attention provided to customers. While automation inherently risks depersonalization, empathetic automated communication can be achieved through personalized greetings, reference to specific properties or previous interactions, acknowledgment of individual circumstances, and tone that conveys warmth and understanding.

**Tangibles** encompass the physical facilities, equipment, and appearance of service providers. In digital contexts, tangibles translate to the visual design of communication interfaces, the professionalism of email templates, the clarity of chat window layouts, and the overall aesthetic quality of digital touchpoints.

### 2.3.5 Appointment Management Systems

Appointment systems theory provides structured frameworks for designing information systems that manage the booking, scheduling, and allocation of resources, services, or appointments. While extensively developed in the hospitality, healthcare, and transportation sectors, these principles apply directly to real estate where clients schedule property viewings, reserve properties for purchase consideration, and book consultation appointments with agents.

The theory of online appointment scheduling emphasizes several critical success factors: real-time availability display that shows current open slots and prevents double-booking; instant confirmation that provides users with immediate feedback that their appointment has been successfully recorded; automated reminder notifications sent via email or SMS to reduce no-show rates; easy rescheduling and cancellation options that respect user autonomy while capturing freed slots for reallocation; and wait list functionality that allows interested clients to register for fully booked time slots or properties.

Research in healthcare appointment systems, while serving a different industry, provides robust evidence for the benefits of online scheduling. Studies consistently demonstrate that online appointment systems reduce administrative workload for staff, decrease no-show rates by 20-40% through automated reminders, improve patient satisfaction by offering scheduling convenience outside business hours, and optimize resource utilization by filling gaps in schedules more efficiently. These benefits translate directly to real estate contexts, where agent time is a valuable resource, missed viewings represent lost opportunities, and client convenience directly influences engagement levels.

Database design theory for appointment systems emphasizes the importance of maintaining data integrity through proper entity relationships, transaction management, and concurrency control. In relational database design, appointment systems typically involve several interconnected entities: resources being reserved (properties, time slots, agent availability); customers making appointment (clients, buyers, investors); appointment records linking customers to resources with times tamps and status indicators; and potentially payment or deposit records associated with appointment.

Critical database design principles for appointment systems include:

Normalization to eliminate data redundancy and prevent update anomalies. For example, customer contact information should be stored in a dedicated customers table and referenced through foreign keys in appointment records, rather than duplicated in each appointment entry.

Foreign key constraints to enforce referential integrity, ensuring that appointment records cannot reference non-existent properties or customers, and that deletion of referenced records is handled appropriately (either prevented or cascaded based on business rules).

Atomic transactions to ensure that multi-step appointment processes complete entirely or not at all. If a client submits a appointment request, the system must record the appointment, update property status, notify the agent, and send confirmation to the client as a single atomic unit. If any step fails, the entire transaction should roll back to prevent inconsistent states.

Concurrency control to prevent race conditions when multiple clients attempt to reserve the same property or time slot simultaneously. Database locking mechanisms or optimistic concurrency control techniques ensure that only one client successfully claims a resource, while others receive clear notification that the resource is no longer available.

Calendar integration theory examines how appointment systems can synchronize with external calendar applications such as Google Calendar, Microsoft Outlook, or Apple Calendar. While full calendar integration may exceed the scope of the proposed capstone system, the underlying principles inform the design of internal scheduling displays: clear date and time formatting, timezone awareness for international clients (particularly important for OFWs), visual indicators of availability and booked slots, and conflict detection algorithms that prevent overlapping appointments.

## 2.3.6 Property Management and Real Estate Operations Theory

Property management theory encompasses the comprehensive processes, systems, and strategies used to oversee residential, commercial, and industrial real estate properties on behalf of owners or investors. Effective property management spans the entire property lifecycle, from initial acquisition and marketing through leasing or sale, ongoing occupancy or ownership, maintenance and improvement, and eventual disposition or resale.

The property lifecycle model identifies distinct stages that digital systems must support:

Acquisition involves identifying, evaluating, and purchasing properties for the management portfolio. While the proposed system focuses primarily on marketing and transaction management, understanding acquisition helps contextualize how properties enter the system and what information must be captured initially.

Marketing and leasing/sale encompasses all activities required to attract and secure tenants or buyers. This stage includes property preparation, photography and description creation, listing distribution across channels, inquiry management, viewing scheduling, application processing, and lease or sale agreement execution. The proposed

system concentrates heavily on this stage, providing tools for listing creation, inquiry handling, appointment management, and appointment tracking.

Occupancy and ownership management involves rent collection, lease enforcement, tenant relations, and ongoing communication during the tenancy period. While not the primary focus of the proposed system, understanding occupancy management informs the design of post-appointment workflows and client relationship maintenance features.

Maintenance and improvement addresses the physical upkeep and enhancement of properties. Digital systems may track maintenance requests, schedule repairs, and document property condition over time.

Disposition involves the eventual sale or transfer of properties from the portfolio. Accurate records of property history, improvements, and financial performance support disposition decisions and valuation.

Listing management theory addresses how properties are presented in digital platforms to maximize visibility, engagement, and conversion. Key principles derived from e-commerce and real estate research include:

Comprehensive information provision ensures that listings answer the most common buyer questions without requiring additional inquiry. This includes not just basic specifications (bedrooms, bathrooms, floor area,lot area) but also detailed descriptions of features, neighborhood characteristics, nearby amenities, utility arrangements, and any restrictions or special conditions.

High-quality visual content significantly influences buyer interest and engagement. Research consistently demonstrates that listings with professional photographs receive substantially more views and inquiries than those with poor-quality, limited, or no images. The number of images also matters; studies suggest that listings with 10-20 high-quality photos perform better than those with fewer images, while excessive images (50+) may overwhelm viewers. Virtual tours, drone footage, and video walkthroughs further enhance engagement, particularly for remote buyers such as OFWs.

Search optimization requires structuring listing data to support filtering and sorting by the criteria buyers actually use. Common search parameters include location (city, neighborhood, barangay), price range, property type (house, lot, condominium, commercial), number of bedrooms and bathrooms, lot size, floor area, and specific features (garage, garden, security). The database schema and search interface must accommodate these parameters efficiently.

Status accuracy is critical for maintaining trust and operational efficiency. Listings must immediately reflect current availability status. Properties that are reserved or sold should be promptly updated to prevent wasted inquiries on unavailable properties, which frustrate buyers and create administrative burden for agents. Automated status updates triggered by appointment system events help maintain accuracy without requiring manual intervention for every change.

Categorization and tagging organize properties into logical groups that match how clients think about real estate. Beyond basic property types, useful categorizations might include investment properties, family homes, retirement properties, vacation homes, or development opportunities. Tags can identify specific features such as "near school," "flood-free", "gated community" or "ready for occupancy" that attract specific buyer segments.

Lead management theory in real estate contexts examines how potential clients (leads) are captured, qualified, nurtured, and converted into transactions. The lead management process typically includes:

Lead capture involves collecting contact information and inquiry details through web forms, chat interactions, phone calls, or walk-in visits. Digital systems should minimize friction in capture processes while ensuring data quality through validation rules. Progressive profiling, which collects additional information over multiple interactions rather than requiring extensive forms upfront, can improve conversion rates.

Lead qualification assesses lead quality and readiness based on factors such as stated budget, desired location, timeline for purchase, financing pre-approval status, and level of engagement with the platform. Qualified leads can be prioritized for agent attention, while unqualified leads may be nurtured through automated content until they become ready to proceed.

Lead distribution assigns qualified leads to appropriate agents based on expertise, workload, geographic specialization, or performance metrics.Automated distribution rules ensure equitable allocation while matching leads with agents best positioned to serve their specific needs.

Lead nurturing maintains regular, valuable communication with leads who are not yet ready to transact. Automated nurture sequences might include market updates, new listings matching their criteria, educational content about buying processes, and periodic check-ins. Effective nurturing keeps the agency top-of-mind until the lead's circumstances or readiness changes.

Conversion tracking monitors which lead progress through successive stages of the sales funnel: initial inquiry, property viewing, formal offer, appointment, and final transaction. Conversion metrics enable performance evaluation, process optimization, and accurate forecasting.

### 2.3.7 Security Frameworks and Defensive Programming for Web Applications

Web application security theory addresses the systematic protection of web-based systems from threats, vulnerabilities, and malicious attacks. The field has matured significantly, with standardized frameworks identifying common risks and established methodologies for mitigating them.

The Open Web Application Security Project (OWASP), an open community dedicated to enabling organizations to develop, purchase, and maintain trustworthy applications, maintains regularly updated lists of the most critical web application security risks. The OWASP Top 10 identifies common vulnerabilities including:

Injection attacks, where malicious data is sent as part of commands or queries to trick the interpreter into executing unintended commands or accessing unauthorized data. SQL injection, where attackers insert malicious SQL code into input fields to manipulate database queries, is particularly dangerous for systems like the proposed real estate platform that rely heavily on database interactions.

Broken authentication occurs when authentication mechanisms are implemented incorrectly, allowing attackers to compromise passwords, keys, session tokens, or exploit implementation flaws to assume other users' identities. In real estate systems, broken authentication could allow unauthorized access to client data, property financial details, or administrative functions.

Sensitive data exposure results from inadequate protection of data such as financial information, personal identifiers, or authentication credentials. Real estate systems handle sensitive client contact information, property pricing, and potentially financial documentation that requires strong encryption and access controls.

Broken access control happens when restrictions on authenticated users are not properly enforced, allowing attackers to access unauthorized functionality or data. For example, a client might be able to access other clients' appointment records or an agent might access administrative functions beyond their role.

Security mis-configuration represents the most commonly seen vulnerability, resulting from insecure default configurations, incomplete configurations, open cloud storage, misc-onfigured HTTP headers, or verbose error messages containing sensitive information. Cross-site scripting (XSS) allows attackers to execute scripts in victims'
browsers to hijack user sessions, deface websites, or redirect users tomalicious sites. XSS vulnerabilities in real estate systems could be used to steal client session cookies or present fraudulent property listings.

Using components with known vulnerabilities occurs when software components such as libraries, frameworks, or other software modules run with privileges that attackers can exploit. Keeping all components updated with security patches is essential. Insufficient logging and monitoring prevents timely detection and response to security breaches. Effective systems must log security-relevant events and establish monitoring processes to detect suspicious patterns.

Authentication theory examines methods for verifying user identity before granting system access. Password-based authentication, while common, requires careful implementation to be secure. Best practices include requiring minimum password length and complexity, implementing password hashing using strong algorithms such as bcrypt or Argon2 (never storing plaintext passwords or using weak hashing like MD5 or SHA1), enforcing periodic password changes, and preventing password reuse.

Session management maintains authenticated state between HTTP requests, which are inherently stateless. Secure session management requires generating cryptographically strong session identifiers, storing session data server-side rather than in client cookies, implementing session expiration and idle timeout, providing secure logout functionality
that invalidates server-side session data, and protecting against session fixation attacks.Input validation ensures that all user-provided data meets expected formats, types, lengths, and ranges before processing.

Validation should occur at multiple levels: client-side validation for immediate user feedback, server-side validation as the authoritative security control (since client-side validation can be bypassed), and database-level constraints to enforce data integrity. For real estate systems, input validation prevents injection attacks, ensures that property prices are numeric, verifies that email addresses follow proper formats, and restricts file uploads to expected image types and sizes.

Output encoding transforms data before rendering it in web pages to prevent XSS attacks. When displaying user-generated content such as client names, inquiry messages, or property descriptions, the system must encode special characters so that browsers interpret them as text rather than executable code.

Authorization theory, distinct from authentication, governs what authenticated users are permitted to do within a system. Role-Based Access Control (RBAC) is the most widely adopted authorization model for business applications. In RBAC, permissions are associated with roles rather than individual users, and users are assigned to roles based on their job functions. This simplifies administration because permissions can be managed at the role level, and users automatically inherit appropriate permissions when assigned to roles.For the proposed real estate system, RBAC would typically define roles such as:

**Administrator:** Full system control including user management, property listing approval, system configuration, and access to all reports and data.

**Agent:** Ability to manage assigned properties, respond to client inquiries, view their own appointment schedules, and update client interaction records. Agents cannot access other agents' private client data or administrative functions.

**Client:** Ability to browse properties, submit inquiries, make appointment requests, and view their own inquiry and appointment history. Clients cannot access other clients' data, administrative functions, or unapproved property drafts.

**Guest/Unauthenticated User:** Ability to browse public property listings and access general information, but inability to submit inquiries or appointment without registration. The principle of least privilege, a foundational security concept, dictates that users should be granted only the minimum permissions necessary to perform their legitimate functions. This principle minimizes the potential damage from compromised accounts, insider threats, or accidental misuse. RBAC implementations should regularly review role assignments and remove unnecessary

permissions. Data protection theory emphasizes the confidentiality, integrity, and availability (CIA triad) of information:

**Confidentiality** ensures that data is accessible only to authorized individuals. In real estate systems, this means encrypting sensitive data in transit (using HTTPS/TLS) and at rest (encrypting database files or specific sensitive columns), implementing strict access controls, and preventing unauthorized disclosure.

**Integrity** guarantees that data is accurate, complete, and unaltered by unauthorized parties. Mechanisms ensuring integrity include database transaction controls, input validation, check sums or hashes for critical data, and audit logs tracking changes to sensitive records.

**Availability** ensures that systems and data are accessible when needed by authorized users. Availability is protected through redundant infrastructure, backup and recovery procedures, denial-of-service protection, and disaster planning. Defensive programming theory advocates for writing code that continues to function correctly even under unexpected conditions, invalid inputs, or malicious attacks. Defensive programming techniques include: Fail-safe defaults: Systems should default to the most restrictive, secure state when errors occur. For example, if authorization checking fails due to a database error, the system should deny access rather than grant it.

**Complete mediation:** Every access to every resource must be checked for authorization, not just during initial authentication. Each page request, API call, and database query should verify that the requesting user has appropriate permissions.

**Economy of mechanism:** Simple designs with fewer components and interactions are easier to secure and verify. Unnecessary complexity increases vulnerability surface area.

**Separation of privilege:** Critical operations should require multiple independent conditions or approvals. For example, deleting a property listing might require both an administrative role and confirmation of intent.

**Least common mechanism:** Shared resources and communication channels should be minimized to prevent information leakage between users or processes.

**Psychological acceptability:** Security mechanisms must be user-friendly enough that legitimate users will actually use them correctly. Overly burdensome security measures encourage users to circumvent them, creating greater vulnerability.

## 2.3.8 Technology Stack Analysis: PHP, MySQL, HTML, CSS, and JavaScript

The selection of specific technologies for system development carries significant implications for development efficiency, system performance, security, maintainability, and long-term viability. The proposed system's technology stack—PHP for server-side logic, MySQL for data persistence, HTML for document structure, CSS for presentation, and JavaScript for client-side interactivity—represents a time-tested, widely supported, and academically validated combination for web application development.

PHP (Hypertext Preprocessor) is a server-side scripting language specifically designed for web development, created by Rasmus Lerdorf in 1994 and continuously evolved through active open-source community contributions. PHP powers a substantial portion of the World Wide Web, with estimates suggesting that PHP runs on over 75% of all websites andserves as the foundation for major platforms including WordPress (which powers approximately 40% of all websites), Facebook, Wikipedia, and numerous e-commerce and content management systems.

PHP's advantages for web-based real estate systems are numerous and well-documented. As a server-side language, PHP executes on the web server before delivering HTML to client browsers, enabling dynamic

content generation based on database queries, user authentication status, and business logic. This server-side execution model protects sensitive code and database credentials from exposure to end users. PHP offers native, robust support for MySQL and other database systems through extensions such as PDO (PHP Data Objects) and MySQLi, providing both object-oriented and procedural interfaces for database operations. PDO's prepared statement support is particularly valuable for preventing SQL injection attacks, one of the most critical security concerns for database-driven web applications.

PHP runs across various operating systems (Linux, Windows, macOS, Unix variants) and web servers (Apache, Nginx, Microsoft IIS), providing deployment flexibility. The language benefits from decades of widespread use that have produced comprehensive documentation, extensive libraries and frameworks (Laravel, Symfony, CodeIgniter),
large developer communities, and abundant hosting options ranging from inexpensive shared hosting to enterprise cloud infrastructure. As open-source software, PHP incurs no licensing costs, making it economically accessible for academic projects and small business applications.

Modern PHP (versions 7.x and 8.x) has addressed earlier criticisms regarding performance and language consistency. PHP 8 introduced significant performance improvements through the JIT (Just-In-Time) compiler, union types, named arguments, attributes, match expressions, and nullsafe operator, bringing the language's capabilities closer to contemporary programming standards while maintaining backward compatibility.

MySQL is an open-source relational database management system (RDBMS) originally developed by MySQL AB, now owned by Oracle Corporation. As a central component of the LAMP stack (Linux, Apache, MySQL, PHP/Python/Perl), MySQL is one of the most widely deployed database systems globally, powering applications ranging from small personal projects to massive enterprise systems handling millions of transactions per second.
MySQL's relational data model organizes data into tables with defined schemas, supporting complex queries across multiple related tables through JOIN operations. This relational structure is ideally suited for real estate systems where entities such as properties, clients, agents, appointments, and messages have defined relationships that must be maintained consistently. MySQL's ACID compliance (Atomicity, Consistency, Isolation, Durability) for transactions using the InnoDB storage engine ensures that database operations complete reliably even in the face of system failures or concurrent access.

Security features in MySQL include SSL/TLS encryption for data in transit, user authentication with password policies, fine-grained access control through privilege management, and audit logging capabilities. The system supports replication for data redundancy and read scaling, backup utilities for disaster recovery, and import/exporttools for data migration. For the proposed real estate system, MySQL provides robust data integrity, reliable transaction processing, and sufficient performance for anticipated user loads.

HTML (HyperText Markup Language) is the standard markup language for creating web pages and applications. The current version, HTML5, developed through cooperation between the World Wide Web Consortium (W3C) and the Web Hypertext Application Technology Working Group (WHATWG), introduces semantic elements (header, nav, article, section, footer, aside), native multimedia support (video and audio elements without plugins), improved form controls (date pickers, email validation, range sliders), canvas for dynamic graphics, and enhanced accessibility features.

For real estate systems, HTML provides the structural foundation for all user interface components: property listing pages with organized sections for images, descriptions, specifications, and contact forms; inquiry forms with appropriate input types and validation attributes; administrative dashboards with tabular data displays; and navigation structures that help users move efficiently between system areas. Semantic HTML5 elements improve both accessibility for users with disabilities and search engine optimization by clearly indicating content structure and importance.

CSS (Cascading Style Sheets) controls the visual presentation of HTML documents, separating content structure from visual design. CSS3, the current standard, enables responsive layouts through Flexbox and Grid systems, sophisticated animations and transitions, custom fonts, shadows, gradients, and transforms. Media queries allow styles to adapt based on device characteristics, enabling responsive design that works across desktop, tablet, and mobile screens.In real estate web design, CSS ensures that property images are displayed attractively with appropriate sizing, aspect ratios, and hover effects; that forms are user-friendly with clear labels, focus indicators, and error styling; that navigation menus are intuitive and accessible; and that the overall aesthetic projects professionalism, trustworthiness, and brand identity. The visual design of a real estate website significantly influences user trust and engagement, making CSS expertise essential for system success.

JavaScript is a versatile, high-level programming language that enables dynamic, interactive web experiences. Originally developed for client-side browser scripting, JavaScript has evolved into a full-stack language capable of server-side execution (Node.js), mobile application development, and even desktop software creation. For browser-based applications, JavaScript manipulates the Document Object Model (DOM), handles user events, validates form input, makes asynchronous server requests, and creates interactive visual effects.

Modern JavaScript (ECMAScript 6 and later) supports advanced features including arrow functions, template literals, destructuring, modules, promises for asynchronous operations, and the async/await syntax for cleaner asynchronous code. These features enable sophisticated client-side functionality without requiring heavy frontend frameworks.

For real estate systems, JavaScript enables critical interactive features: dynamic property search filters that update results without full page reloads, improving user experience and reducing server load; client-side form validation that provides immediate feedback before submission, reducing errors and server processing; interactive image galleries with thumbnail navigation, zoom capabilities, and slideshow functionality; real-time chat interfaces for client-agent communication; AJAX requeststhat load property data, submit inquiries, and update listings seamlessly; and interactive maps that display property locations with clickable markers and information pop ups.

### 2.3.9 Agile Software Development Methodology

Agile software development methodology represents a paradigm shift from traditional, plan-driven development approaches toward adaptive, repeating, and human-centered methods. First formally articulated in the Agile Manifesto [32], the methodology emerged from recognition that rigid, sequential development models often failed to deliver satisfactory software in rapidly changing business environments.

The Agile Manifesto establishes four foundational values: individuals and interactions over processes and tools; working software over comprehensive documentation; customer collaboration over contract negotiation; and responding to change over following a plan. These values do not reject processes, documentation, contracts, or planning, but rather prioritize the former elements when conflicts arise.

Twelve principles elaborate these values, including: satisfying the customer through early and continuous delivery of valuable software; welcoming changing requirements even late in development; delivering working software frequently (weeks rather than months); maintaining daily collaboration between business stakeholders and developers; building projects around motivated individuals with proper support and trust; face-to-face conversation as the most efficient communication method; working software as the primary measure of progress; sustainable development maintaining constant pace indefinitely; continuous attention to technical excellence and good design; simplicity maximizing work not done; self-organizing teams producing best architectures and designs; and regular reflection on team effectiveness with corresponding adjustment.

Scrum, the most widely adopted Agile framework, organizes work into time-boxed iterations called sprints, typically lasting 1-4 weeks. Scrum defines three core roles: the Product Owner, responsible for maximizing product value by managing the product backlog and prioritizing features based on business value; the Scrum

Master, responsible for facilitating the Scrum process, removing impediments, and coaching the team; and the Development Team, a cross-functional group responsible for delivering potentially releasable product increments at the end of each sprint.

Scrum ceremonies provide structured opportunities for planning, synchronization, review, and improvement: Sprint Planning establishes the sprint goal and selects backlog items for the sprint; Daily Stand-ups (15-minute time-boxed meetings) synchronize team members on progress, plans, and impediments; Sprint Review demonstrates completed work to stakeholders and gathers feedback; and Sprint Retrospective reflects on the sprint process and identifies improvements for subsequent sprints.

Scrum artifacts include: the Product Backlog, an ordered list of all desired product features, functions, requirements, and fixes; the Sprint Backlog, the subset of product backlog items selected for the current sprint plus the plan for delivering them; and the Increment, the sum of all completed product backlog items at the end of a sprint,representing a potentially releasable product version.

For capstone projects such as the proposed real estate system, a simplified Agile approach offers several advantages over traditional waterfall methodologies:

Repeating delivery allows for early demonstration of core functionalities such as property browsing, user registration, and basic inquiry submission, providing tangible progress that can be evaluated and refined based on advisor feedback.

Requirement flexibility accommodates the learning curve inherent in academic projects, where understanding of technical possibilities and user needs evolves throughout development. Agile's welcoming attitude toward changing requirements prevents the project from being locked into early, potentially flawed specifications.

Risk reduction through frequent delivery ensures that technical challenges are identified and addressed early rather than discovered during final integration phases. If a particular feature proves technically infeasible within project constraints, Agile allows for scope adjustment while preserving core functionality.

Stakeholder engagement through regular reviews keeps academic advisors, potential users, and team members aligned on project direction and progress, preventing misalignment that often derails waterfall projects.

Quality focus through continuous testing integrates quality assurance throughout development rather than compressing it into a final testing phase, resulting in more robust, better-tested systems.

For the proposed real estate system, Agile development will proceed through repeating cycles: Sprint 1 focusing on database design, basic HTML structure, and user authentication; Sprint 2 implementing property listing creation, image uploads, and basic browsing; Sprint 3 developing search and filter functionality; Sprint 4 creating the inquiry and chat system; Sprint 5 building the appointment management module; Sprint 6 developing the administrative dashboard; and subsequent sprints refining features, improving security, enhancing responsive design, and conducting comprehensive testing.

### 2.3.10 User Experience Design and Usability Principles

User Experience (UX) design theory encompasses the comprehensive set of practices, processes, and principles used to create products that provide meaningful, relevant, and satisfying experiences to users. UX extends beyond visual design to encompass the entire user journey, including discovery, learning, usage, troubleshooting, and ongoing relationship maintenance. In web-based real estate systems, UX design determines whether clients can efficiently find suitable properties, confidently submit inquiries, and successfully complete appointment without frustration, confusion, or abandonment.

User-centered design (UCD), the foundational philosophy of effective UX, emphasizes understanding the needs, goals, limitations, and contexts of target users through systematic research before and during design. UCD

methods include user interviews, surveys, persona development, task analysis, usability testing, and analytics review. For the proposed real estate system, UCD requires understanding how local buyers in San Miguel search for properties, what information OFWs need to make remote decisions, how agents currently manage client interactions, and what administrative tasks consume excessive time.

Information architecture (IA) theory addresses the organization, structuring, and labeling of content to support find ability and usability. For real estate platforms, IA encompasses how properties are categorized (by type, location, price range, features), how search and filter functions are structured, how navigation menus organize system areas, and how individual property pages present information hierarchically. Effective IA aligns with users' mental models, matching how they think about real estate rather than imposing alien organizational schemes.

Usability, a core UX attribute, refers to the effectiveness, efficiency, and satisfaction with which users can achieve specific goals in a given context. The International Organization for Standardization (ISO 9241-11) defines usability as the extent to which a product can be used by specified users to achieve specified goals with effectiveness, efficiency, and satisfaction in a specified context of use. For real estate systems, effectiveness means users successfully find properties matching their criteria; efficiency means they do so quickly without unnecessary steps; and satisfaction means they feel confident, comfortable, and pleased with the experience.

Accessibility ensures that systems can be used by people with diverse abilities, including those with visual, auditory, motor, or cognitive impairments. Web Content Accessibility Guidelines (WCAG) 2.1 provides internationally recognized standards for web accessibility, organized around four principles: content must be perceivable (available to senses through text alternatives, captions, adaptable layouts, and distinguishable elements); operable (interface components must be navigable and usable with various input methods including keyboards and assistive technologies); understandable (information and interface operation must be clear, predictable, and assist users in avoiding and correcting mistakes); and robust (content must work reliably with current and future assistive technologies).

Findability, a specialized UX concern for information-rich systems, ensures that users can locate specific properties or information efficiently. Findability techniques include prominent search functionality with auto complete suggestions, faceted filtering that allows progressive refinement of results, clear category navigation, breadcrumb trails showing current location within site hierarchy, and related content recommendations that surface relevant properties users might otherwise miss.

User Interface (UI) design principles address the visual and interactive elements through which users engage with the system. Jakob Nielsen's [11] ten usability heuristics, while developed decades ago, remain remarkably relevant for contemporary web design:

Visibility of system status keeps users informed about what is happening through appropriate feedback within reasonable time. Examples include loading indicators when property images are fetching, confirmation messages after inquiry submission, and progress indicators during multi-step appointment processes.

Match between system and real world ensures that the system speaks users' language with familiar concepts rather than system-oriented technical terms. Real estate systems should use standard property terminology (bedrooms, bathrooms, floor area, lot area) rather than database field names or internal codes.

User control and freedom provides clearly marked exits from unwanted states, supporting undo and redo functionality. Users who accidentally submit incomplete inquiries should be able to edit or withdraw them; clients who reserve a property but change their minds need clear cancellation procedures.

Consistency and standards maintain consistent terminology, layout, and interaction patterns throughout the system. If property cards on search results pages display price below the image, detail pages should follow the same convention. Navigation menus should remain in fixed locations across all pages.

Error prevention designs interfaces that prevent problems from occurring, through careful constraint design, confirmation dialog for destructive actions, and input validation that catches errors before submission.

Recognition rather than recall minimizes memory load by making objects, actions, and options visible. Users should not need to remember property IDs or agent names from previous pages; information should be readily available where needed.

Flexibility and efficiency of use accommodates both inexperienced and experienced users through accelerators, shortcuts, and customization options. While novice users may navigate through step-by-step wizards, experienced agents might benefit from keyboard shortcuts or bulk operations.

Aesthetic and minimalist design avoids unnecessary elements that distract from primary tasks. Property pages should highlight essential information and images without cluttering the interface with peripheral content or excessive advertisements.

Help users recognize, diagnose, and recover from errors through clear, plain-language error messages that precisely indicate problems and suggest constructive solutions. Rather than generic "Error 404" messages, the system should explain that a property listing may have been removed and suggest browsing similar properties.

Help and documentation provides accessible, searchable, concise assistance that guides users through tasks without overwhelming them with excessive information. Context-sensitive help, tool tips, and brief explanatory text can reduce the need for separate documentation.

For real estate systems specifically, UI design must prioritize property imagery, as visual content significantly influences buyer interest and engagement. Research consistently demonstrates that listings with professional photographs receive substantially more views and inquiries than those with poor-quality or absent images. The interface must therefore showcase property images prominently, support high-resolution displays, enable easy navigation through photo galleries, and potentially integrate virtual tour or video viewing capabilities.

Mobile-first design theory advocates for designing the mobile version of websites before creating desktop versions, ensuring that core functionality works effectively on smaller screens with touch interfaces and potentially slower network connections. Given the widespread use of smartphones in the Philippines for internet access, mobile-first design is essential for reaching the broadest possible audience, including OFWs accessing the platform from mobile devices abroad and local users who may not own desktop computers.

### 2.3.11 Database Design Theory and Normalization

Database design theory provides systematic frameworks for structuring data to minimize redundancy, ensure integrity, support efficient querying, and accommodate future growth. The database design process typically involves three stages: conceptual design using entity-relationship modeling to identify entities and their relationships independent of specific database technologies; logical design translating the conceptual model into a schema appropriate for the chosen database type (relational, document, graph); and physical design optimizing storage, indexing, and access paths for performance.

Entity-Relationship (ER) modeling, developed by Peter Chen [33], identifies three core components: entities representing objects of interest (things about which data is stored), attributes describing properties of entities, and relationships representing associations between entities. For the proposed real estate system, core entities and their attributes include:

Users/Clients entity stores individual property seekers with attributes such as unique identifier, full name, email address, phone number, password hash, registration date, and account status. This entity supports authentication, personalization, and communication functions.

Administrators/Agents entity represents system operators with attributes including unique identifier, username, email, password hash, role designation (administrator or agent), account creation date, and activity status. Role designation enables RBAC implementation by distinguishing between full administrators and agents with limited permissions.

A properties entity contains real estate listings with attributes such as unique identifier, title, description, price, location address, geographic coordinates, property type, number of bedrooms, number of bathrooms, floor area, lot area, availability status, listing date, and associated agent identifier. The availability status attribute is particularly critical for preventing inquiries about already-reserved or sold properties.

Property Images entity stores visual assets with attributes including unique identifier, property identifier (foreign key linking to the Properties entity), image file path or URL, upload time stamp, display order, and caption text. Separating images into a distinct entity supports multiple images per property while normalizing the database structure.

Appointments entity records client requests with attributes such as unique identifier, client identifier (foreign key), property identifier (foreign key), requested date, requested time, appointment status (pending, confirmed, completed, cancelled), submission timestamp, and any special notes or requirements.

Messages/Chat Logs entity stores communication records with attributes including unique identifier, sender identifier, recipient identifier, property identifier (optional, for context), message content, timestamp, and read status. This entity enables both client-to-agent inquiries and system-generated automated messages.

Contact Messages entity captures general inquiries from website visitors who may not be registered users, storing attributes such as unique identifier, sender name, sender email, sender phone, message content, submission time stamp, and read/response status.

Relationships between these entities define how data connects across the system:

Properties to Property Images: One-to-Many relationship, as each property can have multiple associated images, but each image belongs to exactly one property. This relationship is implemented through the property identifier foreign key in the Property Images entity.

Properties to Appointments: One-to-Many relationship, as each property can be the subject of multiple appointment requests over time, but each appointment references exactly one property.

Users to Appointments: One-to-Many relationship, as each client can submit multiple appointments, but each appointment is submitted by one client.

Users to Messages: One-to-Many relationship for both senders and recipients, as each user can send and receive multiple messages.

Properties to Messages: One-to-Many relationship, as multiple messages may reference the same property for context, though messages can also exist without property references for general inquiries.

Normalization theory organizes relational databases to reduce data redundancy and prevent update, insertion, and deletion anomalies. The normal forms provide progressive rules for table structure:

First Normal Form (1NF) requires that tables have a primary key, contain only atomic (indivisible) values in each column, and have no repeating groups. For example, a property listing should not store multiple image file name in a single column separated by commas; instead, each image should have its own row in a separate Property Images table.

Second Normal Form (2NF) builds on 1NF by requiring that all non-key attributes depend on the entire primary key, not just part of it. This primarily applies to tables with composite keys. In the proposed system, most tables use simple surrogate keys (auto-incrementing integers), making 2NF automatically satisfied if 1NF is achieved.

Third Normal Form (3NF) requires that all non-key attributes depend only on the primary key, not on other non-key attributes (eliminating transitive dependencies). For example, if an agent's email address were stored in both the Administrators/Agents table and the Properties table (assuming properties reference their listing agent), this would violate 3NF because the agent's email depends on the agent identifier, not directly on the property identifier. Proper 3NF design stores agent emails only in the Administrators/Agents table, with Properties referencing agents through foreign keys.

While normalization reduces redundancy, practical database design sometimes intentionally denormalizes specific tables to improve query performance, particularly for read-heavy applications like property search where joining multiple normalized tables for every query might introduce unacceptable latency. The proposed system will primarily follow 3NF principles while considering strategic denormalization if performance testing identifies bottlenecks.

### 2.3.12 Geographic Information Systems and Location-Based Services

Geographic Information Systems (GIS) theory addresses the comprehensive set of technologies, processes, and methods for capturing,storing, analyzing, managing, and presenting spatial or geographic data. In real estate, GIS enables property mapping, neighborhood analysis, proximity calculations, and location-based search functionality that significantly enhances property discovery and evaluation.

The integration of web mapping services such as Google Maps, OpenStreetMap, or Mapbox into real estate platforms allows developers to leverage sophisticated geographic infrastructure without building proprietary mapping systems. These services provide APIs that enable embedding interactive maps, geocoding addresses to geographic coordinates, reverse geocoding coordinates to addresses, calculating distances and routes, and displaying satellite or street view imagery.

For the proposed real estate system, GIS integration supports several valuable features: map-based property search displaying available properties as interactive markers on zoomable, pannable maps; location auto complete suggesting addresses and neighborhood names as users type location queries; proximity filtering finding properties within specified distances of landmarks, schools, or business districts; and static map images showing property locations in listing previews.

Location-based services (LBS) theory examines how applications can use geographic positioning to deliver contextually relevant information and services. While primarily associated with mobile applications using GPS positioning, LBS principles apply to web-based real estate systems through user-specified location preferences and map-based interactions. Key LBS capabilities for real estate include geofencing (defining geographic boundaries for targeted notifications or search results), location history analysis (understanding which areas users search most frequently to personalize recommendations), and proximity alerts (notifying users when new listings appear in their areas of interest).

For the San Miguel, Iloilo City context, location-based features must account for local geographic conventions including barangay boundaries, proximity to key landmarks (plazas, churches, markets, schools), transportation access, and neighborhood characteristics that influence property values. While full GIS implementation may exceed capstone project scope, understanding GIS principles ensures that property location data is structured effectively for future enhancements and that any map integrations follow established usability patterns.

## 2.4 Synthesis of the Reviewed Literature and Studies

This synthesis section integrates findings from foreign studies, local studies, and related literature to establish convergent themes, identify critical research gaps, articulate the theoretical framework guiding the proposed system, and justify the study's specific contributions to knowledge and practice. The synthesis is organized thematically to build a coherent argumentative structure that moves from empirical validation through gap analysis to integrated justification.

### 2.4.1 Convergence of Findings on Digital Transformation in Real Estate

The reviewed literature demonstrates remarkable consistency across international and local sources regarding the ongoing transformation of real estate marketing and management from traditional, manual methods to digital, automated, web-based systems. This convergence provides strong empirical and theoretical validation for the proposed system's fundamental premise: that digital transformation is essential for real estate competitiveness in contemporary markets.

Foreign studies uniformly confirm the benefits of web-based real estate platforms. Al Ghawail [1] demonstrated that PHP and MySQL-based systems successfully replace traditional methods while improving communication and data management. The online platform research [10] confirmed that web-based systems significantly reduce time, paperwork, and communication delays. The cloud-based Salesforce system (2025) showed that automated workflows improve lead management efficiency. Digital marketing research [2], [14], [21] consistently found that digital strategies outperform traditional methods in reach, engagement, and conversion. AI and automation studies [15], [20], [25] provided quantitative evidence of dramatic improvements in response times, conversion rates, and operational efficiency.

Local studies similarly establish the digital imperative. The Philippine Statistics Authority [17] documented nationwide digital adoption trends. The Department of Trade and Industry [7] formally mandated SME digital transformation. Market analysis [13] revealed a rapidly expanding digital real estate ecosystem. The Tokkatok case study [6] exposed the fragmentation and informality of current Philippine property marketing channels. Government initiatives [16] explicitly support digital platforms for OFW investors. OFW investment research [4], [19], [22] consistently identified digital accessibility as a critical need for overseas investors.

The convergence of foreign and local findings across diverse methodologies, contexts, and sources establishes that digital transformation in real estate is not a niche trend or luxury option but a global and national imperative. Businesses that fail to adopt digital systems face documented disadvantages in customer reach, operational efficiency, and competitive positioning. This convergence is particularly significant because it spans both developed and developing markets, suggesting that the benefits of real estate digitalization are not conditional on high-income economies but are achievable in contexts similar to San Miguel, Iloilo City.

### 2.4.2 Validation of Proposed System Features

The reviewed literature provides robust validation for each major feature planned in the proposed web-based real estate management system:
Property browsing and listing management is validated by Al Ghawail [1], the online platform study [10], the advanced PMS research [8], and the Tokkatok case study [6], all of which successfully implemented and tested similar features. These studies confirm that users can effectively navigate property catalogs, view detailed information, and make informed decisions through web interfaces.

Online appointment management is supported by appointment system research (Jurnal Galaksi, 2025; IJEAIS, 2023) demonstrating that digital booking reduces administrative burden, prevents double-booking, and

improves user convenience. While these studies focused on hospitality applications, their findings regarding scheduling integrity, confirmation processes, and reminder notifications directly apply to real estate contexts.

Automated communication and chat functionality receives particularly strong validation from AI-CRM research [20], which documented conversion rate improvements from 20% to 40% and query handling improvements from 40% to 85% after chatbot implementation. Whippy [25] and Nylas [15] further confirmed that automated follow-ups, real-time alerts, and integrated messaging significantly improve client engagement and agent productivity.

Administrative dashboards for system management are validated by multiple studies including the cloud-based Salesforce research (2025) and the advanced PMS study [8], which both implemented comprehensive dashboards for monitoring metrics, managing listings, and tracking client activities. ISO/IEC 25010 evaluation of the PMS dashboard achieved a 4.53 security rating and 4.62 usability rating, demonstrating that well-designed dashboards meet high quality standards.

Security measures including authentication, input validation, and defensive programming are extensively validated by the advanced PMS research [8], which implemented bcrypt hashing, MFA, OAuth 2.0, RBAC, and ISO 25010 compliance. OWASP principles and security frameworks reviewed in the related literature provide additional
theoretical foundations for the proposed system's security architecture.

The cumulative validation across all feature categories suggests that the proposed system is not experimental in its feature selection but rather assembles proven, empirically supported functionalities into an integrated package tailored to a specific local context.

### 2.4.3 Identification of Critical Research Gaps

Despite extensive literature on real estate digital platforms, several significant gaps justify and position the current study:

Geographic specificity gap: Existing Philippine real estate platforms (Lamudi, Property24, Carousell) serve national or regional markets with generic interfaces that lack deep localization for specific municipalities. These platforms treat San Miguel, Iloilo City as merely one of thousands of locations, providing no community-specific context, neighborhood-level insights, or locally relevant features. The current study addresses this gap by focusing specifically on properties in San Miguel, Iloilo City, ensuring that content, navigation, and features reflect local geographic conventions, community characteristics, and market conditions.

Small agency integration gap: Major platforms target large developers and established agencies with substantial marketing budgets and technical resources. Small, local real estate businesses lack access to integrated, affordable systems that combine property management, digital marketing, automated communication, and appointment management in customizable solutions. The current study addresses this gap by designing a system specifically for a local real estate business with limited technical resources, using cost-effective open-source technologies (PHP, MySQL) and straightforward deployment options.

OFW-specific functionality gap: While general platforms serve various user segments, limited research and development focus specifically on the unique needs of OFW property investors. OFWs require timezone-aware communication, comprehensive property documentation for remote decision-making, simplified appointment processes that accommodate international schedules, and interfaces optimized for mobile access from abroad. The current study explicitly identifies OFWs as primary beneficiaries and incorporates features designed to serve their specific circumstances.

Academic-practical integration gap: Many existing studies either focus on theoretical marketing strategies without accompanying system implementation, or present commercial system development without rigorous academic grounding. The current study bridges this gap by applying established information systems theories

(TAM, UTAUT, DeLone and McLean's Success Model) to guide the development of a practical, functional system within an academic capstone framework. This integration produces both a working software solution and a theoretically informed scholarly contribution.

Technology stack validation gap: While numerous studies demonstrate the viability of various technology stacks (MERN, Python Flask, Salesforce, Java Spring Boot), limited recent research validates the continued effectiveness of PHP and MySQL for real estate applications in the Philippine context. Al Ghawail [1] provides valuable validation, but additional research in different geographic and market contexts strengthens the evidence base for this technology choice. These gaps are not merely abstract academic deficiencies but represent real market failures that affect actual buyers, sellers, and agents in San Miguel, Iloilo City. By addressing these gaps, the current study contributes both to scholarly knowledge and to practical market improvement.

### 2.4.4 Integrated Theoretical Framework

The proposed study synthesizes multiple theoretical frameworks into an integrated model that guides system design, implementation, and evaluation:

From Information Systems theory, the study adopts the Socio-Technical Systems perspective to ensure the system balances technical capabilities with social and organizational factors. The system must fit within existing real estate business workflows, respect established client-agent relationships, and accommodate the learning curves of users with varying digital literacy levels. The Technology Acceptance Model and Unified Theory of Acceptance and Use of Technology guide interface design and feature prioritization to maximize perceived usefulness, ease of use, performance expectancy, and facilitating conditions. DeLone and McLean's IS Success Model provides the evaluation framework, ensuring assessment across system quality, information quality, service quality, use, user satisfaction, and net benefits dimensions.

From Digital Marketing theory, the 7Ps Marketing Mix Model structures the system's marketing features, ensuring comprehensive coverage of product presentation (property listings), pricing display, digital placement (web platform), promotional tools, people (agent profiles and communication), process (inquiry and appointment workflows), and physical evidence (visual design and imagery quality). SEO and content marketing principles inform the structure of property listings and website content to maximize organic visibility for location-specific searches.

From Web Development theory, three-tier architecture guides the system's technical organization, separating presentation, business logic, and data layers for maintainability and scalability. Responsive web design principles ensure accessibility across the diverse devices used by Philippine consumers and OFWs abroad. Security frameworks (OWASP, RBAC, defensive programming) protect system integrity and user data against common web application threats.

From CRM and Communication theory, the CRM value chain model structures client interaction workflows from initial inquiry through post-appointment follow-up. Automated communication principles guide the design of chat functionality and inquiry response systems, ensuring immediate acknowledgment and systematic follow-up. Service quality dimensions (responsiveness, reliability, assurance, empathy, tangibles) inform the design of client communication interfaces to maintain professional standards even in automated interactions.

From Software Engineering theory, Agile development methodology provides the project management framework, enabling repeating delivery, continuous feedback incorporation, and adaptability to evolving requirements. Database normalization principles ensure data integrity and efficiency, while defensive programming techniques protect against security vulnerabilities and system failures.This integrated framework does not merely list theories but actively combines them into a coherent design philosophy. For example, when designing the chat functionality, the system draws simultaneously from TAM (ensuring perceived usefulness and ease of use), SERVQUAL (ensuring responsiveness and assurance), security frameworks (ensuring input

validation and output encoding), and Agile methodology (delivering chat functionality incrementally for early testing).

### 2.4.5 Technological and Methodological Justification

The choice of PHP, MySQL, HTML, CSS, and JavaScript for system development is thoroughly justified by the reviewed literature. Al Ghawail [1] successfully implemented a real estate system using this exact technology stack, demonstrating its viability for property management applications in a comparable context. The LAMP stack remains one of the most widely deployed web development platforms globally, with extensive documentation, community support, and hosting availability that reduce development risks and deployment costs.

PHP's native database integration capabilities, combined with MySQL's relational data model and ACID compliance, provide robust foundations for managing property listings, user accounts, appointment and message logs. The combination has been refined over decades of widespread use, resulting in mature, stable, well-understood technologies with abundant learning resources suitable for academic projects.

HTML5, CSS3, and JavaScript enable modern, responsive user interfaces without requiring specialized frontend frameworks that might increase development complexity beyond capstone project scope. These foundational web technologies are universally supported, ensuring compatibility across all modern browsers and devices without requiring plugin installations or specialized run time environments.

The Agile development methodology, supported by software engineering literature and validated in multiple studies, provides an appropriate project management framework for repeating development within academic timeline. The methodology's emphasis on working software, continuous feedback, and adaptability aligns with the exploratory and
learning-oriented nature of capstone projects while ensuring regular deliverables and demonstrable progress. The technological choices also reflect practical constraints and opportunities specific to the deployment context. PHP and MySQL hosting is widely available and inexpensive in the Philippines, making ongoing system operation feasible for a small local agency. The technologies require no proprietary licensing fees, aligning with the resource constraints of small business users. The large pool of PHP developers in the Philippines ensures that future maintenance and enhancement can be sourced locally if needed.

### 2.4.6 Synthesis of Local Market Needs and Opportunities

The local studies collectively establish clear market needs and opportunities that the proposed system directly addresses:

Fragmented market structure: The Philippine real estate market outside major metropolitan areas lacks structured, trustworthy digital platforms for property transactions. Informal channels such as Facebook groups dominate, creating verification challenges, inconsistent listing quality,and unreliable communication. The proposed system provides a dedicated, professional platform with structured listings, verified information, and organized inquiry management. OFW market importance and underserved needs: Multiple studies confirm that OFWs represent a critical, high-value market segment for Philippine real estate, yet they face significant challenges in remote property research and investment. The system's web-based accessibility, comprehensive property information, automated communication, and mobile-responsive design directly address documented OFW pain points including time zone differences, inability to visit properties physically, and need for immediate, reliable information.

Government and industry alignment: Government backing for digital real estate platforms [16] and DTI encouragement for SME digital transformation [7] create favorable policy conditions for system adoption. The proposed system aligns with national digitalization trends, government initiatives supporting OFW investment, and industry movement toward online property marketing.

Competitive differentiation: While national platforms serve broad markets, localized systems can offer deeper community context, more personalized service, and stronger local trust relationships. The San Miguel, Iloilo City focus enables the system to capture market share among buyers specifically interested in this area, who may prefer a
specialized local platform over generic national alternatives.

The synthesis of market needs also reveals a temporal opportunity: the post-pandemic acceleration of digital adoption, combined with sustained OFW investment flows and government digitalization support, creates a window where the introduction of a localized digital platform is likely to encounter receptive users and favorable market conditions.

### 2.4.7 Summary of Justification and Study Positioning

The comprehensive synthesis of reviewed literature and studies provides multi-faceted justification for the proposed web-based real estate management system:

Empirical justification derives from multiple foreign and local studies demonstrating that similar systems have successfully improved real estate operations, client satisfaction, and business performance across diverse contexts. The consistency of positive outcomes suggests high probability of success for the proposed system in the San Miguel, Iloilo City context. Theoretical justification is established through the integration of established information systems, digital marketing, CRM, and web development theories that provide robust frameworks for system design, implementation, and evaluation. These theories ensure the system is grounded in proven principles rather than temporary development decisions.

Technological justification is validated by existing real estate system implementations using the chosen technology stack and supported by extensive development resources. The PHP/MySQL/HTML/CSS/JavaScript combination offers proven capability, development efficiency, and deployment flexibility suitable for academic and small business contexts. Market justification emerges from Philippine market research, government initiatives, documented OFW investment patterns, and identified gaps in existing digital platforms. The specific focus on San Miguel, Iloilo City addresses an underserved local market with distinct characteristics and needs. Gap justification is established through the identification of geographic specificity gaps, small agency integration gaps, OFW functionality gaps, academic-practical integration gaps, and technology validation gaps that the current study explicitly addresses. These gaps ensure the study contributes meaningfully to existing knowledge and practice rather than merely replicating prior work in identical contexts.

In conclusion, the reviewed literature and studies collectively establish that the development of a web-based real estate management system with integrated digital marketing and automated communication features is theoretically sound, empirically supported, technologically feasible, market-justified, and academically warranted. The system addresses documented inefficiencies in traditional real estate operations while serving the specific needs of the San Miguel, Iloilo City market and its significant OFW client base. The convergence of international best practices, local market insights, and robust theoretical frameworks provides a solid foundation upon which the subsequent methodology and system development chapters will build. The following chapter will detail the research methodology through which this theoretically grounded and empirically informed system will be developed, tested, and evaluated against established quality criteria.

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# CHAPTER III

# METHODOLOGY

## 3.1 Research Design

The research design is the overall plan that guides how the study was conducted. It acts like a road map that shows the researchers what to do, when to do it, and how to do it. For this study, the researchers used a combination of descriptive research and development research. These two approaches were chosen because they work well together when the goal is to understand a real-world problem and then create a practical solution .

Descriptive research is a type of study that aims to describe the current situation, condition, or problem in a detailed way. It does not try to change anything or prove cause-and-effect relationships. Instead, it focuses on collecting facts and information about what is happening right now. In this study, the researchers used descriptive research to understand the real problems that real estate businesses in San Miguel, Iloilo City are facing. They looked at how these businesses handle property listings, how they communicate with clients, and what difficulties they experience in their daily operations.

The researchers chose descriptive research because it allowed them to gather real and accurate information about the existing problems before trying to solve them. By understanding the problem deeply, the researchers could design a system that truly meets the needs of the users. For example, through descriptive research, the researchers found out that many real estate agents still use manual methods to record client inquiries. They also learned that property listings are often outdated and that communication with clients is slow. Without this descriptive phase, the researchers might have built a system that does not address the real needs of the business.

Development research, on the other hand, is a type of study that focuses on creating a new product, system, or solution to solve a specific problem. It is also called Research and Development or R&D. In this study, the researchers used development research to build the web-based real estate management system. This approach involves designing, building, testing, and improving a product until it becomes useful and effective.

The combination of descriptive and development research is very effective for capstone projects because it follows a logical flow: first understand the problem, then build the solution. The researchers did not just jump into coding and designing. Instead, they took time to study the situation, interview potential users, review existing systems, and analyze the data they collected. Only after this understanding phase did they begin the actual development of the system.

The research design also includes the overall strategy for collecting and analyzing information. The researchers used both primary and secondary data sources. Primary data refers to information that is collected directly from the source. In this study, primary data came from interviews with real estate business owners, agents, and potential clients. Secondary data refers to information that already exists, such as books, articles, and studies from other researchers. The researchers reviewed related literature and studies to support their understanding of the problem and to learn from the experiences of others.

The researchers also decided to follow a structured and organized approach to avoid confusion and wasted effort. They created a timeline, set clear milestones, and divided the work among team members. This organized approach is important in any research project because it helps the team stay focused and finish the work on time.

Another important part of the research design is the setting or location of the study. This study focused on real estate properties located in San Miguel, Iloilo City. The researchers chose this location because it is a growing area with active real estate development. It is also a place where the researchers can easily access real estate businesses for interviews and testing. By limiting the study to one specific area, the researchers wereable to focus their efforts and create a system that is relevant to that location.

The target users of the system were also clearly defined in the research design. The system is intended for three main groups of users: clients who want to buy or rent properties, real estate agents who manage listings and communicate with clients, and administrators who oversee the entire system. By identifying these users early in the research design, the researchers were able to include features that are useful for each group.

In summary, the research design of this study is a combination of descriptive and development research. It follows a clear and logical process: understand the problem first, then develop the solution. The researchers collected both primary and secondary data, focused on a specific location, and identified the target users. This design ensures that the final system is based on real needs and real data, making it more likely to be useful and effective in solving the identified problems.

## 3.2 System Development Methodology

The system development methodology is the step-by-step process that the researchers followed to build the web-based real estate management system. It is like a recipe that tells the researchers what ingredients to use, what steps to take, and in what order. Choosing the right methodology is very important because it affects how fast the system is built, how well it works, and how easy it is to fix problems along the way.

For this study, the researchers chose to use the Agile Development Model. Agile is a popular method for building software because it is flexible, adaptive, and focused on delivering working features quickly. Unlike older methods that require all planning to be done before any coding starts, Agile allows the team to build the system in small parts,test each part, get feedback, and then improve it. This makes Agile ideal for student projects like capstones because it allows for continuous learning and adjustment.

The Agile Development Model is based on several important ideas. **First,** it values people and communication more than strict processes and tools. This means that the research team worked closely together, talked often, and made decisions as a group. **Second,** it values working software more than detailed documentation. While documentation is still important, the main goal is to create a system that actually works and can be used. **Third,** it values customer collaboration more than fixed contracts. The researchers worked with their client, asked for feedback, and made changes based on what the client needed. **Fourth,** it values responding to change more than following a rigid plan. If the researchers discovered a better way to do something, they were free to change their approach.

The Agile method works in cycles called iterations or sprints. Each sprint is a short period of time, usually a few weeks, during which the team plans, designs, builds, and tests a small part of the system. At the end of each sprint, the team has a working piece of software that can be shown to the client. The client gives feedback, and the team uses that feedback to plan the next sprint. This cycle continues until the entire system is complete.

The researchers chose Agile for several reasons. First, real estate business needs can change quickly. A client might ask for a new feature or realize that an old feature is not needed. Agile makes it easy to handle these changes without starting over. Second, Agile reduces risk. By building and testing small parts of the system at a time, the researchers could find and fix problems early instead of discovering them at the end. Third,

Agile improves quality. Because each sprint includes testing, the final system is more reliable and has fewer bugs. Fourth, Agile keeps the client involved. The real estate business owner and agents were able to see progress regularly and suggest improvements.

The Agile Development Model used in this study consisted of five main phases. Each phase is explained in detail below.

### 3.2.1 Phase 1: Requirements Analysis

The first phase of the Agile Development Model is Requirements Analysis. This phase is all about understanding what the system needs to do. The researchers spent time talking to the client, observing how the real estate business currently operates, and writing down all the things that the new system should be able to do.

During this phase, the researchers conducted interviews with the real estate business owner and several agents. They asked questions like: How do you currently list your properties? How do clients contact you? What problems do you face every day? What features would make your work easier? The researchers also looked at how the business handles appointments, how they update property information, and how they communicate with potential buyers.

The researchers divided the requirements into two types: functional requirements and non-functional requirements. Functional requirements are the specific features and functions that the system must have. For this study, the functional requirements included:

**Property browsing:** Users should be able to view a list of available properties with images, descriptions, prices, and locations.

**Property search and filter:** Users should be able to search for properties by location, price range, or property type.

**Online appointment:** Users should be able to select a property and make a appointment to view it.

**Automated chat:** Users should be able to send messages and receive quick responses about property details.

**Admin dashboard:** Administrators should be able to add, edit, and delete property listings.

**Inquiry management:** Administrators should be able to view and respond to client messages and appointment.

**User authentication:** The system should have login and registration features to protect user accounts.

**Database management:** The system should store all property data, user data, and transaction records securely.

Non-functional requirements are the qualities that the system should have, rather than specific features. These include: Usability: The system should be easy to use, even for people who are not tech-savvy. Security: The system should protect user data and prevent unauthorized access. Performance: The system should load quickly and handle multiple users at the same time. Reliability: The system should work correctly most of the time without crashing. Compatibility: The system should work on different devices such as computers, tablets, and smartphones.

The researchers documented all these requirements in a requirements specification document. This document served as a guide throughout the development process. Whenever the team was unsure about what to build,they could look back at the requirements document to remind themselves of the client's needs.

Requirements analysis is a very important phase because mistakes made here can cause big problems later. If the researchers did not understand what the client needed, they might build the wrong system. That is why they spent extra time on this phase, asking many questions and confirming their understanding with the client before moving forward.

### 3.2.2 Phase 2: System Design

The second phase of the Agile Development Model is System Design. In this phase, the researchers planned how the system would look and how it would work behind the scenes. This phase is like drawing the blueprints for a house before construction begins. Without good design, the system might be confusing to use or difficult to build.

The system design phase was divided into two main parts: front-end design and back-end design. The front-end is the part of the system that users see and interact with. It includes the colors, buttons, menus, images, and overall layout of the website. The back-end is the part of the system that users do not see. It includes the server, the database, and the logic that processes user actions.

For the front-end design, the researchers created wireframes and mockups. A wireframe is a simple sketch that shows where different elements will be placed on a web page. It does not include colors or images; it just shows the structure. A mockup is a more detailed design that includes colors, fonts, and sample images. The researchers used these tools to visualize the user interface before writing any code.

The researchers designed several key pages for the system:

Home Page: This is the first page that visitors see. It includes a welcome message, featured properties, and navigation links.

Property Listings Page: This page shows all available properties in a grid or list format. Each property card includes an image, title, price, and brief description.

**Property Details Page:** When a user clicks on a property, they see this page. It includes large images, full descriptions, location details, and a appointment button.

**Appointment Page:** This page contains a form where users can enter their name, contact information, preferred date, and time for viewing the property.

**Chat Page:** This page allows users to send messages and receive automated or manual responses.

**Login and Registration Pages:** These pages allow users to create accounts and log in to the system.

**Admin Dashboard:** This is a special area for administrators. It includes menus for managing properties, viewing appointment, reading messages, and generating reports.

For each page, the researchers considered user experience principles. They made sure that the navigation was simple and intuitive. They used consistent colors and fonts throughout the site. They placed important buttons in easy-to-find locations. They also made sure that the design was responsive, meaning that the pages would look good on both large computer screens and small smartphone screens.

For the back-end design, the researchers planned the system architecture, the database structure, and the flow of data. They decided to use a client-server architecture, which is explained in detail in Section 3.3. They also designed the database tables, which is explained in detail in Section 3.8.

The researchers created several design documents during this phase. These included:

System Architecture Diagram: This shows how the different parts of the system connect to each other. Database Schema: This shows all the tables in the database and how they relate to each other. Data Flow Diagrams: These show how information moves through the system from the user to the database and back. Entity Relationship Diagram: This shows the relationships between different types of data, such as users, properties, and

appointment. Use Case Diagram: This shows the different actions that users and administrators can perform in the system.

All these design documents were reviewed by the team and the client before moving to the next phase. This review process helped catch errors early and ensured that everyone agreed on the plan.

### 3.2.3 Phase 3: Development

The third phase of the Agile Development Model is Development. This is the phase where the researchers actually built the system by writing code, creating the database, and connecting all the parts together. In Agile, development is done in small pieces called iterations. Each iteration focuses on building one or a few features, testing them, and then adding them to the growing system.

The researchers planned their development work into several iterations. Each iteration lasted about two to three weeks. At the start of each iteration, the team held a planning meeting where they decided which features to build. They chose features based on priority, with the most important features being built first. This approach is called prioritization, and it ensures that even if the project runs out of time, the most critical parts of the system are already working.

**Iteration 1** focused on setting up the project environment and building the basic structure. The researchers installed the necessary software, configured the development server, and created the main folders and files for the project. They also built the database and created the basic login and registration system. This foundation was important because all other features depend on it.

**Iteration 2** focused on property management features for the admin side. The researchers built the admin dashboard and created the pages where administrators can add new properties, upload images, edit property details, and delete listings. They also built the database queries needed to store and retrieve property information.

**Iteration 3** focused on the client-side property browsing features. The researchers built the home page, the property listings page, and the property details page. They made sure that property information was pulled correctly from the database and displayed in an attractive way. They also added search and filter features so that users could find properties based on their preferences.

**Iteration 4** focused on the appointment system. The researchers built the appointment form, created the database table for appointments, and developed the logic for recording and displaying appointment. They also built the admin pages for viewing and managing appointments.

**Iteration 5** focused on the communication features. The researchers built a chat interface where users can send messages. They also built the contact form and the admin message management page. Automated responses were programmed for common questions about property availability and pricing.

**Iteration 6** focused on security and polishing. The researchers added input validation to prevent users from entering harmful data. They improved the login system with password hashing. They tested all features together and fixed any bugs they found. They also improved the visual design based on feedback from the client.

During each iteration, the researchers followed good coding practices. They wrote clean and organized code with comments explaining what each part does. They used meaningful names for variables and functions so that the code is easy to understand. They also kept backups of their work to prevent data loss.

Communication was very important during the development phase. The team met regularly to discuss progress, share problems, and help each other. When a team member was stuck on a difficult programming task, the others would help find a solution. This teamwork made the development process faster and more enjoyable.

### 3.2.4 Phase 4: Testing

The fourth phase of the Agile Development Model is Testing. Testing is the process of checking if the system works correctly and if it meets the requirements. In Agile, testing is not done only at the end. Instead, it is done continuously throughout the development process. Every time a new feature is built, it is tested before being added to the main system.

The researchers used several types of testing to make sure their system was reliable and effective. These testing methods are explained in detail in Section 3.6, but a brief overview is provided here.

**Unit Testing** was done by the developers themselves. In unit testing, individual parts of the code are tested separately. For example, if the researchers built a function that calculates the total price of a property appointment, they would test that function with different numbers to make sure it always gives the correct answer. Unit testing helps find small errors in the code before they become big problems.

**System Testing** was done to check if all the parts of the system work together correctly. Even if each individual feature works well on its own, problems can appear when they are combined. System testing involves going through the entire system from start to finish, performing all possible actions, and checking if everything flows smoothly.

**User Acceptance Testing** was done with the help of the client and some volunteer users. In this type of testing, real users try the system and give feedback. The researchers asked the users to perform specific tasks, such as searching for a property, making a appointment, and sending a message. The researchers watched how the users interacted with the system and asked them questions about their experience. This feedback was very valuable because it showed the researchers how actual users feel about the system.

During testing, the researchers kept a record of all the problems they found. This record is called a bug list or issue tracker. For each problem, they wrote down what went wrong, where it happened, and how serious it was. They then fixed the problems one by one, starting with the most serious ones. After fixing a problem, they tested again to make sure the fix worked and did not cause new problems.

Testing is a critical phase because it directly affects the quality of the final system. A system that has not been tested properly might have hidden errors that cause frustration for users or even data loss. The researchers took testing very seriously and spent a significant amount of time on it.

### 3.2.5 Phase 5: Deployment (Prototype)

The fifth phase of the Agile Development Model is Deployment. Deployment means making the system available for use. In this study, the researchers deployed a prototype of the system. A prototype is an early version of the system that is used for demonstration and testing purposes. It may not have all the features of the final product, but it shows how the system will work.

The researchers deployed the prototype on a local server and also made it accessible through a web hosting service. This allowed the client and test users to access the system from their own devices. The deployment process involved several steps.

First, the researchers prepared the server environment. They made sure that the server had all the necessary software installed, including the web server software, the PHP programming language, and the MySQL database system.

They also configured the security settings to protect the system from unauthorized access
.
Second, the researchers transferred all the system files to the server. This included the HTML, CSS, JavaScript, and PHP files, as well as the images and other media files. They made sure that all files were placed in the correct folders and that the file permissions were set correctly.

Third, the researchers set up the database on the server. They created the database, created all the necessary tables, and imported any sample data that was needed for testing. They also tested the connection between the system files and the database to make sure data could be read and written correctly.

Fourth, the researchers performed final checks before announcing that the prototype was ready. They tested the login system, checked all the main pages, and verified that the system loaded properly on different browsers and devices.

After deployment, the researchers collected feedback from users. They asked questions like: Is the system easy to use? Do the pages load quickly? Are there any features missing? Is there anything confusing? This feedback was used to plan improvements for the next iteration of development.

The Agile approach means that deployment is not the final step. Even after the prototype is live, the researchers continue to improve the system based on user feedback. This cycle of building, testing, deploying, and improving continues until the system meets all the requirements and satisfies the users.

In summary, the Agile Development Model provided a flexible and effective framework for building the web-based real estate management system. It allowed the researchers to understand requirements, design carefully, build in small steps, test continuously, and deploy a working prototype. The iterative nature of Agile ensured that the system evolved based on real feedback and real needs, making it more likely to succeed in solving the problems identified in Chapter 1.

## 3.3 System Architecture

System architecture refers to the overall structure of the system. It shows how the different parts of the system are organized and how they communicate with each other. Think of system architecture like the floor plan of a building. It shows where each room is located, how big it is, and how it connects to other rooms. Without a good architecture, a system can become messy, slow, and difficult to maintain.

The web-based real estate management system uses a client-server architecture. This is one of the most common and reliable ways to build web applications. In a client-server architecture, there are two main sides: the client side and the server side. The client side is what the user sees and interacts with. The server side is where data is stored and processed. These two sides communicate with each other through the internet.

### 3.3.1 The Client Side (Front End)

The client side, also called the front end, is the part of the system that runs on the user's device. When a person opens the real estate website on their computer or smartphone, the front end is what appears on their screen. The front end includes all the visual elements such as text, images, buttons, forms, and menus.

The front end of this system was built using three main technologies: HTML, CSS, and JavaScript. HTML stands
for HyperText Markup Language. It is the basic building block of web pages. HTML is used to define the structure of a page, such as where the headings go, where the paragraphs are, and where images are placed. Every web page on the internet uses HTML.

CSS stands for Cascading Style Sheets. CSS is used to make web pages look attractive. It controls colors, fonts, spacing, sizes, and layouts. Without CSS, web pages would look plain and boring, like plain text documents.

With CSS, the researchers were able to create a professional and visually appealing design for the real estate system.

JavaScript is a programming language that adds interactivity to web pages. It allows the page to respond to user actions. For example, when a user clicks a button to view more property images, JavaScript makes that action happen. JavaScript can also validate form inputs, show pop up messages, and create animations. In this system, JavaScript was used to make the user experience smooth and interactive.

The front end communicates with the back end by sending requests. For example, when a user searches for properties in San Miguel, the front end sends a request to the server asking for properties that match the search. The server processes the request, looks in the database, and sends back the results. The front end then displays those results to the user.

The researchers made sure that the front end was responsive. A responsive design means that the website automatically adjusts its layout to fit the screen size of the device being used. Whether the user is on a large desktop monitor, a tablet, or a small smartphone, the website should still look good and be easy to use. This was achieved by using CSS techniques such as media queries and flexible grid layouts.

### 3.3.2 The Server Side (Back End)

The server side, also called the back end, is the part of the system that runs on the web server. The user does not see the back end, but it is responsible for all the important work that happens behind the scenes. The back end processes user requests, runs business logic, and communicates with the database.

The back end of this system was built using PHP. PHP stands for PHP: Hypertext Preprocessor. It is a popular programming language for web development. PHP is especially good for creating dynamic web pages, which are pages that change based on user input or database content. For example, when a user views a property details page, PHP retrieves the property information from the database and inserts it into the HTML page before sending it to the user's browser.

PHP was chosen for this project because it is widely used, well-documented, and works very well with MySQL databases. It is also supported by most web hosting services, which makes deployment easier. The researchers had previous experience with PHP, which allowed them to develop the system more efficiently.

The back end handles many important tasks in the system. It checks if users have entered correct login information. It validates form data to make sure users are not sending harmful or incorrect information. It processes appointment requests and stores them in the database. It handlesfile uploads when administrators add property images. It also manages the automated chat responses by checking the user's message and returning an appropriate reply.

Security is a major concern on the server side. The back end must protect the database and user information from attacks. The researchers implemented several security measures, which are discussed in detail in Section 3.11. These include input validation, password hashing, and protection against SQL injection attacks.

### 3.3.3 The Database

The database is the storage area of the system. It is where all the information is kept in an organized way. The database stores user accounts, property listings, images, appointment records, chat messages, and contact form submissions. Without a database, the system would not be able to remember anything from one visit to the next.

The database for this system was built using MySQL. MySQL is a relational database management system. It organizes data into tables, and each table contains rows and columns. For example, the properties table has columns for id, title, price, location, description, and image. Each row in the table represents one property.

MySQL was chosen because it is free, reliable, and works very well with PHP. It is also easy to learn and has a large community of users who can provide help and resources. The researchers used a tool called phpMyAdmin to manage the database. phpMyAdmin provides a visual interface for creating tables, adding data, and running queries.

The database design is very important because a poorly designed database can cause problems such as duplicate data, slow performance, and data loss. The researchers carefully planned the database structure to make sure it was efficient and scalable. They used primary keys to uniquely identify each record, foreign keys to link related tables, and proper data types to save storage space.

The relationship between the front end, back end, and database can be summarized as follows: The user interacts with the front end. The front end sends requests to the back end. The back end processes the requests and talks to the database. The database returns the requested data. The back end formats the data and sends it back to the front end. The front end displays it to the user. This cycle happens every time a user clicks a button, fills out a form, or loads a page.

This three-layer architecture is strong, flexible, and easy to maintain. If the researchers want to change how the website looks in the future, they only need to update the front end. If they want to change how data is processed, they only need to update the back end. If they need to store more types of information, they only need to update the database. This separation of concerns makes the system easier to develop and improve over time.

## 3.4 Tools and Technologies Used

To build the web-based real estate management system, the researchers used several tools and technologies. Each tool was chosen for a specific purpose, and together they form the complete development environment. This section explains each tool in detail, including what it does and why it was selected for this project.

### 3.4.1 HTML (HyperText Markup Language)

HTML is the standard language for creating web pages. It provides the basic structure and content of a website. Every web page on the internet is built using HTML. HTML uses tags to define different elements such as headings, paragraphs, images, links, tables, and forms.

In this project, HTML was used to create all the pages of the real estate system. The home page, property listings page, property details page, appointment page, chat page, login page, registration page, and admin dashboard were all built with HTML. HTML tags were used to organize the content in a logical way so that users can easily understand and navigate the information.

The researchers used HTML5, which is the latest version of HTML. HTML5 includes new features that make web development easier and more powerful. For example, HTML5 has new form input types such as email, date, and number, which help validate user input automatically. It also supports video and audio elements, though these were not heavily used in this project.

HTML is essential because it is the foundation of every web page. Without HTML, there would be no structure or content. Even though HTML alone cannot create beautiful or interactive websites, it is the starting point upon which CSS and JavaScript are added.

### 3.4.2 CSS (Cascading Style Sheets)

CSS is the language used to style and layout web pages. While HTML provides the content, CSS provides the appearance. CSS controls colors, fonts, sizes, spacing, borders, backgrounds, and the overall layout of the

page.In this project, CSS was used to create an attractive and professional design for the real estate system. The researchers chose a color scheme that looks clean and trustworthy, which is important for a real estate business. They used consistent fonts that are easy to read. They created layouts that organize information clearly, such as grid layouts for property listings and centered layouts for login forms.

The researchers used CSS3, which is the latest version of CSS. CSS3 includes powerful features such as animations, transitions, gradients, and flexible box layouts. These features were used to create smooth hover effects on property cards and responsive layouts that adapt to different screen sizes.

A special technique called media queries was used to make the website responsive. Media queries allow the CSS to change based on the width of the screen. For example, on a large screen, property listings might be displayed in three columns. On a small smartphone screen, the same listings might be displayed in one column for easier reading. This adaptability is very important because many users access websites from mobile devices.

CSS was also used to create the admin dashboard layout. The dashboard includes a sidebar menu, a top navigation bar, and a main content area. CSS ensured that these elements stayed in place and looked organized regardless of the amount of content being displayed.

### 3.4.3 JavaScript

JavaScript is a programming language that adds behavior and interactivity to web pages. While HTML and CSS create static pages, JavaScript makes them dynamic and responsive to user actions.

In this project, JavaScript was used for many purposes. It was used to validate form inputs before they are sent to the server. For example, if a user tries to submit a appointment form without entering their name, JavaScript can show an error message immediately without needing to reload the page. This saves time and reduces server load.

JavaScript was also used to create interactive elements such as image sliders on property details pages. When a property has multiple images, JavaScript allows users to click through the images or swipe on mobile devices. This creates a better viewing experience than simply showing one static image.

Another important use of JavaScript was in the search and filter features. When users type in the search box or select filter options, JavaScript can update the displayed results instantly without reloading the entire page. This technique is called AJAX (Asynchronous JavaScript and XML), and it makes the website feel faster and more modern.

JavaScript was also used for the chat feature. When a user sends a message, JavaScript sends it to the server in the background and displays the response immediately. This creates a smooth chat experience similar to popular messaging applications.

The researchers used plain JavaScript without heavy frameworks, which kept the system lightweight and easy to understand. They also made sure that the JavaScript code was well-organized and commented for future maintenance.

### 3.4.4 PHP (PHP: Hypertext Preprocessor)

PHP is a server-side programming language designed specifically for web development. It runs on the web server and generates dynamic content that is sent to the user's browser.

In this project, PHP was the main back-end language. It was used to handle all server-side tasks, including user authentication, database queries, form processing, file uploads, and session management. Every time a user interacts with the system in a way that requires data storage or retrieval, PHP handles that request.

PHP was chosen for several reasons. First, it is free and open-source, which means there are no licensing costs. This is important for a student project with a limited budget. Second, it has excellent support for MySQL databases, which was the chosen database system. Third, it has a large community and extensive documentation, making it easy to find solutions to problems. Fourth, it is supported by almost all web hosting providers, making deployment simple.

The researchers wrote PHP scripts to perform specific tasks. For example, there are scripts for logging in users, registering new accounts, adding properties, editing properties, deleting properties, making appointment, retrieving chat messages, and updating appointment statuses. Each script was designed to be secure and efficient.

PHP also handles the inclusion of common page elements. For example, the navigation menu and footer appear on almost every page. Instead of copying the same HTML code to every file, the researchers created separate PHP files for the menu and footer, and then included them in each page. This makes maintenance easier because changes only need to be made in one place.

### 3.4.5 Maria DB

Maria DB is a database management system that stores and organizes data. It is one of the most popular database systems in the world and is commonly used with PHP in web development.

In this project, Maria DB was used to create and manage the database. The database contains several tables, each designed to store a specific type of information. The main tables include:

**Users table:** Stores client account information such as names, emails, phone numbers, and passwords.

**Admin users table:** Stores administrator and agent account information.

**Properties table:** Stores property listings including titles, descriptions, prices, locations, images, and availability status.

**Property images table:** Stores additional images for each property, linked to the main properties table.

**Appointments table:** Stores appointment requests including client information, property details, preferred dates and times, and status.
**Chat logs table:** Stores messages exchanged between users and the system.

**Contact messages table:** Stores messages sent through the contact form.

MySQL uses a language called SQL (Structured Query Language) to interact with the database. The researchers wrote SQL queries to insert new data, update existing data, delete old data, and retrieve information. For example, when a user searches for properties in a specific location, the system runs a SQL query that looks for matching records in the properties table.

MySQL was chosen because it is reliable, fast, and free. It also integrates seamlessly with PHP through built-in functions. The researchers used prepared statements in their SQL queries to prevent security vulnerabilities such as SQL injection attacks.

### 3.4.6 Visual Studio Code

Visual Studio Code, often called VS Code, is a code editor used by developers to write and manage their code. It is free, lightweight, and supports many programming languages through extensions. In this project, Visual Studio Code was the main development tool. The researchers used it to write HTML, CSS, JavaScript, and PHP code. VS Code provides helpful features such as syntax highlighting, which colors different parts of the code to make it easier to read. It also provides code completion, which suggests possible code as the developer types, saving time and reducing errors.

VS Code has a built-in file explorer that makes it easy to navigate between different project files. The researchers could quickly switch between the front-end files, back-end scripts, and configuration files. VS Code also has an integrated terminal, which allowed the researchers to run commands and manage the local server without leaving the editor.

Extensions were installed to enhance productivity. For example, extensions for PHP provided better code suggestions and error detection. Extensions for HTML and CSS helped with formatting and previewing changes. A live server extension allowed the researchers to see their web pages update automatically as they made changes. VS Code was chosen because it is modern, free, and widely used by developers around the world. Its user-friendly interface and powerful features made the development process faster and more organized.

### 3.4.7 XAMPP

XAMPP is a software package that provides a complete local server environment for web development. The name XAMPP stands for Cross-Platform, Apache, MariaDB, PHP, and Perl. It includes everything needed to run a PHP and MySQL application on a local computer.

In this project, XAMPP was used during the development and testing phases. It allowed the researchers to build and test the system on their own computers before deploying it to a live web server. XAMPP includes Apache, which is the web server software that handles HTTP requests. It also includes MySQL (through MariaDB) for the database, and PHP for processing scripts.

The researchers installed XAMPP on their development computers. They placed their project files in the htdocs folder, which is the root directory for web files in XAMPP. They started the Apache and MySQL services from the XAMPP control panel, and then they could access their website by typing localhost in their web browser.

XAMPP was chosen because it is free, easy to install, and provides all the necessary components in one package. It eliminated the need to install and configure each component separately, which saved time and reduced technical difficulties.

### 3.4.8 phpMyAdmin

phpMyAdmin is a web-based tool for managing MySQL databases. It provides a visual interface that makes it easy to create databases, create tables, add data, run queries, and manage users.

In this project, phpMyAdmin was used extensively during development. The researchers used it to create the database schema, define table structures, set primary keys, and establish relationships between tables. They also used it to insert sample data for testing purposes and to verify that their PHP scripts were storing data correctly.

phpMyAdmin was especially useful for debugging. When a database query did not work as expected, the researchers could use phpMyAdmin to test the query directly and see the results. They could also browse the tables to check if data was being saved correctly.

phpMyAdmin comes included with XAMPP, so no additional installation was needed. Its user-friendly interface made database management accessible even for team members who were less experienced with SQL commands.

### 3.4.9 Web Browser (Google Chrome and Mozilla Firefox)

Web browsers were essential tools for testing and viewing the system. The researchers used Google Chrome and Mozilla Firefox to open the website, test features, and check the appearance of pages.

Different browsers can sometimes display the same website slightly differently. Therefore, the researchers tested the system on both Chrome and Firefox to ensure consistency. They also used the browser's developer tools, which allow developers to inspect HTML elements, view CSS styles, debug JavaScript, and monitor network requests.

The developer tools were very helpful for fixing layout problems, identifying JavaScript errors, and measuring page load times. The responsive design mode in Chrome's developer tools allowed the researchers to see how the website would look on different screen sizes without needing multiple physical devices.

### 3.4.10 Git and GitHub

Git is a version control system that tracks changes to code over time. GitHub is a web-based platform that hosts Git repositories and provides collaboration features.

In this project, the researchers used Git to manage their source code. Every time a team member made significant changes, they committed those changes with a descriptive message explaining what was done. This created a history of all changes, which made it easy to see who did what and when.

GitHub was used to store the project repository online. This provided a backup of the code and allowed team members to access the latest version from any computer. GitHub also made it easy to merge changes from different team members and resolve conflicts when two people edited the same file.

Version control is very important in team projects because it prevents loss of work and helps coordinate efforts. The researchers could experiment with new features on separate branches without affecting the main code base, and then merge the successful changes when ready.

## 3.5 Data Gathering Techniques

Data gathering is the process of collecting information that helps the researchers understand the problem and design the right solution. Without accurate and relevant data, the researchers might build a system that does not solve the real needs of the users. This section explains the techniques used to gather data for this study.

### 3.5.1 Review of Existing Real Estate Platforms

The first data gathering technique was the review of existing real estate platforms. The researchers studied popular real estate websites and applications both locally and internationally. They looked at how these platforms display property listings, how they handle user searches, how they process appointment, and what communication features they offer.

Some of the platforms reviewed included international sites like Zillow and Realtor.com, as well as local Philippine real estate websites. The researchers paid attention to the user interface design, the types of information shown for each property, the filtering and sorting options, and the overall user experience.

This review provided several benefits. First, it showed the researchers what features are considered standard in the industry. For example, most real estate platforms include high-quality images, detailed descriptions, and map locations. Second, it revealed what features make a platform stand out. Third, it helped the researchers identify common mistakes or annoying features that they should avoid in their own system.

The researchers documented their findings from each platform review. They took screenshots, listed features, and noted both positive and negative aspects. This documentation was used during the requirements analysis and design phases to guide decision-making.

### 3.5.2 Interviews with the Client

The second and most important data gathering technique was conducting interviews with the client. The client in this study is the real estate business owner or manager who will use the system. Interviews allowed the researchersto ask direct questions and get detailed answers about the business operations, challenges, and needs.

The researchers prepared an interview guide with a list of questions organized by topic. The questions covered areas such as:

**Current business processes:** How do you currently list properties? How do clients find you? How do you handle inquiries? Challenges and problems: What difficulties do you face every day? Where do you lose potential clients? What tasks take too much time? Desired features: What would you like the system to do? What information should be included for each property? How should appointment work?

**User expectations:** Who will use the system? What level of computer skills do they have? What devices will they use?

The interviews were conducted in person at the real estate office in San Miguel, Iloilo City. The researchers recorded the answers (with permission) and took detailed notes. They asked follow-up questions when something
was unclear. The interviews were conversational and friendly, which helped the client feel comfortable sharing honest information.

The information gathered from interviews was extremely valuable because it came directly from the people who would use the system. For example, the client explained that they often receive inquiries from OFWs who are abroad and cannot visit properties in person. This insight led the researchers to prioritize high-quality images and detailed descriptions. The client also mentioned that they sometimes forget to follow up with potential buyers, which led to the development of the inquiry tracking feature.

After the interviews, the researchers analyzed the responses. They looked for common themes, identified the most frequently mentioned problems, and ranked the requested features by importance. This analysis was used to create the requirements specification document that guided the entire development process.

### 3.5.3 Observation

In addition to interviews, the researchers also used observation as a data gathering technique. Observation means watching how the real estate business operates in its natural setting. The researchers visited the office and observed how agents handle phone calls, how they write down client information, how they update property boards or lists, and how they manage their daily tasks.

Observation provided information that interviews alone could not capture. Sometimes people say they do things one way, but observation shows that they actually do things differently. Observation also revealed small details that the client might not think to mention, such as how often the phone rings, how many clients walk in without appointments, or how agents search through paper files to find property information.

The researchers took notes during observation sessions and later discussed what they saw. These observations helped them understand the real workflow of the business and design a system that fits naturally into that workflow rather than forcing the business to change completely.

### 3.5.4 Document Analysis

The researchers also analyzed existing documents used by the real estate business. These included property brochures, listing sheets, appointment logs, and any existing spreadsheets or databases. Document analysis showed what information the business currently collects and how it is organized.

For example, looking at a property listing sheet reveals what details are typically recorded for each property: address, lot area, floor area, number of bedrooms, number of bathrooms, price, and special features. This information was used to design the database fields for the properties table. Looking at appointment logs showed what information is needed when booking a viewing: client name, contact number, preferred date and time, and property of interest.

Document analysis ensured that the digital system would capture all the information that the business already considers important. It also helped identify information that was missing or inconsistently recorded, which the new system could improve.

## 3.6 Testing Procedures

Testing is one of the most important steps in building a software system. It is the process of checking if the system works correctly, if it meets the requirements, and if it is ready for real users. Testing helps find and fix problems before the system is deployed. A system that has not been tested properly can have errors that frustrate users, lose data, or create security risks.

The researchers used three main types of testing for the web-based real estate management system: Unit Testing, System Testing, and User Acceptance Testing. Each type of testing serves a different purpose and was conducted at different stages of the development process.

### 3.6.1 Unit Testing

Unit testing is the process of testing individual parts of the code to make sure each part works correctly on its own. A unit is the smallest testable part of a program, such as a single function, method, or procedure. Unit testing is usually done by the developers themselves as they write the code.

In this project, the researchers performed unit testing throughout the development phase. Every time they wrote a new function or feature, they tested it immediately before moving on to the next task. This approach is sometimes called test-driven development or continuous testing.

For example, when the researchers built the login function, they tested it with different combinations of usernames and passwords. They tested what happens when the correct credentials are entered, what happens when the wrong password is entered, what happens when an empty form is submitted, and what happens when special characters are used. They check if the function correctly validates the input, queries the database, starts a user session, and redirects the user to the right page.

Another example is the property search function. The researchers tested it with various search terms: exact matches, partial matches, empty searches, searches with no results, and searches with many results. They verified that the function returns the correct properties, sorts them properly, and handles errors gracefully.

Unit testing was also performed on database operations. The researchers tested insert operations to make sure new records are added correctly.They tested update operations to confirm that changes are saved. They tested delete operations to ensure that records are removed without affecting other data. They tested select operations to verify that the right data is retrieved.

The researchers kept a log of all unit tests performed. For each test, they recorded what was being tested, what input was used, what the expected result was, what the actual result was, and whether the test passed or failed. Failed tests were investigated immediately, and the code was fixed before proceeding.

Unit testing is important because it catches errors early when they are still small and easy to fix. If a coding error is not discovered until the end of the project, it might be buried under many layers of other code, making it much harder to find and fix. By testing each unit as it is built, the researchers maintained a higher quality code base throughout the project.

### 3.6.2 System Testing

System testing is the process of testing the entire system as a whole. While unit testing checks individual parts, system testing checks if all the parts work together correctly. It verifies that the complete system meets the requirements and performs all intended functions properly.

System testing was conducted after several iterations of development were completed and the major features were built. The researchers performed system testing by following test cases that covered all the main functions of the system. A test case is a detailed description of a specific test scenario. It includes the steps to perform, the expected outcome, and the actual outcome. The researchers created test cases for every major feature of the system.

For the user registration feature, the test case included steps like: open the registration page, fill in the name field, fill in the email field, fill in the password field, click the register button, and check if the account is created successfully. The expected outcome was that a new user record appears in the database and the user is redirected to the login page.

For the property appointment feature, the test case included: log in as a user, browse properties, select a property, click reserve, fill in the appointment form, submit the form, and check if the appointment is recorded. The expected outcome was that the appointment appears in the admin dashboard and the database.

For the admin features, test cases covered adding a property, uploading images, editing property details, deleting a property, viewing appointment, and responding to messages. Each test case was performed step by step, and the results were recorded.

System testing also included testing for error conditions. The researchers tested what happens when invalid data is entered, when required fields are left blank, when users try to access pages without logging in, and when the system receives unexpected requests. The system should handle these situations gracefully by showing appropriate error messages instead of crashing.

Performance was also checked during system testing. The researchers measured how long pages take to load, how quickly search results appear, and how the system behaves when multiple users are active at the same time. While this system was not expected to handle thousands of simultaneous users, basic performance testing ensured that normal usage would be smooth.

System testing was repeated multiple times. After bugs were fixed, the researchers ran the tests again to make sure the fixes worked and did not break anything else. This repetition is called regression testing, and it is important because sometimes fixing one problem accidentally creates another problem.

### 3.6.3 User Acceptance Testing

User Acceptance Testing, often called UAT, is the final phase of testing. In UAT, real users test the system to determine if it meets their needs and if they are satisfied with it. UAT is important because developers might think a system works well, but the actual users might find it confusing or missing important features.

For this project, the researchers invited the client, several real estate agents, and a few potential clients to participate in User Acceptance Testing. The testing sessions were conducted at the real estate office and at the school computer laboratory. Each participant was given a set of tasks to complete using the system.

The tasks were designed to cover all the main features from the user's perspective. For regular users, the tasks included:

Create a new account. Log in to the system. Browse available properties. Use the search feature to find a specific type of property. View the details of a property. Make a appointment for a property viewing. Send a message through the chat feature. Log out.

For administrators and agents, the tasks included:

Log in to the admin dashboard. Add a new property listing with images. Edit an existing property. View all appointment. Respond to a client message. Update the status of an appointment. Generate or view reports.

The researchers observed the participants as they performed these tasks. They watched where users clicked, where they hesitated, where they made mistakes, and what questions they asked. They did not help the users during the tasks because the goal was to see if the system was intuitive enough to use without assistance.

After completing the tasks, each participant filled out a feedback form. The form asked questions about the overall experience, the ease of use, the visual design, the usefulness of features, and any problems encountered. It also included space for open comments where participants could suggest improvements or mention anything they liked or disliked.

The feedback collected during UAT was very valuable. Some participants suggested adding more filter options for property searches. Others requested clearer confirmation messages after submitting forms. Some mentioned that certain buttons were hard to find. The real estate agents suggested improvements to the admin dashboard layout to make their daily tasks faster.

The researchers analyzed all the feedback and prioritized the suggested changes. They fixed the most critical issues immediately and noted less urgent improvements for future development. After making changes, they invited some participants to test again and confirm that the problems were solved.

UAT gave the researchers confidence that the system was ready for real use. It also helped build trust with the client because the client could see that their input was valued and incorporated into the final product.

## 3.7 Evaluation Criteria

Evaluation criteria are the standards or measures used to judge whether the system is successful. They answer the question: How do we know if the system is good? The researchers established clear evaluation criteria before testing began so that they could objectively measure the system's performance. The system was evaluated based on five main criteria: Functionality, Usability, Accuracy of Real Estate Properties, Security, and Efficiency. Each criterion is explained in detail below.

### 3.7.1 Functionality

Functionality refers to whether the system does what it is supposed to do. It measures if all the required features are present and if they work correctly. A system with good functionality meets all the requirements specified during the requirements analysis phase.

To evaluate functionality, the researchers checked each feature against the requirements list. They asked questions such as:

Can users successfully register and create accounts? Can users log in and log out securely? Can users browse and search for properties? Can users view detailed property information including images? Can users make appointment for property viewings? Can users send messages through the chat or contact form? Can administrators add, edit, and delete property listings? Can administrators view and manage appointment? Can administrators respond to client messages? Does the automated chat provide appropriate responses?

For each question, the researchers tested the feature and recorded whether it worked as expected. They also checked if there were any missing features or incomplete functions. The functionality criterion was considered satisfied if all major features worked correctly and matched the requirements.

Functionality is the most basic and important criterion because a system that does not function properly cannot serve its purpose no matter how beautiful or fast it is.

### 3.7.2 Usability

Usability refers to how easy the system is to use. A system with good usability can be used effectively by people with different levels of technical skill. Usability is important because even a fully functional system will fail if users find it too confusing or frustrating.

To evaluate usability, the researchers considered several factors:

**Learnability:** How quickly can a new user learn to use the system? Can they complete basic tasks without training? Efficiency: Once learned, can users perform tasks quickly?

**Memorability:** If a user stops using the system for a while, can they remember how to use it when they return?

**Error prevention and recovery:** Does the system help users avoid mistakes? If users make mistakes, can they easily recover?

**Satisfaction:** Do users enjoy using the system? Are they satisfied with the experience?

The researchers measured usability primarily through User Acceptance Testing. They observed how long it took participants to complete tasks, how many errors they made, and whether they needed help. They also analyzed the feedback forms where participants rated the ease of use.

Specific usability checks included:

Is the navigation menu clear and consistent on all pages? Are buttons and links easy to find and click? Is the text readable and appropriately sized? Do forms have clear labels and helpful instructions? Are error messages easy to understand? Does the system work well on mobile devices? Are page layouts clean and uncluttered?

The researchers aimed for a system that users could understand within a few minutes of first use. They made improvements based on usability feedback, such as adding clearer labels, improving button placement, and simplifying complex processes.

### 3.7.3 Accuracy of Real Estate Properties

Accuracy of real estate properties refers to whether the property information displayed in the system is correct, complete, and up to date. This criterion is especially important for a real estate system because clients make

important decisions based on the information they see. If the information is wrong, clients might lose trust in the business.

To evaluate accuracy, the researchers checked:

Are property details such as price, location, and description correct? Do property images match the actual properties? Is the availability status accurate? Are sold or reserved properties marked correctly? Are there any duplicate listings? Is property information updated promptly when changes occur? Are location details specific enough for clients to understand where the property is?

The researchers worked with the client to verify that the sample data entered into the system was accurate. They also designed the admin dashboard to make it easy for administrators to update information quickly, which helps maintain accuracy over time.

Accuracy also depends on how the system handles data input. The researchers added validation rules to prevent administrators from entering impossible values, such as negative prices or invalid dates. They also included image upload requirements to ensure that property listings always have visual representations.

### 3.7.4 Security

Security refers to how well the system protects data and prevents unauthorized access. A secure system keeps user information private, prevents hackers from stealing data, and ensures that only authorized users can perform sensitive actions.

To evaluate security, the researchers implemented and tested several security measures:

**Authentication:** Does the login system correctly verify user identities? Are passwords stored securely using hashing?

**Authorization:** Can regular users access admin pages? Can one user access another user's private data? Input validation: Does the system reject harmful input such as SQL injection attempts or script tags?

**Session management:** Are user sessions properly managed? Do sessions expire after a period of inactivity?

**Data protection:** Is sensitive information such as passwords encrypted? Is the database protected from direct access?

**Error handling:** Does the system hide technical error details from users? Do error messages reveal too much information?

The researchers tested security by attempting common attacks on their own system. They tried SQL injection by entering special characters in form fields. They tried to access admin pages without logging in. They checked if passwords were visible in the database. They verified that error messages were user-friendly and did not expose database structure or file paths.

Security is a continuous concern. While the researchers implemented best practices for a student project, they also acknowledged that a live production system would need additional security measures such as SSL certificates, regular backups, and server-level protections.

### 3.7.5 Efficiency

Efficiency refers to how well the system uses resources such as time, memory, and processing power. An efficient system responds quickly to user actions and does not waste server resources.

To evaluate efficiency, the researchers measured:

**Page load time:** How long does it take for pages to fully load? The researchers aimed for load times under three seconds for most pages.

**Search response time:** How quickly do search results appear when users type queries?

**Database query performance:** Are database queries optimized? Do they return results quickly even as the database grows?

**Resource usage:** Does the system use reasonable amounts of server memory and CPU?

**Scalability:** Can the system handle an increasing number of users and properties without significant slowdown?

The researchers optimized efficiency by:

Writing efficient SQL queries that only retrieve necessary data. Using indexes on database tables to speed up searches. Optimizing images so they load quickly without losing quality. Minimizing the use of heavy scripts that slow down page rendering. Using pagination for long lists so that only a few items are loaded at a time.

Efficiency testing was done by monitoring page load times using browser developer tools. The researchers tested the system with various amounts of data to simulate real-world usage. They fixed any bottlenecks that caused slow performance.

## 3.8 Database Design and Implementation

The database is the heart of the web-based real estate management system. It stores all the information that the system needs to function. A well-designed database makes the system fast, reliable, and easy to maintain. A poorly designed database can cause data duplication, slow performance, and errors.

### 3.8.1 Database Planning

Before creating the database tables, the researchers carefully planned what information needed to be stored. They reviewed the requirements from Phase 1 and identified all the data entities in the system. A data entity is any person, place, thing, or concept about which information is stored. In this system, the main entities are users, properties, appointments, messages, and administrators.

For each entity, the researchers listed the attributes or details that needed to be stored. For example, the property entity needed attributes such as title, description, price, location, number of bedrooms, number of bathrooms, and images. The user entity needed attributes such as full name, email address, phone number, and password.

The researchers also identified the relationships between entities. For example, one property can have many appointments. One user can make many appointments. One property can have many images. These relationships helped determine how the tables should be connected using primary keys and foreign keys.

### 3.8.2 Database Tables

The database was implemented using MySQL and contains the following main tables:

Users Table (customers). This table stores information about clients who register on the website. The fields include: id: A unique number that identifies each user. This is the primary key. full name: The complete name of the user. email: The email address of the user, used for login and communication. phone: The contact number

of the user. password: The hashed password for secure login. Admin Users Table (admin_users). This table stores information about administrators and agents who manage the system. The fields include: id: A unique identifier and primary key. username: The login name of the administrator. g mail: The email address used for admin login. password: The hashed password for security.

Properties Table (properties). This table stores all the property listings. The fields include: id: A unique identifier for each property. Primary key. title: The name or headline of the property listing. display_image: The main image shown in property previews. price: The selling or rental price of the property. location: The address or area where the property is located. description: A detailed explanation of the property features. bedrooms: The number of bedrooms. bathrooms: The number of bathrooms. property_page: Additional categorization or page assignment. availability_status: Indicates if the property is available, sold, or reserved.

Property Images Table (property_images). This table stores additional images for each property. It supports a one-to-many relationship with the properties table. The fields include: id: Unique identifier. property_id: A foreign key linking to the properties table. image_path: The file path or URL of the additional image.

Appointments Table (appointments). This table stores appointment requests from clients. The fields include: id: Unique identifier. fullname: The name of the client making the appointment. email: The email address of the client. phone: The contact number of the client. property_id: A foreign key linking to the properties table. preferred_date: The date requested for property viewing. preferred_time: The time requested for property viewing. status: The current status of the appointment (pending, confirmed, completed, cancelled). notes: Additional comments from the client or admin.

Chat Logs Table (chat_logs). This table stores messages exchanged through the automated chat system. The fields include: id: Unique identifier. user_message: The message sent by the user. bot_response: The automated response generated by the system. timestamp: The date and time when the message was sent. user_id: Optional link to identify the user if logged in.

Contact Messages Table (contact_messages). This table stores messages sent through the contact form. The fields include: id: Unique identifier. name: The name of the sender. email: The email address of the sender. phone: The phone number of the sender. message: The content of the message. date_sent: When the message was submitted. is_read: Whether the admin has viewed the message.

### 3.8.3 Database Relationships

The tables in the database are connected through relationships that ensure data integrity and prevent duplication.

**One-to-Many Relationships:** One property can have many images. The property_images table has a foreign key (property_id) that references the id in the properties table. One property can have many appointments. The appointments table has a foreign key (property_id) that references the properties table. One user can make many appointments. The appointments table can also include a user_id field referencing the customers table.

These relationships are enforced using foreign key constraints in MySQL. Foreign key constraints ensure that a record cannot be deleted if other records depend on it. For example, if a property has appointments, the system will prevent accidental deletion of that property unless the appointments are handled first.

### 3.8.4 Database Normalization

Normalization is the process of organizing database tables to reduce redundancy and improve integrity. The researchers normalized the database to at least the third normal form (3NF).

**First Normal Form (1NF)** was achieved by ensuring that each table has a primary key and that each column contains only atomic (indivisible) values. For example, instead of storing multiple phone numbers in one field, the database uses one phone field per user.

**Second Normal Form (2NF)** was achieved by removing partial dependencies. All non-key attributes depend fully on the primary key. For example, property details depend only on the property id, not on any other column.

**Third Normal Form (3NF)** was achieved by removing transitive dependencies. Non-key attributes do not depend on other non-key attributes. For example, user details are stored in the users table rather than being duplicated in the appointments table. The appointments table only stores the user_id, and the full user information is retrieved through
the relationship. Normalization helps prevent data anomalies such as update anomalies, insert anomalies, and delete anomalies. It also saves storage space by avoiding unnecessary duplication.

## 3.9 System Features and Functionalities

This section provides a detailed description of all the features and functions included in the web-based real estate management system. Each feature is explained in terms of what it does, who uses it, and why it is important.

### 3.9.1 User Registration and Authentication

The user registration feature allows new clients to create accounts on the system. When a visitor wants to make a appointment or use the chat feature, they are encouraged to register. The registration form asks for basic information: full name, email address, phone number, and password.

The system validates the registration form to ensure that all required fields are filled and that the email address is in the correct format. It also checks if the email address is already registered to prevent duplicate accounts. Passwords are hashed using a secure algorithm before being stored in the database. Hashing transforms the password into a random-looking string that cannot be easily reversed, which protects user accounts even if the database is compromised.

The authentication feature allows registered users to log in using their email and password. The system checks the credentials against the database, starts a user session if they match, and redirects the user to their intended page. If the credentials do not match, the system shows an error message without revealing which part was wrong, which adds a layer of security.

Session management keeps users logged in as they navigate between pages. Sessions expire after a period of inactivity to protect accounts on shared computers. Users can also manually log out to end their session immediately

### 3.9.2 Property Browsing and Search

The property browsing feature is the main attraction for visitors. It displays all available properties in an organized and visually appealing format. Each property is shown as a card containing the main image, title, location, price, and a brief description. Users can click on any property card to see more details.

The search feature allows users to find specific properties based on their preferences. Users can search by keyword, location, price range, number of bedrooms, or number of bathrooms. The system processes the search query, looks through the properties table, and returns matching results.

The search results are displayed dynamically without requiring a full page reload, thanks to JavaScript and AJAX. This makes the search experience fast and smooth. If no properties match the search criteria, the system displays a friendly message suggesting that the user try different keywords or browse all listings.

Sorting options allow users to arrange results by price (low to high or high to low), by date added (newest first), or alphabetically by title. These options help users find the most relevant properties quickly.

### 3.9.3 Property Details Viewing

The property details page provides comprehensive information about a specific property. When a user clicks on a property from the listings page, they are taken to this detailed view.

The page includes: A large image gallery showing all photos of the property. Users can click through images or view them in a larger format. The full property title and price prominently displayed. A complete description including lot area, floor area, features, and nearby amenities. Location information, sometimes including a map or landmarks for reference. Property specifications such as number of bedrooms, bathrooms, and parking spaces. Availability status clearly indicated (Available, Reserved, or Sold). A appointment button that allows interested users to book a viewing appointment. A contact or chat option for immediate inquiries.

The property details page is designed to answer as many client questions as possible without requiring them to contact the agent. This is especially important for OFW clients who cannot visit the property in person and rely entirely on digital information for their initial assessment.

### 3.9.4 Online Appointments System

The online appointment system is one of the most important features of the platform. It allows clients to request appointments to view properties without making phone calls or visiting the office.

When a user clicks the book appointment button on a property details page, they are taken to the appointment form. The form asks for: Full name. Email address. Phone number. Preferred date for viewing. Preferred time for viewing. The system validates all inputs to ensure completeness and correctness. It checks that the selected date is not in the past and that the time is within reasonable business hours. Once submitted, the appointment is saved in the appointments table with a status of "Pending."

The user receives a confirmation message on their email indicating that their request has been received. Meanwhile, the administrator sees the new appointment in the admin dashboard and can confirm or cancel it. The status updates are reflected in the system, and if the client is logged in, they can view their appointment history.

This feature solves the major problem of manual appointment tracking. Instead of writing appointments in notebooks or relying on memory, agents have a centralized digital record of all viewing requests. The system also prevents double-booking by showing which time slots are already taken.

### 3.9.5 Automated Communication and Chat

The automated communication feature improves the speed and consistency of responses to client inquiries. It includes two main components: the contact form and the chat system.

The contact form is a simple form on the website where visitors can send messages without needing an account. It collects name, email, phone, and message. Submissions are stored in the contact_messages table and displayed in the admin dashboard for response.

The chat system provides instant interaction. When a user opens the chat interface, they can type questions about properties, availability, or procedures. The system uses predefined responses for common questions. For

example, if a user asks "What are your office hours?" the system can automatically reply with the business hours. If a user asks about a specific property's availability, the system can check the database and respond instantly.

For questions that the automated system cannot answer, the message is flagged for a human agent to respond. The chat logs are saved so that agents can review past conversations and provide consistent follow-up. This feature significantly reduces the workload of agents by handling repetitive questions automatically. It also ensures that clients receive immediate responses at any time of day, even outside business hours. This 24/7 availability improves client satisfaction and prevents potential buyers from losing interest while waiting for a response.

### 3.9.6 Administrative Dashboard

The administrative dashboard is the control center of the system. Only authorized administrators and agents can access it after logging in. The dashboard provides all the tools needed to manage the real estate business digitally.

The main sections of the dashboard include:

**Property Management:** Add new properties with detailed information and image uploads. Edit existing properties to update prices, descriptions, or availability. Delete properties that are no longer available. View all properties in a list or grid format with quick status indicators.

**Appointment Management:** View all incoming appointment in chronological order. Filter appointments by status (pending, confirmed, completed, cancelled). Update appointment status and add internal notes. Contact clients directly regarding their appointments.

**Message Management:** View all messages from the contact form and chat system. Mark messages as read or unread. Reply to messages or assign them to specific agents. Archive old messages for record-keeping.
**User Management:** View registered client accounts. Manage admin and agent accounts. Control access permissions.

**Reports and Analytics:** View summary statistics such as total properties, total appointments, and total messages. See which properties are generating the most interest. Monitor response times and appointment conversion rates.

The dashboard is designed to be intuitive and efficient. Common tasks are accessible within one or two clicks. The interface uses clear labels, color coding for statuses, and responsive tables that work on both desktop and mobile devices.

### 3.9.7 Content Management

The content management features allow administrators to control the information displayed on the public-facing website. This includes not only property listings but also static content such as the business description, contact information, and any announcements.

Administrators can update the home page content to feature specific properties or promotions. They can manage the about page to reflect current business information. They can also update footer information such as address, phone numbers, and social media links.

While the system does not include a full blog management feature as originally considered, the content management capabilities ensure that the website remains current and relevant without requiring technical knowledge to update.

## 3.10 User Interface Design Approach

The user interface, often called UI, is the visual and interactive part of the system that users see and touch. A good user interface makes the system enjoyable and easy to use. A bad user interface can make even the most powerful system frustrating and confusing. The researchers followed several important principles when designing the user interface for the real estate system.

### 3.10.1 Simplicity

Simplicity means keeping the design clean and avoiding unnecessary complexity. The researchers aimed to present information clearly without overwhelming the user. Each page has a clear purpose and a clear next step. The home page introduces the business and highlights featured properties. The listings page focuses on helping users find properties. The details page focuses on providing comprehensive information about one property.Unnecessary elements were removed. The researchers avoided cluttering pages with too many advertisements, pop-ups, or distracting animations. Every element on the page serves a purpose.

### 3.10.2 Consistency

Consistency means using the same design patterns throughout the system. When users learn how to do something on one page, they should be able to do the same thing on another page without relearning. The researchers maintained consistency by: Using the same color scheme across all pages. Using the same fonts and text sizes for similar types of content. Placing the navigation menu in the same location on every page. Using the same button styles for primary actions. Using the same form layout patterns for all input fields. Maintaining the same footer information across all pages. Consistency reduces the mental effort required to use the system. Users feel more confident and comfortable when they know what to expect.

### 3.10.3 Feedback

Feedback means telling users what is happening in response to their actions. When users click a button, submit a form, or perform any action, the system should confirm that the action was received and indicate what will happen next.

The researchers included feedback mechanisms such as: Success messages after form submissions. Error messages when something goes wrong, with suggestions for how to fix it. Loading indicators when the system is processing a request. Confirmation dialogs before destructiveactions like deleting a property. Visual changes when hovering over clickable elements. Status indicators showing if a property is available, reserved, or sold.

Good feedback prevents users from feeling confused or uncertain. It assures them that the system is working and that their actions have effects.

**3.10.4 Accessibility** Accessibility means designing the system so that it can be used by as many people as possible, including those with disabilities or limitations. While the researchers did not pursue formal accessibility certification, they incorporated basic accessibility practices.

These practices included: Using sufficient color contrast so that text is readable against backgrounds. Using descriptive labels for form fields. Providing alternative text for images (where technically implemented). Ensuring that the system can be navigated using a keyboard. Avoiding design elements that might cause seizures or physical discomfort. Using clear and simple language that is easy to understand.

Accessibility is important because real estate clients come from all backgrounds and may have varying levels of technical ability, vision quality, or device capabilities.

### 3.10.5 Responsive Design

Responsive design means that the website adapts to the screen size of the device being used. The researchers recognized that many users would access the system from smartphones, as mobile internet usage is very high in the Philippines.

The responsive design approach used flexible layouts, flexible images, and media queries. On large screens, property listings might display in multiple columns with large images. On small screens, the same listings stack vertically in a single column with appropriately sized images. The navigation menu collapses into a hamburger icon on mobile devices to save space. Forms and buttons resize to remain tappable on touch screens.

The researchers tested the responsive design by re-sizing their browser windows and using the device simulation tools in Chrome and Firefox. They also tested on actual smartphones and tablets to ensure real-world usability.

## 3.11 Security Measures and Data Protection

Security is a critical aspect of any web-based system that handles personal information and business data. The researchers implemented multiple layers of security to protect the system from common threats and to ensure that user data remains confidential.

### 3.11.1 Input Validation

Input validation is the process of checking data entered by users to ensure it is correct, complete, and safe. The researchers validated all form inputs on both the client side and the server side.

Client-side validation uses JavaScript to check inputs before the form is submitted. It provides immediate feedback to users and reduces unnecessary server requests. For example, it checks if an email field contains the @ symbol, if required fields are filled, and if phone numbers contain only numbers.

Server-side validation uses PHP to check inputs again after they reach the server. This is essential because client-side validation can be bypassed by malicious users. Server-side validation checks for: Required fields: Ensuring that important information is not missing. Data type: Ensuring that numbers are actually numbers and dates are valid dates. Length limits: Preventing excessively long inputs that might cause errors. Format patterns: Using regular expressions to validate email addresses and phone numbers. Harmful content: Stripping or escaping special characters that might be used in attacks.

### 3.11.2 Password Hashing

Password hashing is a security technique that transforms passwords into unreadable strings before storing them in the database. The researchers used a strong hashing algorithm to protect user passwords.

When a user creates an account or changes their password, the system hashes the password immediately. The hash, not the original password, is stored in the database. When the user logs in, the system hashes the password they entered and compares it to the stored hash. If they match, the login is successful.

Hashing is important because even if an attacker gains access to the database, they cannot easily determine the actual passwords. The researchers also encouraged users to choose strong passwords by requiring a minimum length.

### 3.11.3 SQL Injection Prevention

SQL injection is a common attack where malicious users insert harmful SQL code into form inputs to manipulate the database. The researchers prevented SQL injection by using prepared statements in all database queries. Prepared statements separate the SQL command from the data. The database knows which parts are commands and which parts are data, so even if a user enters SQL code as input, it is treated as plain text rather than executable commands. This technique effectively eliminates the risk of SQL injection attacks.

### 3.11.4 Cross-Site Scripting (XSS) Prevention

Cross-site scripting is an attack where malicious scripts are injected into web pages viewed by other users. The researchers prevented XSS by escaping all user-generated content before displaying it on web pages. When users submit messages, comments, or any text that will be shown to others, the system converts special HTML characters into their safe equivalents. For example, the less-than symbol < becomes < and the greater-than symbol > becomes >. This ensures that any script tags entered by users are displayed as plain text rather than being executed by the browser.

### 3.11.5 Session Security

Session security ensures that user sessions cannot be hijacked or impersonated. The researchers implemented several session security measures:

**Session regeneration:** When a user logs in, the system generates a new session ID to prevent session fixation attacks.

**Session timeout:** Sessions expire automatically after a period of inactivity, requiring the user to log in again.

**Secure session storage:** Session data is stored on the server, not in the user's browser, and session IDs are transmitted securely.

**Logout functionality:** Users can manually destroy their session by clicking the logout button.

### 3.11.6 Access Control

Access control ensures that users can only access the parts of the system they are authorized to use. The researchers implemented role-based access control with two main roles: regular users and administrators.

Regular users can browse properties, make appointment, send messages, and manage their own accounts. They cannot access the admin dashboard, edit properties, view other users' appointment, or delete records.

Administrators can access all management features. The system checks the user's role before allowing access to restricted pages. If a non-admin user tries to access an admin page by typing the URL directly, they are redirected to the home page or login page.

### 3.11.7 Data Backup

While the prototype did not include automated backup features, the researchers manually backed up the database and project files regularly during development. Backups were stored in multiple locations to prevent loss due to hardware failure or accidental deletion. For a production deployment, the researchers recommended implementing automated daily backups.

## 3.12 Project Timeline and Development Schedule

A project timeline is a schedule that shows when each phase of the project will be completed. It helps the team stay organized, meet deadlines, and allocate time properly. The researchers created a timeline at the beginning of the project and updated it as needed.

The development of the web-based real estate management system was completed over the course of one semester, approximately four to five months. The timeline was divided into the following major periods:

**Weeks 1 to 2:** Research and Planning. During this period, the researchers finalized their proposal, conducted interviews with the client, reviewed existing platforms, and gathered all necessary data. They wrote the first chapters of the documentation and created the requirements specification.

**Weeks 3 to 4:** System Design. During this period, the researchers created wireframes, mockups, and design documents. They designed the database schema, drew system diagrams, and planned the system architecture. They presented the design to their adviser and the client for approval.

**Weeks 5 to 10:** Development (Iterations 1 to 4). During this period, the researchers built the core features of the system. The first iteration established the project foundation and user authentication. The second iteration built the property management features. The third iteration built the client browsing and search features. The fourth iteration built the appointment system. Each iteration included coding, unit testing, and team review.

**Weeks 11 to 12:** Development (Iterations 5 to 6). During this period, the researchers built the communication features and the admin dashboard. They also implemented security measures and optimized performance. Iteration 5 focused on chat and contact forms. Iteration 6 focused on polishing, security, and responsive design improvements.

**Weeks 13 to 14:** System Testing and Bug Fixing. During this period, the researchers conducted comprehensive system testing. They tested all features together, fixed bugs, and performed regression testing. They also optimized database queries and page load times.

**Weeks 15 to 16:** User Acceptance Testing and Documentation. During this period, the researchers invited users to test the system. They collected feedback, made final improvements, and completed the system documentation. They prepared the deployment environment and created user manuals.

**Weeks 17 to 18:** Deployment and Final Presentation. During this period, the researchers deployed the prototype, conducted final demonstrations for the client and panel, and prepared for the oral defense. They also compiled all project materials and ensured that the documentation was complete.

The researchers used a Gantt chart to visualize the timeline. A Gantt chart is a bar chart that shows the start and end dates of each task. It makes it easy to see which tasks overlap and which tasks depend on others. The team reviewed the Gantt chart during weekly meetings to track progress and adjust the schedule if any delays occurred.

Time management was crucial for the success of the project. The researchers avoided procrastination by setting mini-deadlines for each iteration. They also built buffer time into the schedule to handle unexpected problems or additional requests from the client.

## 3.13 Supporting Diagrams and Visual Representations

Diagrams and visual representations are important tools for understanding how a system works. They provide a clear and concise way to show complex information that would be difficult to explain with words alone. This section describes the main diagrams used in this study.

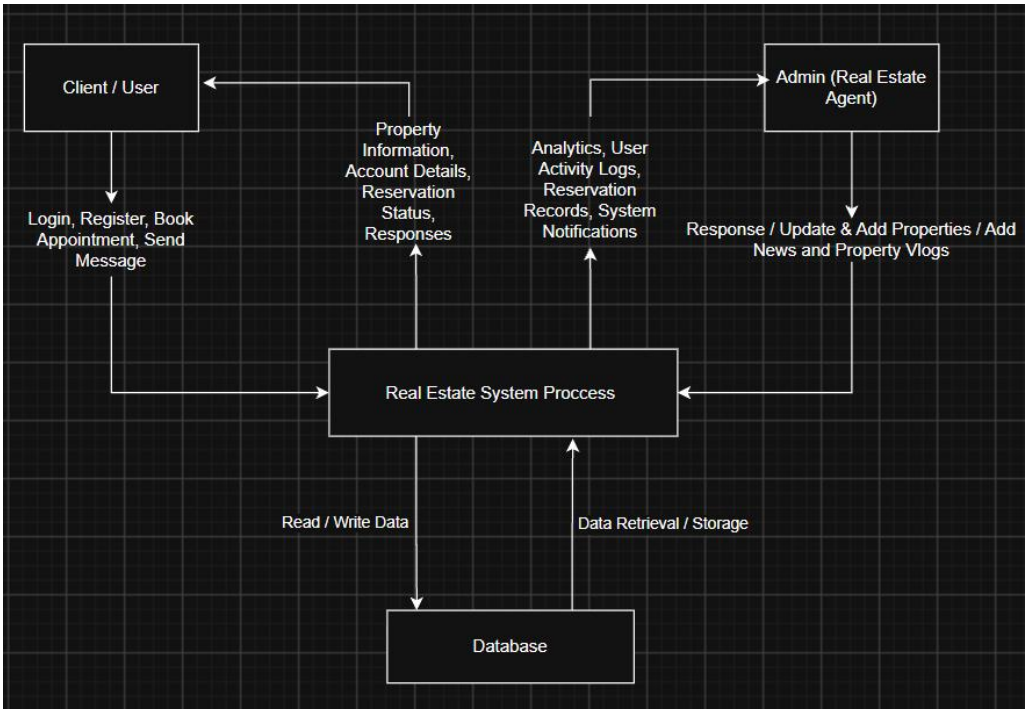
### 3.13.1 Data Flow Diagram (DFD) Level 0

The Data Flow Diagram Level 0, also called the Context Diagram, shows the system as a single process and illustrates how external entities interact with it. The external entities in this system are the Client (user) and the Administrator.

The Context Diagram shows that clients send inputs such as registration information, login credentials, search queries, appointment requests, and messages. The system processes these inputs and returns outputs such as property listings, property details, confirmation messages, and chat responses.

Administrators send inputs such as login credentials, property data, and status updates. The system processes these and returns outputs such as admin dashboard views, property management confirmations, and appointment lists.

The Level 0 DFD provides a high-level overview that is useful for understanding the system's boundaries and its interaction with the outside world.




### 3.13.2 Data Flow Diagram (DFD) Level 1

The Data Flow Diagram Level 1 breaks down the single process from Level 0 into major sub-processes. It shows more detail about how data moves within the system.

The main processes in the Level 1 DFD include:

**User Authentication Process:** Handles registration, login, and session management.

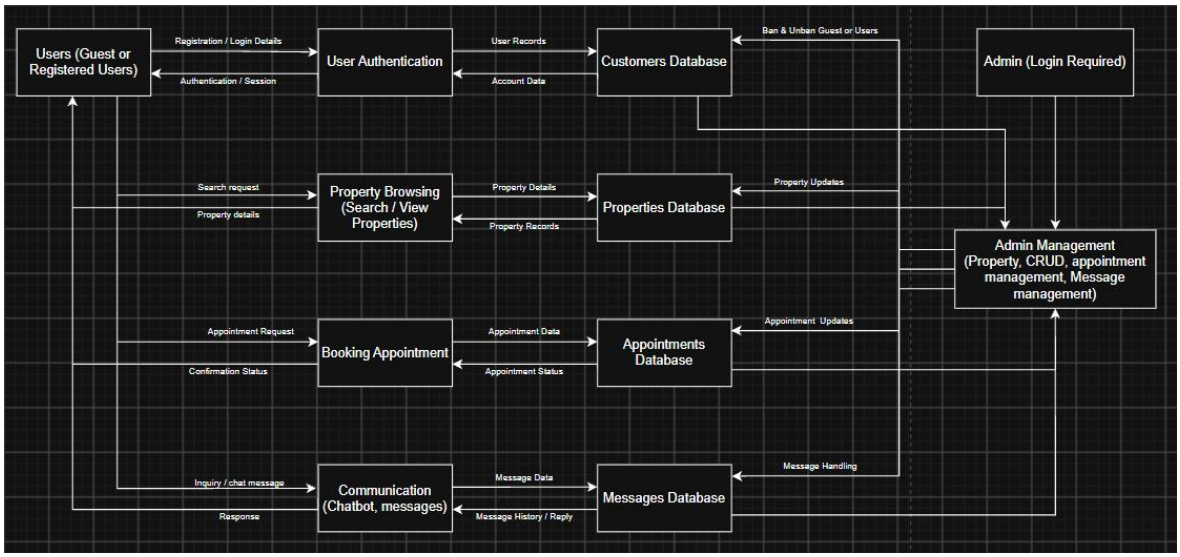
**Property Browsing Process:** Handles search queries and returns property information.

**Appointment Process:** Handles appointment requests and updates the database.
**Communication Process:** Handles messages and chat interactions.

**Admin Management Process:** Handles property CRUD operations, appointment management, and message handling.

Data stores shown in the DFD include the Users Database, Properties Database, Appointments Database, and Messages Database. The diagram shows how data flows from external entities to processes, from processes to data stores, and back to external entities.

The Level 1 DFD is useful for understanding the main functional components of the system and how they relate to each other.




## 3.13.3 Entity Relationship Diagram (ERD)

The Entity Relationship Diagram shows the logical structure of the database. It identifies the entities (tables), their attributes (fields), and the relationships between them.

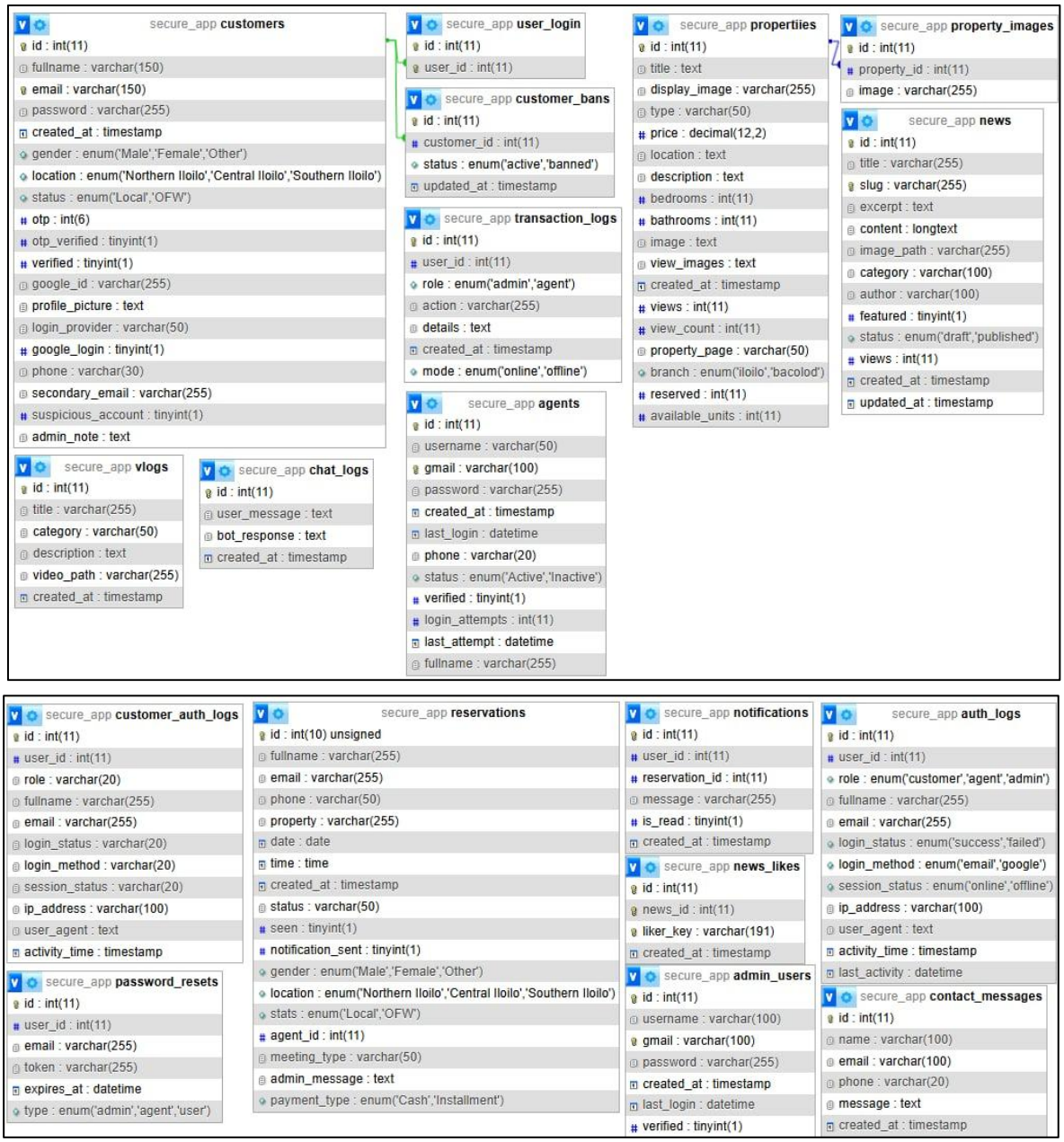
The main entities in the ERD are:

- Customer: Represents clients who register on the website.
- Admin User: Represents administrators and agents.
- Property: Represents real estate listings.
- Property Image: Represents additional photos linked to properties.
- Appointment: Represents appointment requests.
- Chat Log: Represents automated chat conversations.
- Contact Message: Represents messages from the contact form.

The relationships shown in the ERD include: One Customer can make Many Appointments. One Property can have Many Property Images. One Property can have Many Appointments. One Admin User can manage Many Properties and Many Appointments.

The ERD uses crow's foot notation to show the cardinality of relationships. Lines connect related entities, and symbols at the ends of lines indicate whether the relationship is one-to-one, one-to-many, or many-to-many.

The ERD is essential for database design because it ensures that all necessary data is captured and that relationships are properly structured.




### 3.13.4 Use Case Diagram

The Use Case Diagram shows the different ways that users interact with the system. It identifies the actors (types of users) and the use cases (actions they can perform).

The actors in this system are:

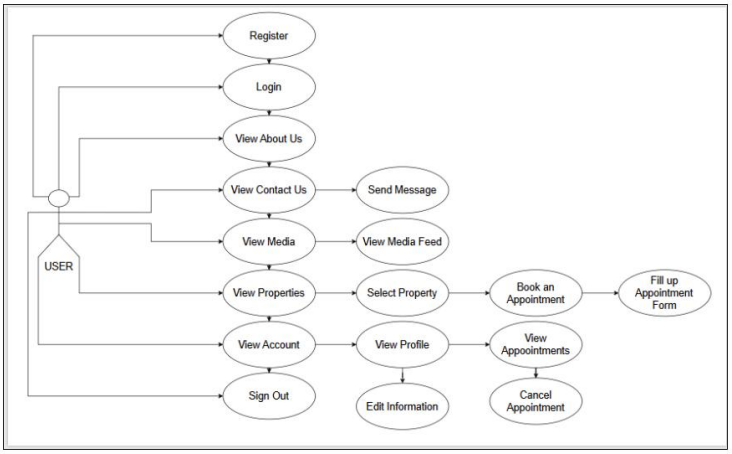
**User (Client):** A person who visits the website to browse properties and make appointment.

**Administrator:** A person who manages the system, properties, and client interactions.

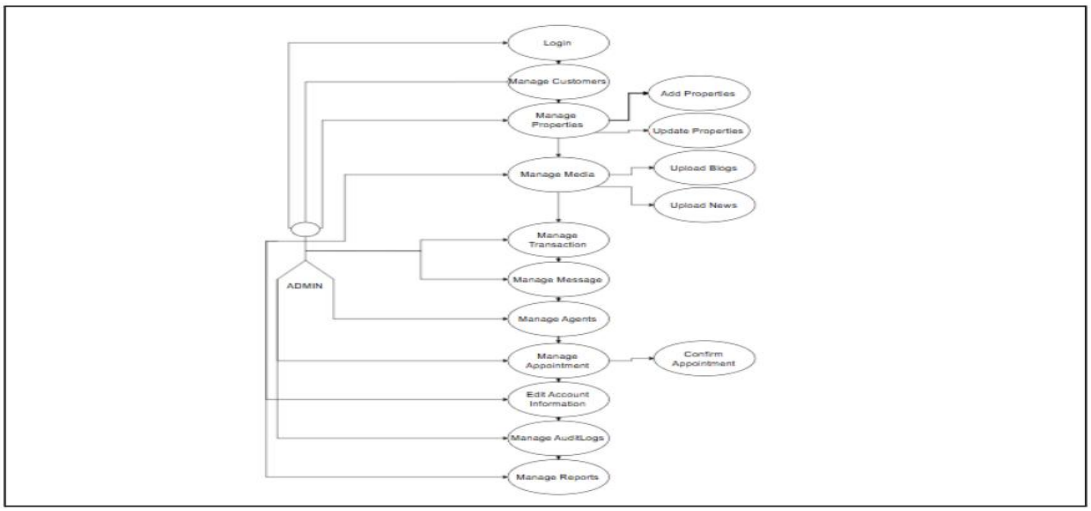
**Agents:** A real estate representative who assists clients by responding to inquiries, providing property details, scheduling property viewings and coordinating with the administrator to deliver efficient client service.

The use cases for the User actor include:

●**Register**: Creates a new user account to access the system.
● **Login:** Allows the user to securely access their account.
● **View About Us:** Displays information about the company and its services.
● **View Contact Us:** Shows the company's contact information.
● **Send Message:** Allows the user to send inquiries or messages.
● **View Media:** Displays media content such as blogs, news, photos, and videos.
● **View Media Feed:** Shows the latest media updates and promotional posts.
● **View Properties:** Displays the list of available real estate properties.
● **Select Property:** Allows the user to choose a property to view its details.
● **Book an Appointment:** Allows the user to schedule a property viewing.
● **Fill Up Appointment Form:** Collects the required information for the appointment.
● **View Account:** Displays the user's account information.
● **View Profile:** Shows the user's personal profile details.
● **Edit Information:** Allows the user to update their personal information.
● **View Appointments:** Displays the user's scheduled appointments.
● **Cancel Appointment:** Allows the user to cancel a scheduled appointment.
● **Sign Out:** Logs the user out of the system.

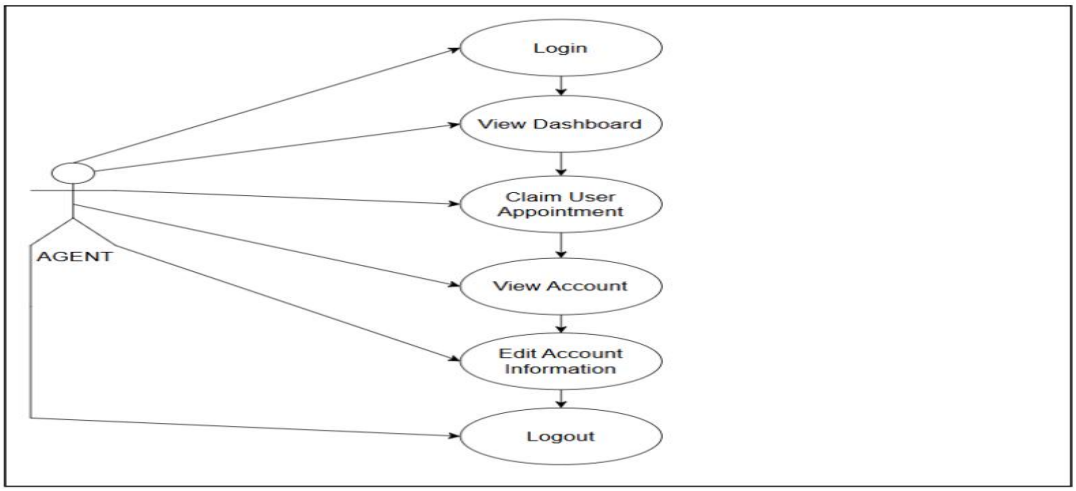



The use cases for the Administrator actor include:
● **Login:** Allows the administrator to access the system.
● **Manage Customers:** Manages customer accounts and information.
● **Manage Properties:** Manages all property listings.
● **Add Properties:** Adds new property listings to the system.
● **Update Properties:** Updates existing property information.
● **Manage Media:** Manages the website's media content.
● **Upload Blogs:** Publishes blog articles on the website.
● **Upload News:** Publishes news and announcements.
● **Manage Transaction:** Monitors and manages property transactions.
● **Manage Message:** Views and responds to customer messages.
● **Manage Agents:** Manages agent accounts and information.
● **Manage Appointment:** Manages customer appointment requests.
● **Confirm Appointment:** Approves and confirms appointments.
● **Edit Account Information:** Updates the administrator's account details.
● **Manage Audit Logs:** Monitors system activities and user actions.
● **Manage Reports:** Generates and manages system report.




The use cases for the Agents actor include:

- **Login:** Allows the agent to securely access the system.
- **View Dashboard:** Displays the agent's dashboard with assigned tasks and appointments.
- **Claim User Appointment:** Allows the agent to accept and manage a customer's appointment request.
- **View Account:** Displays the agent's account information.
- **Edit Account Information:** Allows the agent to update their personal and account details.
- **Logout:** Logs the agent out of the system securely.




The Use Case Diagram uses ovals to represent use cases, stick figures to represent actors, and lines to show which actors participate in which use cases. This diagram is helpful for verifying that all required features are included and that each user type has appropriate access.

## 3.14 Summary of Methodology

This chapter has presented a comprehensive explanation of the methods and approaches used to develop the web-based real estate management system. The research design combined descriptive research to understand the problem and development research to create the solution. The Agile Development Model guided the entire process through its five phases: Requirements Analysis, System Design, Development, Testing, and Deployment.

The system architecture follows a proven client-server model with clear separation between the front end, back end, and database. The tools and technologies were carefully selected for their suitability, availability, and the team's familiarity with them. HTML, CSS, and JavaScript power the user interface. PHP handles server-side processing. MySQL stores all data securely. Visual Studio Code, XAMPP, and phpMyAdmin support the development workflow.

Data was gathered through multiple techniques including platform reviews, client interviews, observation, and document analysis. These techniques ensured that the system was built on real needs rather than assumptions.

Testing was thorough and multi-layered, including unit testing, system testing, and user acceptance testing. Evaluation criteria covered functionality, usability, accuracy, security, and efficiency, ensuring a well-rounded assessment of the system's quality.

The database was carefully designed with proper normalization and relationships. System features were developed to address the specific problems identified in Chapter 1, including property browsing, online appointment, automated communication, and administrative management.

The user interface was designed with simplicity, consistency, feedback, accessibility, and responsiveness in mind. Security was addressed through input validation, password hashing, SQL injection prevention, XSS prevention, session security, and access control. The project followed a realistic timeline with clear milestones and iterative development cycles. Supporting diagrams provided visual clarity for the system's structure and behavior.

All of these methodological choices work together to ensure that the final system is functional, user-friendly, secure, and effective in addressing the challenges of digital marketing and client communication for real estate properties in San Miguel, Iloilo City. The systematic approach documented in this chapter provides a re-plicable framework and a solid foundation for the results and discussions presented in the following chapters.

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# CHAPTER IV

## SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

## 4.2 Overview of the Developed System

The developed system, Iloilo Top Property House System, is a centralized, data-driven platform designed to modernize and streamline traditional real estate operations in San Miguel, Iloilo City. The system addresses the limitations of manual property management, such as inefficient record-keeping, delayed communication, and time-consuming transactions, by providing a secure, responsive, and automated environment for both property seekers and administrators.

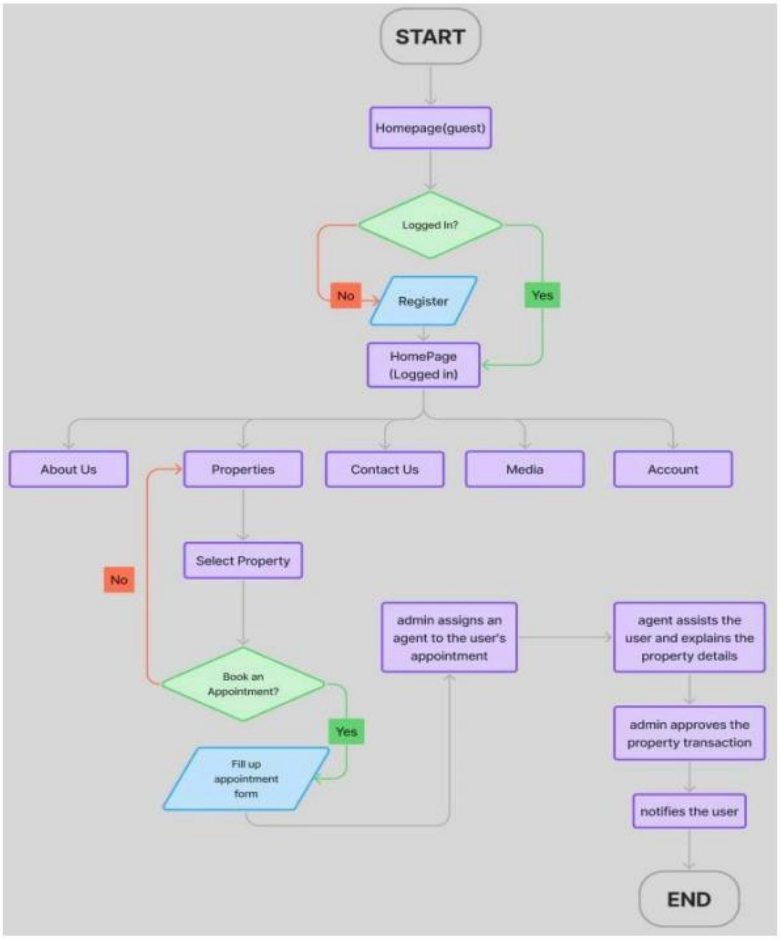
The platform serves as a comprehensive solution for managing property listings, client information, inquiries, appointment, and transaction records in a single, organized database. Through its user-friendly web interface, prospective buyers and renters can conveniently browse available properties, view detailed descriptions, compare listings, and submit inquiries anytime and anywhere using an internet-connected device. This improves accessibility and provides users with a more efficient and convenient property search experience.

For administrators, the system simplifies daily operations by offering tools to manage property information, monitor inquiries, update listing availability, organize client records, and generate reports for better decision-making. The centralized database minimizes data redundancy, reduces the risk of lost or inaccurate records, and ensures that information is consistently updated and readily available.

In addition, the system incorporates essential security features such as user authentication, role-based access control, and secure data storage to protect sensitive information and maintain data integrity. Its responsive design also ensures compatibility across desktop computers, laptops, tablets, and mobile devices, allowing users to access the platform conveniently regardless of the device they are using.

Overall, the Web-Based Real Estate Management System enhances operational efficiency, improves customer engagement, and supports faster, more accurate real estate transactions. By replacing traditional manual processes with a digital and automated solution, the system contributes to a more reliable, transparent, and efficient real estate management process for both clients and administrators.

**4.1 Illustrate the overall workflow of the developed system:**




## 4.3 System Development Methodology

The development of the **Iloilo Top Property Homes** website was accomplished using the **Agile Software Development Methodology**. Agile is a collaborative, iterative, and flexible software development approach that emphasizes continuous planning, development, testing, evaluation, and improvement throughout the project lifecycle. Unlike traditional software development methodologies that follow a fixed sequence of phases, Agile allows developers to adapt to changing requirements by delivering the system in smaller and functional increments called sprints. Each sprint produces a working version of the software that can be evaluated by stakeholders, allowing immediate feedback and continuous enhancement until the desired objectives are achieved.

The researchers selected the Agile Software Development Methodology because the proposed website required continuous improvements based on the evolving needs of the real estate industry. The real estate sector is highly dynamic, with frequent changes in property listings, customer preferences, marketing strategies, and business processes. These changes require a software solution that can easily accommodate modifications without disrupting the overall development process. Agile provides the flexibility needed to implement new requirements, improve existing features, and respond quickly to stakeholder feedback while maintaining the quality and stability of the system.

Another significant reason for selecting Agile was its strong emphasis on stakeholder collaboration. Throughout the development of the proposed website, the researchers maintained regular communication with the property administrators, real estate agents, and prospective users. Their participation allowed the researchers to gain a better understanding of the organization's operational requirements and identify areas where technology could improve existing processes. Stakeholders actively evaluated completed features after every development iteration and provided recommendations for improvement. Their continuous involvement ensured that the

system remained aligned with business objectives and user expectations, resulting in a website that effectively addressed the identified problems of the organization.

Agile methodology also enabled the researchers to divide the project into smaller, manageable development tasks organized into multiple sprints. Instead of attempting to complete the entire website in a single development cycle, each sprint focused on implementing a specific group of functionalities such as user authentication, property listing management, inquiry management, messaging features, administrative controls, and reporting modules. This incremental approach made the development process more organized and allowed the researchers to monitor progress more effectively. Completing smaller tasks within each sprint also reduced project complexity and allowed problems to be identified and resolved before they affected other parts of the system.

At the end of every sprint, the researchers conducted evaluations to determine whether the completed features met the established functional and non-functional requirements. Sprint reviews were performed to verify that each module functioned correctly and complied with the project's quality standards. Stakeholder feedback gathered during these reviews was carefully analyzed and incorporated into subsequent development cycles whenever necessary. This iterative process ensured continuous improvement of the website while minimizing development risks and preventing the accumulation of unresolved issues.

Another advantage of adopting the Agile methodology was the opportunity for continuous testing throughout the software development process. Rather than postponing testing until the entire application was completed, the researchers tested every module immediately after implementation. Unit testing was conducted to verify the correctness of individual functions, while integration testing ensured that different modules worked together properly. System testing was performed to evaluate the overall functionality, usability, performance, and security of the website before deployment. Continuous testing enabled the researchers to identify software defects at an early stage, reducing the time and effort required to correct errors while improving the overall reliability of the proposed system.

The Agile methodology also promoted continuous improvement through regular evaluation and refinement of the system. As each sprint was completed, the researchers analyzed the strengths and weaknesses of the implemented features and identified opportunities for enhancement. New ideas and recommendations from stakeholders were reviewed during sprint planning sessions and prioritized based on their importance and feasibility. This adaptive approach ensured that the website evolved according to the actual needs of the users instead of relying solely on assumptions made during the initial planning stage.

Risk management was another important benefit of using Agile. Since the project was developed incrementally, potential technical issues, design flaws, and usability concerns were identified much earlier than they would have been in traditional development approaches. Early identification of these issues allowed the researchers to implement corrective actions immediately, reducing the possibility of major system failures during the later stages of development. This proactive approach contributed to a more stable and dependable software product.

Furthermore, Agile encouraged effective teamwork and communication among the members of the development team. Regular meetings, progress discussions, and sprint planning sessions allowed the researchers to monitor the status of ongoing tasks, identify development challenges, and coordinate responsibilities efficiently. Continuous collaboration among team members improved productivity, enhanced problem-solving, and ensured that all development activities remained aligned with the overall objectives of the project.

Comprehensive documentation was likewise maintained throughout every phase of the Agile development process. Although Agile prioritizes working software over excessive documentation, the researchers recognized the importance of maintaining complete records for academic purposes and future system maintenance. Documentation included software requirements, database design, system architecture, interface prototypes, sprint plans, coding standards, testing results, stakeholder feedback, revisions, and deployment procedures.

These documents served as valuable references during development and will facilitate future maintenance, troubleshooting, and system enhancements.

Moreover, the Agile methodology supported the production of a high-quality website by encouraging continuous monitoring of software performance and user satisfaction. Every enhancement made during development was carefully validated to ensure that it improved the usability, efficiency, and reliability of the system. This continuous quality improvement process enabled the researchers to develop a website that not only fulfilled the identified system requirements but also provided an effective platform for managing property listings, promoting real estate services, and improving communication between administrators, agents, and prospective clients.

Overall, the Agile Software Development Methodology proved to be the most appropriate approach for developing the **Iloilo Top Property Homes** website. Its iterative development process, continuous stakeholder involvement, frequent testing, adaptive planning, effective risk management, and emphasis on continuous improvement enabled the researchers to successfully develop a reliable, user-friendly, and efficient web-based real estate management system. By following Agile principles throughout the project, the researchers ensured that the proposed website met the objectives of the study while providing a flexible foundation for future system enhancements and technological advancements.

In addition, the Agile methodology allowed the researchers to conduct frequent testing of the completed aspects of the proposed application before moving on to the subsequent stages. In this way, the researchers ensured that all sections of the designed software were efficient and free from defects and bugs. As such, the proposed website had minimal flaws and had gone through extensive quality assurance tests, making it reliable and effective for the intended purpose.

Finally, Agile methodology was essential for ensuring proper documentation of all aspects of the proposed website. This includes the documentation of all requirements, designs, coding, testing, and feedback that were considered in the development process. Through documentation was critical in ensuring the website met the quality standards and was easy to update and maintain in the future. The documents provided a wealth of information that could be used to expand the current version of

the proposed website in the future.The development process consisted of the following phases:

### 4.3.1 Requirements Gathering

The first phase of the Agile Software Development Methodology focused on gathering comprehensive information regarding the existing practices, challenges and requirements of real estate business related to digital marketing and client communication. The researcher conducted interviews, surveys, observation and document analysis involving real estate agencies, property administrators, marketing personnel and potential clients to gain a clear understanding of their operational processes and existing problems.

During this phase, the researchers identified several issues affecting the efficiency of digital marketing activities, including limited online visibility, inconsistent property promotion, delayed responses to client inquiries, ineffective communication between agents and potential buyers, difficulty managing property listings and the absence of an organized platform for handling customer interactions. The researchers also examined existing workflows to determine how technology could improve marketing efficiency, customer engagement and information accessibility.

In addition, the researchers collected both functional and non-functional requirements necessary for successful development of the proposed system. Functional requirements included user registration and authentication, property listing management, property search functionality, inquiry management, contact forms, messaging features, administrator dashboards and reporting capabilities. Non-functional requirements focused on system security, usability, performance, responsiveness, reliability, maintainability and compatibility across various web browsers and devices.

The information gathered during this phase served as the foundation for planning the system architecture, designing the database structure, developing user interfaces and identifying the core features that would be implemented throughout the Agile development process. By conducting through requirements gathering the researcher ensured that the proposed system would effectively address the real-world needs of real estate businesses while improving customer service, communication efficiency and digital marketing performance.

### 4.3.2 System Planning

After gathering all the necessary requirements the researchers proceeded to the planning phase. During this stage the project scope, objectives, development schedule, deliverables and sprint timeline were established. The researchers identified the major modules of the system and prioritized them based on their importance and dependency.

Tasks were distributed among the members of the development team according to their individual skills and responsibilities. Project milestones were established to monitor the progress and ensure that each sprint would be completed within the specified timeframe.

The planning phase also included selecting the development technologies, designing the database architecture, identifying hardware and software requirements and preparing contingency plans to minimize project risks. Proper planning ensured that the development process remained organized, efficient and aligned with the research objectives.

### 4.3.3 System Design

During the system design phase, the researchers translated the gathered requirements into technical design and system specifications. Wireframes, Flowchart, Entity Relationship Diagram (ERD), Data Flow Diagrams (DFD), Use Case Diagram and database structure were prepared to visualize the overall architecture of the proposed system.

The searchers carefully designed the website interface to ensure that it would be user-friendly, visually appealing and easy to navigate. The layout was organized to allow users to browse property listings, communicate with agents and submit inquiries with minimal effort.

Special attention was given to responsive web design to ensure compatibility across desktop computers, laptops, tablets and smartphones. Database relationships were also designed to securely store user information, property details, inquiries and administrative records while maintaining data integrity and security.

The completed system design served as the blueprint for the succeeding development activities.

### 4.3.4 Software Development

The software development phase involved transforming the approved system design into a fully functional web-based system. The researchers followed the Agile Software Development Methodology, developing the website incrementally through multiple sprints to ensure that each feature met the project’s objectives and user requirements.

The researchers developed both the front-end and back-end components of the website. Core features included property listing management, appointment scheduling, user authentication, contact forms and an administrative dashboard to support digital marketing activities and improve communication between real estate businesses and potential clients.

Throughout the development process, each completed module was integrated, reviewed, and refined to ensure proper functionality and compatibility with the entire system. Identified issues were corrected immediately and continuous testing and user feedback helped improve the website's performance, usability and overall reliability.

### 4.3.5 Testing

Testing was conducted after the completion of every sprint to verify that each module functioned according to its intended purpose. The researchers performed functional testing, integration testing, usability testing, compatibility testing and performance testing to identify software defects and improve system performance.

Users participated in evaluating the website by performing actual tasks such as browsing property listings, searching properties, submitting inquiries and accessing administrative functions. Feedback gathered during testing was documented and incorporated into subsequent improvements.

This phase ensured that the website was reliable, secure, responsive and capable of delivering accurate information to users.

### 4.3.6 Deployment

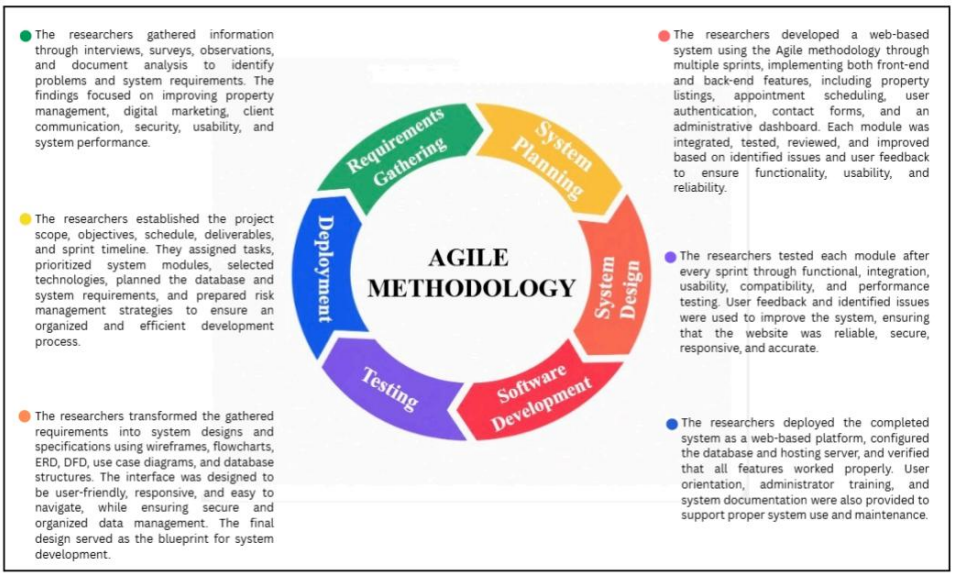
After successful testing and evaluation, the proposed system was deployed as a web-based platform accessible through an internet browser. The deployment process involved publishing the website to a web hosting server, configuring the database, verifying domain accessibility and ensuring that all system components functioned correctly in the production environment.

The researchers conducted final verification to ensure that all features operated smoothly after deployment. Since the proposed system is website-based, users only need an internet connection and compatible web browser to access the platform without installing additional software. The deployment phase also included user orientation, administrator training and preparation of system documentation to facilitate proper system usage and maintenance.

### 4.3.7 Maintenance

The final phase involves continuously monitoring the performance of the web-based system, identifying and resolving errors, applying software updates and improving website features based on user feedback to ensure the system remains reliable, secure and efficient for administrators and clients.

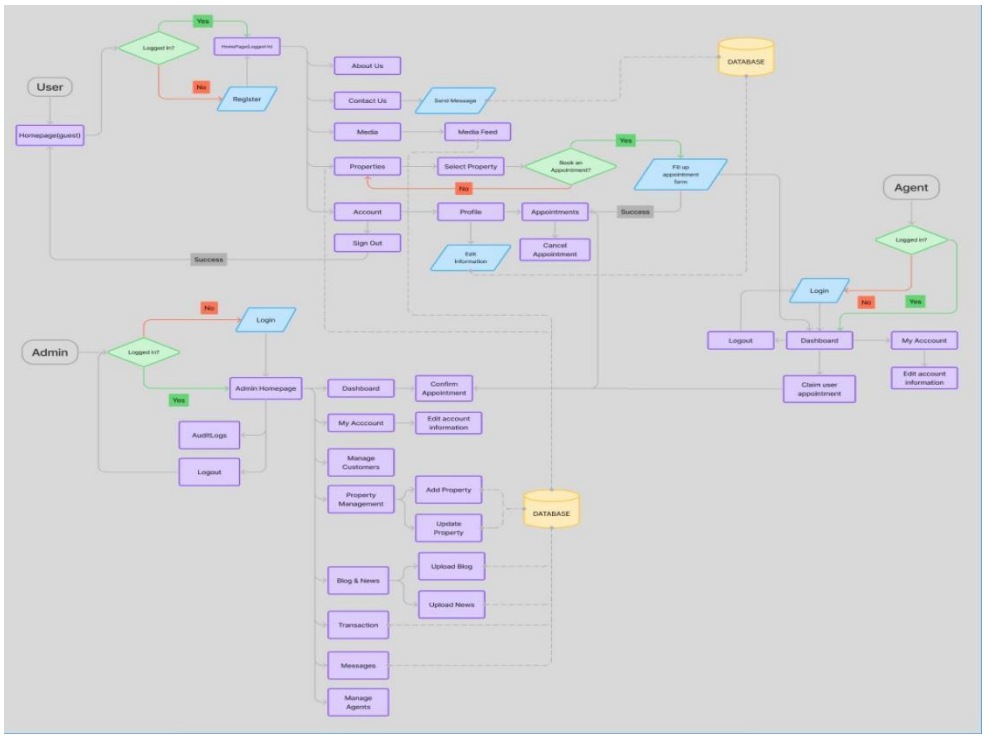
## 4.2 presents the Agile Software Development Life Cycle adopted in the study:



## 4.4 System Architecture

The Iloilo Top Property Home utilizes a **Three-Tier Client-Server Architecture,** which is composed of **Presentation Layer**, **Web Server (Business Logic) Layer** and **Database Layer**. This architecture is designed to organize the website into separate layers, making it easier to develop, maintain and secure. It enables efficient communication between users, the website and the database while ensuring accurate data processing and reliable information management.

## 4.3 illustrates the architecture of the developed system.




The architecture consist of the following components:

### 4.4.1 Presentation Layer

The presentation layer is the front-end interface of the website that allows administrators, property agents and potential clients to access the system through a web browser. It is responsible for displaying web pages that can be accessed using desktop computers, laptop, tablets and smartphones with an internet connection.

Major Website Components include:

- Home Page
- Login Page
- User Registration Page
- Appointment Scheduling
- Property Listings
- Property Details
- Search and Filter Page
- Contact Us Page
- Client Inquiry Form
- Administration Dashboard

### 4.4.2 Web Server (Business Logic) Layer

The **Web Server Layer** processes all requests received from the website interface, It validates user inputs, manages user authentication, processes client requests, retrieves property information and performs all business rules required by the website. This layer also acts as the bridge between the presentation layer and the database layer by processing requests and returning the appropriate information to users.

Its primary responsibilities include:

- Property Listing Management
- Property Search and Filtering
- Appointment Scheduling Management
- Content Management
- Notification Management
- Report Generation

### 4.4.3 Database Layer

The database layer is responsible for storing, organizing and managing all data used by the website. It maintains a centralized database containing user accounts, property listings, client inquiries, communication records and administrative information. This layer ensures that all information is stored securely, retrieved efficiently and protected from unauthorized access while maintaining data accuracy and integrity.

Stored information includes:

- Agent
- Customer
- Properties
- Properties Image
- Contact Message
- Notifications
- Admin User
- Password Reset
- Appointment
- Customer Bans
- User Login
- Authentication Logs
- Customer Authentication Logs
- Transaction Logs
- Vlogs
- Chat Logs
- News

The database server securely stores all website information and supports data backup, recovery and optimization to ensure the long-term reliability, security and performance of the proposed system.

## 4.5 Hardware and Software Requirements

The successful implementation of the Iloilo Top Property Homes requires appropriate hardware and software resources to ensure efficient system performance. These requirements support the development, deployment and operation of the website, allowing users to browse property listings, schedule appointments and enabling administrators to manage property information and client inquiries effectively.

### 4.5.1 Hardware Requirements

Table 4.1 presents the recommended hardware specifications.

| Component | Minimum Specification | Recommended Specification |
|---|---|---|
| Processor | Intel Core i3-1005G1 @ 1.20 GHz | Intel Core i5 or higher |
| Memory (RAM) | 4GB | 8GB Higher |
| Storage | 128GB SSD | 256GB SSD Higher |
| Monitor | 1366 x 768 resolution | 1920 x 1080 resolution |
| Internet Connection | At least 5Mbps | 10Mbps or higher |
| Mobile Device | Android 9/iOS 13 or higher | Android/iOS 15 or higher |

### 4.5.2 Software Requirements

Table 4.2 presents the software requirements.

| Software | Purpose |
|---|---|
| Windows 10/11 | Runs the system software |
| Visual Studio Code | Develops the system code |
| Xampp | Runs the local server |
| PHP | Develops system function |
| MySQL | Manage system database |
| HTML, CSS, JavaScript | Designs interactive web pages |
| phpMyAdmin | Manage the system |
| | |

## 4.6 Database Design

The database serves as the foundation of the Iloilo Top Property Homes by organizing and storing property information, clients records, inquiries, appointments, user accounts and communication data. A well-designed database enables the system to efficiently retrieve, update and manage information while reducing data redundancy and maintaining data accuracy and integrity.

The researchers utilized a **relational database model** implemented using **MySQL**. Database normalization techniques were applied to eliminate duplicate data, establish relationships among related tables and improve the overall performance, consistency and reliability of the system. Figure 4.4 presents the Entity Relationship Diagram (ERD) of the system.

The primary entities of the database include:

● **Agents -** Stores administration and real estate agents account information, including username, email address, password, phone number, account verification, login attempts, account status and last login activity.

● **Customer -** Stores customer information such as full name, email address, password, contact number, gender, location, profile picture, verification status, google login information and account status.

● **Properties -** Contains complete details of real estate properties including property title, type, price, location, description, number of bedrooms and bathrooms, property image, available units, appointment status and total property views.

● **Property Images -** Records appointment request made by customers including customer details, selected property, appointment date and time, assigned agent, meeting type, location and appointment status

● **Notifications -** Stores system-generated notification sent to users regarding appointment, inquiries, updates and other important system activities.

● **Appointment -** Allows users or clients to schedule a meeting or property viewing with a real estate agent at a selected date and time.

● **Contact Message -** Stores messages submitted through the Contact Us page, including the sender's name, email address, phone number, message content and submission date.

● **News -** Manage real estate news articles, announcements, featured content, blog posts, publication status and view count to support digital marketing efforts.

● **Vlogs -** Stores promotional videos, property tours, educational content and other multimedia materials used for online marketing and customer engagements.

● **Chat logs -** Maintains record of conversations between customer and the system chatbot or automated messaging service to improve customer support and communication.

● **Customer Authentication Logs -** Stores login history specifically for customer accounts to monitor authentication activities.

● **Transaction Logs -** Maintains records of administrative actions performed by users including account management, property updates and other system transactions for auditing purposes.

● **User Login -** Tracks currently logged-in users and active sessions within the system.

● **Customer Bans -** Stores information about suspended or banned customer accounts together with the reason and date of action.
● **Password Resets -** Managed password recovery request by storing reset tokens, expiration dates and account type for secure password management.

● **Admin User -** Manage the system by controlling property listings, user accounts and other administrative functions.

### 4.6.1 User Login Table

This table stores active user login sessions.

| Field | Description |
|---|---|
| login_id | Primary Key |
| user_id | Foreign Key |
| login_time | Login Time |
| logout_time | Logout Time |
| session_status | Session Status |

### 4.6.2 Agents

This table stores administrator and real estate agents account information.

| Field | Description |
|---|---|
| agent_id | Primary key |
| fullname | Agent Full Name |
| username | Login Username |
| email | Email Address |
| password | Encrypted Password |
| phone | Contact Number |
| role | User Access Level |
| status | Account Status |
| created_at | Date Created |

### 4.6.3 Customer Table

This table stores customer accounts and personal information.

| Field | Description |
|---|---|
| customer_id | Primary Key |
| fullname | Customer Full Name |
| email | Email Address |
| password | Encrypted Password |
| phone | Contact Number |
| gender | Customer Gender |
| address | Customer Address |
| profile_picture | Customer Profile Image |
| status | Account Status |
| created_at | Date Registered |

### 4.6.4 Properties Table

This table stores real estate property information.

| Field | Description |
|---|---|
| property_id | Primary Key |
| title | Property Title |
| type | Property Type |
| price | Property Price |
| location | Property Location |
| description | Property Description |
| bedrooms | Number of Bedrooms |
| bathrooms | Number of Bathrooms |
| status | Property Availability |
| created_at | Dated Addes |

### 4.6.5 Property Image Table

This table stores images of each property.

| Field | Description |
|---|---|
| image_id | Primary Key |
| property_id | Foreign Key |
| image | Property Image |
| uploaded_at | Upload Date |

### 4.6.6 Appointment Table

This table stores customer appointment records.

| Field | Description |
|---|---|
| appointment_id | Primary Key |
| customer_id | Foreign Key |
| property_id | Foreign Key |
| appointment_date | Appointment Date |
| meeting_type | Meeting Type |
| status | Appointment Status |

### 4.6.7 Notification Table

This table stores system notifications.

| Field | Description |
|---|---|
| notification_id | Primary Key |
| user_id | Foreign Key |
| message | Notification Message |
| status | Notification Status |
| created_at | Date Created |

### 4.6.8 Contact Message Table

This table stores messages submitted through the Contact Us page.

| Field | Description |
|---|---|
| contact_id | Primary Key |
| fullname | Sender Name |
| email | Sender Email |
| phone | Contact Number |
| message | Message Content |
| created_at | Date sent |

### 4.6.9 News Table

This table stores real estate news and announcements.

| Field | Description |
|---|---|
| news_id | Primary Key |
| title | News Title |
| content | News Content |
| image | News Image |
| published_at | Publication Date |

### 4.6.10 Vlog Table

This table stores promotional videos and virtual property tours.

| Field | Description |
|---|---|
| vlog_id | Primary Key |
| title | Video Title |
| video_link | Video URL |
| description | Video Description |
| created_at | Upload Date |

### 4.6.11 Chat Logs Table

This table stores conversations between customers and the chatbot.

| Field | Description |
|---|---|
| chat_id | Primary Key |
| customer_id | Foreign Key |
| message | Chat Message |
| response | Chatbot Response |
| created_at | Date and Time |

### 4.6.12 Authentication Logs Table

This table records administrator and agent login activities.

| Field | Description |
|---|---|
| auth_log_id | Primary Key |
| user_id | Foreign Key |
| login_time | Login Date and Time |
| ip_address | IP Address |
| login_status | Login Status |

### 4.6.13 Customer Authentication Logs Table

This table records customer login activities.

| Field | Description |
|---|---|
| customer_auth_id | Primary Key |
| customer_id | Foreign Key |
| login_time | Login Time and Date |
| ip_address | IP Address |
| login_status | Login Status |

### 4.6.14 Transaction Table

This table stores record system transactions and administrative activities.

| Field | Description |
|---|---|
| transaction_id | Primary Key |
| user_id | Foreign Key |
| action | Performed Action |
| description | Transaction Description |
| created_at | Date and Time |

### 4.6.15 Customer Bans Table

This table stores information about suspended customer accounts.

| Field | Description |
|---|---|
| ban_id | Primary Key |
| customer_id | Foreign Key |
| reason | Reason for Ban |
| banned_at | Date Banned |
| status | Ban Status |

### 4.6.16 Password Resets Table

This table stores password reset requests.

| Field | Description |
|---|---|
| reset_id | Primary Key |
| email | Email Address |
| reset_token | Reset Token |
| expiration_date | Token Expiration Date |
| status | Reset Status |

### 4.6.17 Admin User

This table stores administrator account credentials and login information.

| Field | Description |
|---|---|
| admin_id | Primary Key |
| username | Login Username |
| gmail | Administrator Email Address |
| password | Encrypted Password |
| created_at | Account Creation Date and Time |
| last_login | Last Login Date and Time |
| verified | Account Verification Status |

The relationship among these tables enable the system to manage property listings, customer accounts, appointments , inquiries and communication records efficiently. They also support digital marketing activities by organizing promotional content, tracking customer interactions and generating reports for better decision making. Proper indexing and foreign key constraints maintain data integrity, reduce redundancy and optimize system performance, ensuring reliable and secure management of real estate information.

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# CHAPTER V

# CONCLUSION AND RECOMMENDATIONS

## 5.1 Introduction

This chapter contains the summary of findings, conclusions, and recommendations of the study "**Challenges in Digital Marketing and Client Communication for Real Estate Properties.**" The chapter summarizes the results of the study and provides an overview of the developed web-based real estate digital marketing and client communication website and its ability to address the defined objectives. Moreover, the chapter illustrates how the website resolved the challenges found in real estate business communication processes related to property marketing, client communication, and online appointment.

In addition, this chapter reflects the benefits of the website in terms of effective property marketing through providing an online platform for browsing real estate listings, obtaining the complete information of properties, making inquiries, and requesting appointments.Furthermore, the chapter demonstrates the overall impact of the website, which includes client relationship improvement, efficient administration, and the elimination of limitations related to the manual work process.

Finally, the chapter provides suggestions for future research and developments, which include the addition of new features, improving performance and security, and meeting the changing needs of real estate business to continue the enhancement of the newly designed website.

## 5.2 Summary of Findings

The developed and evaluated proposed ILOILO TOP PROPERTY HOMES has yielded promising results. According to the conducted research, tests, and experiments carried out by the researchers, the following have been determined:

The application provided an easy-to-use and interactive interface through which the clients can access the real estate database and navigate through the information to find their preferred properties. The information about the available properties was comprehensive and included descriptions, prices, locations, images, and availability.

The online appointment system improved the process of scheduling property viewings and client consultations by providing a convenient platform where clients could book appointments online. This feature ensured that appointment details were automatically stored in the database, eliminating the need for paper-based scheduling, reducing manual errors, and preventing duplicate or conflicting appointments.

The online real estate management system improved the marketing strategies as it displayed the latest and most relevant information related to the company on the website. It included informative articles, news, videos, images, and other data needed for marketing and promoting the properties of the real estate companies to the target audience.

The communication system improved the interaction between clients and the company by providing a platform where clients could submit inquiries directly through the website. The system also offered email notifications, in-system notifications, and chat features, enabling fast and efficient communication between the administrator and clients regarding their appointments, property inquiries, and other concerns.

The database improved the organization and management of information about the clients, company properties, appointment, communication system, and other aspects related to the smooth functioning of the real estate company. The database helped avoid errors, repetition, and other potential issues by providing a single platform for storing and managing all of the data. The information in the database was also safe, as physical access to the computer system was limited.

The tests and trials conducted on the proposed real estate management system showed that it was working correctly. The feedback gathered from the respondents revealed that the proposed application was user-friendly, easy to navigate, interact with, and operate. The respondents also confirmed that the developed prototype had a reasonable design to satisfy the demands of the target audience.

Overall, the findings demonstrated that the developed system successfully addresses the challenges in digital marketing and client communication while improving operational efficiency and customer service within the real estate industry.

## 5.3 Conclusion

The Iloilo Top Property Home was a good solution to a problem that many real estate entities usually face when promoting their properties and communicating with clients. The web-based system addressing the issue of property listing management, online appointment and digital marketing and communication was created successfully, making the tasks easier and more organized.

The research conducted fulfilled all the objectives set by the authors. The web-based system was created to help real estate entities offer a user-friendly experience to those browsing their listings and provide the necessary information about the available properties. Moreover, the study met the objectives of developing a system that would improve the online management of property appointment and communication with clients to increase overall customer satisfaction.

In addition, the objective of creating a digital marketing tool was achieved, enabling real estate administrators to post updated information and news about their listings for better promotion and engagement. In a similar manner, the communication tool was created successfully, thus providing a convenient way for clients to get all the information they need about the properties.

Finally, according to the test results, the developed web-based system demonstrated excellent performance and appeared useful to the participants, satisfying all their requirements. Thus, the researchers concluded that the proposed solution could successfully address the initial problem and be beneficial to the entities offering real estate services.

In summary, the study was conducted successfully, addressing the issue of managing property listings, online appointment and communication with clients using a web-based system. The problem was efficiently resolved, which can be useful to the real estate entities and their customers. The researchers hope that the proposed solution will enable real estate administrations to provide better services in the future.

## 5.4 Recommendation

Based on the research findings, the following recommendations are proposed to enhance the sustainability, usability and overall effectiveness of digital marketing and client communication platform for real estate.

### 5.4.1 System Enhancement

● To improve system functionality, the researchers recommend implementing notification features that alert both administrators and clients about appointment updates, responses to inquiries, appointment schedules, and

other important information. These notifications can enhance communication, improve scheduling efficiency, and ensure that clients remain informed throughout the appointment process.

● System security should be strengthened through advanced authentication methods, such as encrypted passwords.

● The user experience can be improved by refining search and filtering capabilities, making it easier for users to find suitable properties.

● Expanding the platform's reach through integrated digital marketing tools such as social media sharing, property advertisements, virtual tours, and client testimonials can increase engagement and visibility.

### 5.4.2 Future Researcher Direction

● Future researchers may consider installing an artificial intelligence or a recommendation system in order to recommend properties to prospective clients based on their preferences and past behavior.

● The future scope may include developing a mobile version of the application or turning it into a progressive web application (PWA).

● Future researchers may consider conducting a long-term study on the effectiveness of the website in the real-estate industry based on the clients' satisfaction level, the impact it has on sales, and general impact analysis on the business using a larger sample size.

● Future researchers may consider adding a payment gateway, document management system and contract management system in order to facilitate easier payment of deposits and clearance of the property by the buyer.

● Future researchers may consider expanding the scope of the study and developing a dashboard that tracks the progress of the website visitors, number of inquiries, and other relevant metrics.

### 5.4.3 Implementation Strategies

● The real-estate companies should conduct proper training of the administrators on how to manage the system in order to ensure efficiency in managing properties and dealing with inquiries using the website.

● The administrators should update the details of the available properties regularly so that the latest information is available to the clients.

● The databases should be backed up on a regular basis in order to avoid loss of valuable data and improve security of the system.

● The real-estate companies should conduct surveys and analysis to understand the needs of their clients in order to ensure that the system is improved based on the users' needs.

● The business should adopt the website to effectively manage property listings and regularly update its features and content to ensure it continues to meet the needs of both administrators and clients.

## 5.5 Impact of the Study

The development and implementation of ILOILO TOP PROPERTY HOMES provide significant contributions to the real estate domain, specifically in digital marketing, client management, and property viewing operations. The overall impact of the study is framed across three main dimensions:

● Impact on Real Estate Operations and Administration: The platform eliminates the reliance on manual, paper-based scheduling and fragmented communication channels. By centralizing property records, inquiry logs, and appointment bookings into a single structured database, the system minimizes human error, prevents double-booking, and reduces administrative workload, allowing staff to focus on closing deals and serving clients effectively.

● Impact on Clients and Prospective Buyers: The web-based system empowers clients by providing round-the-clock access to accurate, transparent, and comprehensive property information (including images, pricing, locations, and real-time availability). The convenience of booking appointments and sending inquiries directly through the platform enhances user experience, fosters trust, and strengthens client-business relationships.
● Impact on Digital Marketing and Industry Standards: The study demonstrates how localized real estate markets can leverage digital marketing tools such as online news updates, informative property blogs, and interactive listings to expand market reach beyond traditional physical advertisements. It serves as a practical model for local real estate firms transitioning toward modern digital transformation.

## 5.6 Summary

This chapter concluded the study on "Challenges in Digital Marketing and Client Communication for Real Estate Properties" through the development and evaluation of ILOILO TOP PROPERTY HOMES. The summary of findings highlighted that the system effectively addressed core operational bottlenecks by providing an interactive property catalog, an automated online appointment module, real-time inquiry management, and a centralized database. Based on user evaluations and testing, the application satisfied all functional requirements, proving to be user-friendly, efficient, and reliable.

Consequently, the study concluded that the platform successfully fulfilled its primary objectives, establishing a modern digital marketing pipeline and streamlining communication between administrators and clients. Recommendations were provided to guide future enhancements including the integration of push notifications, enhanced security, AI-driven property recommendations, mobile application deployment, and online payment gateways to ensure the platform's long-term sustainability and scalability in an evolving market.

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